

L^AT_EX Class for the *Association for Computing Machinery**

Boris Veytsman[†]

2026/08/30, v2.21

Abstract

This package provides a class for typesetting publications of the Association for Computing Machinery.

Contents

1	Introduction	3
2	User's guide	3
2.1	Installation	3
2.2	Invocation and options	5
2.3	Top matter	7
2.4	Top matter of ACM Engage materials	18
2.5	ACM cover page	18
2.6	Internationalization	19
2.7	Tagging	20
2.8	Algorithms	21
2.9	Figures and tables	21
2.10	Descriptions of images	22
2.11	Theorems	23
2.12	Online-only and offline-only material	24
2.13	Note about anonymous mode	24
2.14	Acknowledgments	24
2.15	Bibliography	25
2.15.1	Processing using BibT _E X	25
2.15.2	URL, DOI, eprint	26
2.15.3	Special entry types	27
2.15.4	Dates and sorting	29
2.15.5	Processing using BibL ^A T _E X	30
2.15.6	Manual bibliography	31
2.16	Colors	31
2.17	Other notable packages and typographic remarks	32

*©2016–2026, Association for Computing Machinery

[†]borisv@lk.net, boris@varphi.com

2.18	Counting words	32
2.19	Disabled or forbidden commands	32
2.20	Notes for wizards	33
2.21	Currently supported publications	33
2.22	Samples	35
2.23	A note about sigchi-a format	36
3	Implementation	38
3.1	Identification	38
3.2	Preload hook	38
3.3	Tagging and old kernels	38
3.4	Options	39
3.5	Setting switches	42
3.6	Loading the base class and package	44
3.7	Citations	45
3.8	Internationalization	47
3.9	Sectioning	49
3.10	Hyperxmp and hyperref	50
3.11	Other packages	52
3.12	Paper size and paragraphing	52
3.13	Fonts	56
3.14	Image descriptions	57
3.15	Floats	58
3.16	Lists	61
3.17	Top-matter data	62
3.18	Concepts system	81
3.19	Copyright system	82
3.20	Maketitle hook	88
3.21	ACM Engage top matter	88
3.22	Typesetting top matter	88
3.23	Headers and Footers	104
3.24	Sectioning	110
3.25	TOC lists	114
3.26	Theorems	114
3.27	Balancing columns	117
3.28	Acknowledgments	118
3.29	Conditional typesetting	118
3.30	Additional bibliography commands	119
3.31	Index	119
3.32	End of Class	119

1 Introduction

The Association for Computing Machinery¹ is the world's largest educational and scientific computing society, which delivers resources that advance computing as a science and a profession. It was one of the early adopters of $\text{\TeX}$ for its typesetting.

It provided several different classes for a number of journals and conference proceedings. Unfortunately during the years since these classes were written, the code was patched many times, and supporting different versions of the classes became difficult.

This new consolidated template package replaces all previous independent class files and packages and provides a single up-to-date LaTeX package with optional calls. The package uses only free $\text{\TeX}$ packages and fonts included in $\text{\TeX}$ Live, Mik $\text{\TeX}$ and other popular $\text{\TeX}$ distributions. The new ACM templates use a new font set (libertine) which will need to be installed on your machine before using the templates. Please download and install the libertine font set before writing your paper. Fonts used in the template cannot be substituted; margin adjustments are not allowed.

I am grateful to Michael D. Adams, Leif Andersen, Lawrence Christopher Angrave, Dirk Beyer, Andrew Black, Joachim Breitner, Yegor Bugayenko, Benjamin Byholm, John Collins, Roberto Di Cosmo, Nils Anders Danielsson, Michael Ekstrand, Matthew Fluet, Paolo G. Giarrusso, Ben Greenman, Enrico Gregorio, Jamie Davis, Ulrike Fischer, Jason Hemann, Peter Kemp, David F. Kotz, Luis Leiva, Ben Liblit, Rholais Lii, LianTze Lim, Kuldeep S. Meel, Kai Mindermann, Frank Mittelbach, Serguei Mokhov, Ross Moore, John Owens, Joel Nider, Scott Pakin, Tobias Pape, Henning Pohl, Philip Quinn, Mathias Rav, Andreas Reichinger, Matteo Riondato, Craig Rodkin, Bernard Rous, Feras Saad, Kerry A. Seitz, Jr., M Senthilkumar, David Shamma, Gabriel Scherer, Kartik Singhal, Christoph Sommer, Stephen Spencer, Shin Hwei Tan, Daniel Thomas, Shari Trewin, Zack Weinberg, John Wickerson and many others for their invaluable help.

The development version of the package is available at <https://github.com/borisveytsman/acmart>.

2 User's guide

This class uses many commands and customization options, so it might appear intimidating for a casual user. Do not panic! Many of these commands and options can be safely left with their default values or the values recommended by your conference or journal editors. If you have problems or questions, do not hesitate to ask me directly or the community at <https://github.com/borisveytsman/acmart>, <https://tex.stackexchange.com> or the closest $\text{\TeX}$ Users Group. The world-wide $\text{\TeX}$ Users Group is at <https://tug.org/>; please consider joining us if you use $\text{\TeX}$ regularly.

2.1 Installation

Most probably, you already have this package installed in your favorite $\text{\TeX}$ distribution; if not, you may want to upgrade. You may need to upgrade it anyway since this package uses a number of relatively recent packages, especially the ones related to fonts.

The latest released version of this package can be found on CTAN: <https://www.ctan.org/pkg/acmart>. The development version can be found on GitHub: <https://github.com/borisveytsman/acmart>. At this address you can file a bug report—or even contribute your own enhancement by making a pull request.

¹<http://www.acm.org/>

Please note that the version on Github is a development (or experimental) version: please download it for testing new features. The production version is the one on CTAN and ACM sites.

Most users should not attempt to install this package themselves but should rather rely on their $\text{\TeX}$ distributions to provide it. If you decide to install the package yourself, follow the standard rules:

1. Run `latex acmart.ins`. This will produce the file `acmart.cls`
2. Put the files `acmart.cls`, `acm-jdslogo.png`, and `ACM-Reference-Format.bst` in places where $\text{\TeX}$ can find them (see [1] or the documentation for your $\text{\TeX}$ system).
3. Update the database of file names. Again, see [1] or the documentation for your $\text{\TeX}$ system for the system-specific details.
4. The file `acmart.pdf` provides the documentation for the package. (This is probably the file you are reading now.)

As an alternative to items 2 and 3 you can just put the files in the working directory where your `.tex` file is.

This class uses a number of other packages. They are included in all major $\text{\TeX}$ distributions ($\text{\TeX}$ Live, Mac $\text{\TeX}$, Mik $\text{\TeX}$) of 2015 and later, so you probably have them installed. Just in case here is the list of these packages:

- *amsart*, <https://ctan.org/pkg/amsart>,
- *babel*, <https://ctan.org/pkg/babel>,
- *balance*, <https://ctan.org/pkg/balance>,
- *booktabs*, <https://ctan.org/pkg/booktabs>,
- *caption*, <https://ctan.org/pkg/caption>,
- *cmap*, <https://ctan.org/pkg/cmap>,
- *comment*, <https://ctan.org/pkg/comment>,
- *doclicense*, <https://ctan.org/pkg/doclicense>,
- *draftwatermark*, <https://ctan.org/pkg/draftwatermark>,
- *environ*, <https://ctan.org/pkg/environ>,
- *etoolbox*, <https://ctan.org/pkg/etoolbox>,
- *fancyhdr*, <https://ctan.org/pkg/fancyhdr>,
- *float*, <https://ctan.org/pkg/float>,
- *fontenc*, <https://ctan.org/pkg/fontenc>,
- *framed*, <https://ctan.org/pkg/framed>,
- *geometry*, <https://ctan.org/pkg/geometry>,
- *graphicx*, <https://ctan.org/pkg/graphicx>,

- *hyperref*, <https://ctan.org/pkg/hyperref>,
- *hyperxmp*, <https://ctan.org/pkg/hyperxmp>,
- *iftex*, <https://ctan.org/pkg/iftex>,
- *libertine*, <https://ctan.org/pkg/libertine>,
- *manyfoot*, <https://ctan.org/pkg/manyfoot>,
- *microtype*, <https://ctan.org/pkg/microtype>,
- *natbib*, <https://ctan.org/pkg/natbib>,
- *newtxmath*, <https://ctan.org/pkg/newtxmath>,
- *pbalance*, <https://ctan.org/pkg/pbalance>,
- *pifont*, <https://ctan.org/pkg/pifont>,
- *refcount*, <https://ctan.org/pkg/refcount>,
- *setspace*, <https://ctan.org/pkg/setspace>,
- *totpages*, <https://ctan.org/pkg/totpages>,
- *unicode-math*, <https://ctan.org/pkg/unicode-math>,
- *xcolor*, <https://ctan.org/pkg/xcolor>,
- *xkeyval*, <https://ctan.org/pkg/xkeyval>,
- *xstring*, <https://ctan.org/pkg/xstring>,
- *zi4*, <https://ctan.org/pkg/zi4>,
- *zref-savepos*, <https://ctan.org/pkg/zref-savepos>,
- *zref-user*, <https://ctan.org/pkg/zref-user>.

2.2 Invocation and options

To use this class, put in the preamble of your document

```
\documentclass[<options>]{acmart}
```

There are several options corresponding to the type of the document and its general appearance. They are described below. Generally speaking, the options have key=value forms, for example,

```
\documentclass[format=acmsmall, screen=true, review=false]{acmart}
```

The option `format` describes the format of the output. There are several possible values for this option, for example,

```
\documentclass[format=acmtog]{acmart}
```

Table 1: The possible values for the format option

Value	Meaning
manuscript	A manuscript. This is the default.
acmsmall	Small single-column format. Used for ACMJCSS, ACMJDS, AILET, CIE, CSUR, FAC, GAMES, JACM, JATS, JDIQ, JDS, JEA, JERIC, JETC, JRC, PACMCGIT, PACMHCI, PACMMOD, PACMNET, PACMPL, PACMSE, POMACS, TAAS, TACCESS, TACO, TAIS, TAISAP, TALG, TALLIP (formerly TALIP), TCPS, TDS, TEAC, TECS, TELO, THRI, TIIS, TIOT, TISSEC, TIST, TKDD, TMIS, TOCE, TOCHI, TOCL, TOCS, TOCT, TODAES, TODS, TOIS, TOIT, TOMACS, TOMM (formerly TOMCCAP), TOMPECS, TOMS, TOPC, TOPLAS, TOPML, TOPS, TORS, TOS, TOSEM, TOSN, TQC, TRETS, TSAS, TSC, TSLP, and TWEB, including special issues.
acmlarge	Large single-column format. Used for DLT, DTRAP, HEALTH, IMWUT, JOCCH, and TAP, including special issues.
acmtog	Large double-column format. Used for TOG, including annual conference Technical Papers.
sigconf	Proceedings format for most ACM conferences (with the exception of SIGPLAN) and all ICPS volumes.
sigplan	Proceedings format for SIGPLAN conferences.
acmengage	ACM EngageCSEdu Course materials.
acmcp	ACM cover page.

Actually the words `format=` can be omitted, e.g.,

```
\documentclass[acmtog, review=false]{acmart}
```

The possible formats are listed in Table 1. Note that formats starting with `acm` are intended for journals, transactions, and course materials, while formats starting with `sig` are intended for proceedings published as books.

Sometimes conference proceedings are published as a special issue (or issues) of an ACM journal. In this case, you should use the journal format. The templates `sample-acmsmall-conf.tex` and `sample-acmtog-conf.tex` show how to enter conference information. Note that you need to comment out `\acmJournal{...}` line for such papers to get the conference information in the footers and headers. The `acmtog` format then puts the conference name, date and venue into the footers instead of the journal name, volume, issue and article number. The `acmsmall` format uses the headers and footers of the proceedings formats: the conference name, date and venue in the headers, and no folios, since the page numbers of such a paper are assigned by the proceedings rather than by the article. If you do want the folios, say `\settopmatter{printfolios=true}` after `\acmConference`.

Starting in 2020, ACM retired formats `sigchi` and `sigchi-a`. SIGCHI conferences now use `sigconf` format for their publications. If a file uses `sigchi` format, a warning is issued, and the format is automatically switched to `sigconf`. Format `sigchi-a` can be used for non-ACM documents only (see Section 2.23). The format `acmcp` is used for ACM cover pages discussed in Section 2.5.

There are several Boolean options that can take `true` or `false` values. They are listed

in Table 2. The words `=true` can be omitted when setting a Boolean option, so instead of `screen=true` one can write just `screen`, for example,

```
\documentclass[acmsmall, screen, review]{acmart}
```

The option `review` is useful when combined with the `manuscript` format option. It provides a version suitable for reviewers and copy editors.

Two samples in the `samples` directory, `manuscript` and `acmsmall-submission`, show manuscripts formatted for submission to ACM.

The default for the option `screen` depends on the publication. At present it is `false` for all publications *but* PACM, since PACM is now electronic-only. Thus PACM titles (see Table 5) set this option to `true`. In the future this option may involve additional features suitable for on-screen versions of articles.

The option `natbib` is used when the corresponding $\text{\LaTeX}$ style is based on `natbib`. In most cases you do not need to set it. See Section 2.15.

The option `anonymous` is used for anonymous review processes and causes all author information to be obscured.

The option `timestamp` is used to include a time stamp in the footer of each page. When preparing a document, this can help avoid confusing different revisions. The footer also includes the page range of the document. This helps detect missing pages in hard copies.

The option `authordraft` is intended for authors' drafts that are not intended for distribution. It typesets a copyright block to give the author an idea of its size and the overall size of the paper but overprints it with the phrase "Unpublished working draft. Not for distribution.", which is also used as a watermark. This option sets `timestamp` and `review` to `true`, but these can be overridden by setting these options to `false` *after* setting `authordraft` to `true`.

The option `balance` determines whether the last page in the two column mode has balanced columns. By default it is `true`; however, it may lead to problems for some documents. When there are many figures near the end of the document, the attempts to balance columns may lead to the loss of the figures. Set this option to `false` if you encounter problems. An alternative is the (experimental) option `pbalance`, which uses the new package `pbalance`. You may want to try `pbalance=true` to see if you get better results.

The option `urlbreakonhyphens` determines whether URLs can be split between lines after hyphens. By default it is `true`. Set it to `false` to disallow these breaks.

The option `language` is used to define the languages for the multi-language papers. It is discussed in Section 2.6.

2.3 Top matter

A number of commands set up *top matter* or (in computer science jargon) *metadata* for an article. They establish the publication name, article title, authors, DOI and other data. Some of these commands, like `\title` and `\author`, should be put by the authors. Others, like `\acmVolume` and `\acmDOI`—by the editors. Below we describe these commands and mention who should issue them. These macros should be used *before* the `\maketitle` command. Note that in previous versions of ACM classes some of these commands should be used before `\maketitle`, and some after it. Now they all must be used before `\maketitle`.

Table 2: Boolean options

Option	Default	Meaning
review	false	A review version: lines are numbered and hyperlinks are colored
screen	see text	A screen version: hyperlinks are colored
natbib	true	Whether to use the natbib package (see Section 2.15)
anonymous	false	Whether to make author(s) anonymous
authorversion	false	Whether to generate a special version for the authors' personal use or posting (see Section 2.3)
nonacm	false	Use the class typesetting options for a non-ACM document, which will not include the conference/journal header and footers. Currently such documents allow only a Creative Commons license.
timestamp	false	Whether to put a time stamp in the footer of each page
authordraft	false	Whether author's-draft mode is enabled
acmthm	true	Whether to define theorem-like environments, see Section 2.11
balance	true	Whether to balance the last page in two column mode
pbalance	false	Whether to balance the last page in two column mode using pbalance package
urlbreakonhyphens	true	Whether to break urls on hyphens

This class internally loads the `amsart` class, so many top-matter commands are inherited from `amsart` [2].

`\acmJournal` The macro `\acmJournal{<shortName>}` sets the name of the journal or transaction for journals and transactions. The argument is the short name of the publication *in uppercase*, for example,

```
\acmJournal{TOMS}
```

The currently recognized journals are listed in Table 5. Note that conference proceedings published in *book* form do not set this macro.

`\acmConference` The macro `\acmConference[<short name>]{<name>}{<date>}{<venue>}` is used for conference proceedings published in the book form. The arguments are the following:

short name: the abbreviated name of the conference (optional).

name: the name of the conference.

date: the date(s) of the conference.

venue: the place of the conference.

Examples:

```
\acmConference[TD'15]{Technical Data Conference}{November
12--16}{Dallas, TX, USA}
\acmConference{SA'15 Art Papers}{November 02--06, 2015}{Kobe, Japan}
```

`\acmBooktitle` By default we assume that conference proceedings are published in the book named *Proceedings of CONFERENCE*, where *CONFERENCE* is the name of the conference inferred from the command `\acmConference` above. However, sometimes the book title is different. The command `\acmBooktitle` can be used to set this title, for example,

```
\acmBooktitle{Companion to the first International Conference on the
Art, Science and Engineering of Programming (Programming '17)}
```

An ACM paper should have either `\acmJournal` or `\acmConference` command. If it has both (or more) commands, the last one takes precedence. Note that if you have the command `\acmConference` in a journal format like `acmsmall`, the class will use conference format for bibstrip and reference citation formatting. In the samples directory there is a file `sample-acmsmall-conf.tex` with the example of this usage.

An ACM Engage material should *not* use `\acmJournal` or `\acmConference` command. It may use `\acmBooktitle` to override the default *ACM EngageCSEdu*. It should use `\acmYear` to set the date of the material.

`\editor` In most cases, conference proceedings are edited. You can use the command `\editor{<editor>}` to set the editor of the volume. This command can be repeated, for example,

```
\editor{Jennifer B. Sartor}
\editor{Theo D'Hondt}
\editor{Wolfgang De Meuter}
```

`\title` The command `\title`, as in the `amsart` class, has two arguments: one optional, and one mandatory:

```
\title[<ShortTitle>]{<FullTitle>}
```

The mandatory argument is the full title of the article. The optional argument, if present, defines the shorter version of the title for running heads. If the optional argument is absent, the full title is used instead.

It is expected that this command is inserted by the author of the manuscript.

`\subtitle` Besides title, ACM classes allow a subtitle, set with the `\subtitle{<subtitle>}` macro.

The commands for specifying authors are highly structured. The reason is they serve double duty: the authors' information is typeset in the manuscript *and* is used by the metadata extraction tools for indexing and cataloguing. Therefore it is very important to follow the guidelines exactly.

`\author` The basic commands are `\author`, `\orcid` (for the researchers registered with OR-

`\orcid` CID, <http://www.orcid.org/>), `\affiliation` and `\email`. In the simplest case, you

`\affiliation` enter them in this order:

```
\email
\author{...}
\orcid{...}
\affiliation{...}
\email{...}
```

Do *not* use the $\TeX$ `\and` macro or commas, or `\\` between the authors! Each author deserves their own `\author` command. An attempt to list several authors or their e-mails in one command leads to a warning or an error. This is not a bug, but the expected behavior.

Note that some formats do not typeset e-mails or ORCID identifiers. Do not worry: the metadata tools will get them.

ACM strongly encourages that you include ORCIDs for all authors before compiling or submitting for review and/or production processing.

If you do not have an ORCID, you may get one for free by registering at <http://www.orcid.org/>.

Sometimes an author has several affiliations. In this case, the `\affiliation` command should be repeated:

```
\author{...}
\orcid{...}
\affiliation{...}
\affiliation{...}
\email{...}
```

Similarly you can repeat the `\email` command.

You may have several authors with the same affiliation, different affiliations, or overlapping affiliations (author A_1 is affiliated with institutions I_1 and I_2 , while author A_2 is affiliated with I_2 only, author A_3 is affiliated with I_1 and I_3 , etc.). The recommended solution is to put the `\affiliation` commands after each author, possibly repeating them:

```
\author{...}
\orcid{...}
```

```

\affiliation{...}
\affiliation{...}
\email{...}
\author{...}
\orcid{...}
\affiliation{...}
\email{...}
\author{...}
\orcid{...}
\affiliation{...}
\affiliation{...}
\email{...}

```

In some cases, when several authors share the same affiliation, you can try to save space using the format

```

\author{...}
\email{...}
\author{...}
\email{...}
\affiliation{...}

```

However, this format is not generally recommended.

`\additionalaffiliation` In some cases, too many affiliations can take too much space. The command `\additionalaffiliation{<affiliation>}` creates a footnote after an author's name with the words "Also with {<affiliation>}". You should use this command only as a last resort. An example of usage is:

```

\author{G. Tobin}
\author{Ben Trovato}
\additionalaffiliation{%
  \institution{The Th{\o}rv{\a}ld Group}
  \city{Hekla}
  \country{Iceland}}
\affiliation{%
  \institution{Institute for Clarity in Documentation}
  \city{Dublin}
  \state{Ohio}}

```

Here Trovato and Tobin share their affiliation with the Institute for Clarity in Documentation, but only Ben Trovato is affiliated with The Th{\o}rv{\a}ld Group.

`\position` The `\affiliation` and `\additionalaffiliation` commands are further structured to interact with the metadata extraction tools. Inside these commands you should use `\institution` the `\position`, `\institution`, `\department`, `\city`, `\state`, and `\country` macros to indicate the corresponding parts of the affiliation. Note that in some cases (for example, `\state` journals) these parts are not printed in the resulting copy, but they *are* necessary since `\country` they are used by the XML metadata extraction programs. Do *not* put commas or `\` between the elements of `\affiliation`. They will be provided automatically.

The fields `\institution`, `\city` and `\country` are mandatory. If they are not provided, an error or a warning is issued. Currently the absence of `\country` produces an error; ACM may change this in the future.

Starting 2024, ACM no longer collects or print postal addresses of the authors. Thus the commands `\streetaddress` and `\postcode`, present in the previous versions of the class, now produce a warning.

An example of the author block:

```
\author{A. U. Thor}
\orcid{1234-4564-1234-4565}
\affiliation{%
  \institution{University of New South Wales}
  \department{School of Biomedical Engineering}
  \city{Sidney}
  \state{NSW}
  \country{Australia}}
\email{author@nsw.au.edu}
\author{A. N. Other}
\affiliation{%
  \institution{University of New South Wales}
  \city{Sidney}
  \state{NSW}
  \country{Australia}}
\author{C. O. Respondent}
\orcid{1234-4565-4564-1234}
\affiliation{%
  \institution{University of Pennsylvania}
  \city{Philadelphia}
  \state{PA}
  \country{USA}}
\affiliation{%
  \institution{University of New South Wales}
  \city{Sidney}
  \state{NSW}
  \country{Australia}}
```

Note that the old ACM conference formats did not allow more than six authors and required some effort from authors to achieve alignment. The new format is much better in this.

Sometimes an author works in several departments within the same institution. There could be two situations: the departments are independent, or one department is within another. In the first case, just repeat the command `\department` several times. To handle the second case the command has an optional numerical parameter. The departments with higher numbers are higher in the organizational chart. Compare

```
\affiliation{%
  \department[0]{Department of Lunar Studies} % 0 is the default
  \department[1]{John Doe Institute} % higher than 0
  \institution{University of San Serriffe}
  \country{San Serriffe}}
```

and

```
\affiliation{%
  \department{Department of Lunar Studies} % Not in the John Doe Institute!
  \department{John Doe Institute}}
```

```
\institution{University of San Serriffe}
\country{San Serriffe}}
```

`\correspondingauthor` The command `\correspondingauthor` after an author's name marks the corresponding author. Note that there could be no more than one corresponding author. Make sure that you have the email of the author entered.

`\thanks` Like `amsart` (and unlike standard $\text{\LaTeX}$), we allow `\thanks` only *outside* of the commands `\title` and `\author`. This command is obsolete and should *not* be used in most cases. Do not list your acknowledgments or grant sponsors here. Put this information in the `acks` environment (see Section 2.14).

`\authorsaddresses` In some formats, addresses are printed as a footnote on the first page. By default $\text{\LaTeX}$ typesets them itself using the information you give it. However, you can override its choice using the command `\authorsaddresses{<contact addresses>}`, for example,

```
\authorsaddresses{%
  Authors' addresses: G.~Zhou, Computer Science Department, College of
  William and Mary, 104 Jameson Rd, Williamsburg, PA 23185, US;
  V.~B'eranger, Inria Paris-Rocquencourt, Rocquencourt, France;
  A.~Patel, Rajiv Gandhi University, Rono-Hills, Doimukh, Arunachal
  Pradesh, India; H.~Chan, Tsinghua University, 30 Shuangqing Rd,
  Haidian Qu, Beijing Shi, China; T.~Yan, Eaton Innovation Center,
  Prague, Czech Republic; T.~He, C.~Huang, J.~A.~Stankovic University
  of Virginia, School of Engineering Charlottesville, VA 22903, USA;
  T. F. Abdelzaher, (Current address) NASA Ames Research Center,
  Moffett Field, California 94035.}
```

You can *suppress* printing authors' addresses by setting them to an empty string: `\authorsaddresses{}`. Please note that authors' addresses are mandatory for journal articles.

`\titlenote` While the command `\thanks` generates a note without a footnote mark, sometimes the authors might need notes more tightly connected to the title, subtitle or author. The commands `\titlenote`, `\subtitlenote` and `\authornote` that follow the corresponding commands (`\title`, `\subtitle` and `\author`) generate such notes. For example,

```
\title{This is a title}
\titlenote{This is a titlenote}
\author{A. U. Thor}
\authornote{This is an authornote}
```

Please never use a `\footnote` inside an `\author` or `\title` command since this confuses the metadata extraction software. (Actually these commands now produce errors.)

`\authornotemark` Sometimes one may need to have the same footnote connected to several authors. The command `\authornotemark[<number>]` adds just the footnote mark, for example,

```
\author{A. U. Thor}
\authornote{Both authors contributed equally to the paper}
...
\author{A. N. Other}
\authornotemark[2]
```

The correct numbering of these marks is the responsibility of the user. Note that the corresponding author, if present, is marked with the star. Thus the numbering of footnotes starts with 2 (the dagger symbol).

`\acmVolume` The macros `\acmVolume`, `\acmNumber`, `\acmArticle`, `\acmYear` and `\acmMonth` are
`\acmNumber` inserted by the editor and set the journal volume, issue, article number, year and month
`\acmArticle` correspondingly. The arguments of all these commands, including `\acmMonth`, is numer-
`\acmYear` ical. For example,
`\acmMonth` `\acmVolume{9}`
 `\acmNumber{4}`
 `\acmArticle{39}`
 `\acmYear{2010}`
 `\acmMonth{3}`

Note that `\acmArticle` is used not only for journals but also for some conference proceedings.

`\acmArticleSeq` The articles in the same issue of a journal have a *sequence number*. By default it is the same as the article number, but the command `\acmArticleSeq{<n>}` can be used to change it:

```
\acmArticle{39} % The sequence number will be 39 by default
\acmArticleSeq{5} % We redefine it to 5
```

`\acmSubmissionID` If your paper got a Submission ID from the Conference Management System, put it here:

```
\acmSubmissionID{123-A56-BU3}
```

`\acmISBN` Book-like volumes have ISBN numbers attached to them. The macro `\acmISBN{<ISBN>}` sets it. Normally it is set by the typesetter, for example,

```
\acmISBN{978-1-4503-3916-2}
```

Setting it to the empty string, as `\acmISBN{}`, suppresses printing the ISBN.

`\acmDOI` The macro `\acmDOI{<DOI>}` sets the DOI of the article, for example,

```
\acmDOI{10.1145/9999997.9999999}
```

It is normally set by the typesetter. Setting it to the empty string, as `\acmDOI{}`, suppresses the DOI.

`\acmBadge` Some conference articles get special distinctions, for example, the artifact evaluation for PPOPP 2016 (see <http://ctuning.org/ae/ppopp2016.html>). These articles display special badges supplied by the conference organizers. This class provides command to add these badges: `\acmBadge[<url>]{<graphics>}`. The arguments have the following meaning: [`<url>`], if provided, sets the link to the badge authority in the screen version, while `{<graphics>}` sets the graphics file with the badge image. The file must be a cropped square, which is scaled to a standard size in the output. For example, if the badge image is `ae-logo.pdf`, the command is

```
\acmBadge[http://ctuning.org/ae/ppopp2016.html]{ae-logo}
```

The command can be repeated, if a paper has several badges.

`\startPage` The macro `\startPage{⟨page⟩}` sets the first page of the article in a journal or book. It is used by the typesetter.

`\keywords` The command `\keywords{⟨keyword, keyword,...⟩}` sets keywords for the publication. If present, keywords must be separated by commas, for example,

```
\keywords{wireless sensor networks, media access control,  
multi-channel, radio interference, time synchronization}
```

`CCSXML (env.)` ACM publications are classified according to the ACM Computing Classification Scheme (CCS). CCS codes are used both in the typeset version of the publications *and* in the metadata in various databases. Therefore you need to provide both $\TeX$ commands and XML metadata with the paper.

The tool at <http://dl.acm.org/ccs.cfm> can be used to generate CCS codes. After you select the topics, click on “Generate CCS codes” to get results like the following:

```
\begin{CCSXML}  
<ccs2012>  
  <concept>  
    <concept_id>10010520.10010553.10010562</concept_id>  
    <concept_desc>Computer systems organization~Embedded systems</concept_desc>  
    <concept_significance>500</concept_significance>  
  </concept>  
  <concept>  
    <concept_id>10010520.10010575.10010755</concept_id>  
    <concept_desc>Computer systems organization~Redundancy</concept_desc>  
    <concept_significance>300</concept_significance>  
  </concept>  
  <concept>  
    <concept_id>10010520.10010553.10010554</concept_id>  
    <concept_desc>Computer systems organization~Robotics</concept_desc>  
    <concept_significance>100</concept_significance>  
  </concept>  
  <concept>  
    <concept_id>10003033.10003083.10003095</concept_id>  
    <concept_desc>Networks~Network reliability</concept_desc>  
    <concept_significance>100</concept_significance>  
  </concept>  
</ccs2012>  
\end{CCSXML}  
  
\ccsdesc[500]{Computer systems organization~Embedded systems}  
\ccsdesc[300]{Computer systems organization~Redundancy}  
\ccsdesc[100]{Computer systems organization~Robotics}  
\ccsdesc[100]{Networks~Network reliability}
```

You just need to copy this code and paste it in your paper anywhere before `\maketitle`.

CCS Concepts and user-defined keywords are required for all articles over two pages in length, and are optional for one- and two-page articles (or abstracts).

`\setcopyright` There are several possibilities for the copyright of the papers published by the ACM: the authors may license the paper to the ACM, some or all authors might

Table 3: Parameters for the `\setcopyright` command

Parameter	Meaning
<code>none</code>	The copyright and permission information is not typeset. (This is the option for some ACM conferences.)
<code>acmlicensed</code>	The authors retain the copyright but license the publication rights to ACM.
<code>rightsretained</code>	The authors retain the copyright and publication rights to themselves or somebody else.
<code>usgov</code>	All the authors are employees of the US government.
<code>usgovmixed</code>	Some authors are employees of the US government.
<code>cagov</code>	All the authors are employees of the Canadian government.
<code>cagovmixed</code>	Some authors are employees of the Canadian government.
<code>licensedusgovmixed</code>	Some authors are employees of the US government, and the publication rights are licensed to ACM.
<code>licensedcagov</code>	All the authors are employees of the Canadian government, and the publication rights are licensed to ACM.
<code>licensedcagovmixed</code>	Some authors are employees of the Canadian government, and the publication rights are licensed to ACM.
<code>othergov</code>	Authors are employees of a government other than the US or Canada.
<code>licensedothergov</code>	Authors are employees of a government other than the US or Canada, and the publication rights are licensed to ACM.
<code>iw3c2w3</code>	Special statement for conferences organized by IW3C2.
<code>iw3c2w3g</code>	Special statement for conferences organized by IW3C2, when some authors are approved Google employees.
<code>cc</code>	Creative Commons license. If this key is set, <i>doclicense</i> images are used to typeset the license. See also <code>\setcctype</code> command.
<code>acmcopyright</code>	The authors transfer the copyright to the ACM. This choice is no longer available for the non-commissioned authors and will trigger a warning.

be employees of the US or Canadian governments, etc. Accordingly the command `\setcopyright{...}` is introduced. Its argument is the copyright status of the paper, for example, `\setcopyright{acmcopyright}`. The possible values for this command are listed in Table 3. This command must be placed in the preamble, before `\begin{document}`.

`\setcctype` If Creative Commons license is used, the package by default chooses CC-BY 4.0 Attribution 4.0 International license. You can override this choice by the command `\setcctype[<version>]{<type>}`, where [*<version>*] can be either 3.0 or 4.0 (4.0 by default), and {*<type>*} can be one of zero, by, by-sa, by-nd, by-nc, by-nc-sa, by-nc-nd (see <https://creativecommons.org/licenses/> for the explanation). This command should be used in the preamble only.

Material published under Creative Commons license should include the corresponding icon. A modern $\text{\TeX}$ distribution includes these icons in the package *doclicense*. In case your distribution does not have them, ACM provides a file `ccicons.zip` with these icons. Just unzip it in the same directory where your document is.

The ACM submission software should generate the right command for you to paste into your file.

`\copyrightyear` Each copyright statement must have the year of copyright. By default it is the same as `\acmYear`, but you can override this using the macro `\copyrightyear`, e.g.,

```
\acmYear{2016}
\copyrightyear{2015}
```

There is a special case for a personal copy that the authors may be allowed to generate for their use or a posting on a personal site (check the instructions for the specific journal or conference for the details). The document option `authorversion=true` produces a special form of the copyright statement for this case. Note that you still need the `\setcopyright` command and (optionally) `\copyrightyear` command to tell TeX about the copyright owner and year. Also, you should be aware that due to the different sizes of the permission blocks for the printed version and authors' version, the page breaks might be different between them.

`abstract (env.)` The environment `abstract` must *precede* the `\maketitle` command. Again, this is different from the standard L^AT_EX. Putting `abstract` after `\maketitle` will trigger an error.

`teaserfigure (env.)` A special kind of figure is used for many two-column conference proceedings. This figure is placed just after the authors but before the main text. The environment `teaserfigure` is used for these figures. This environment must be used *before* `\maketitle`, for example,

```
\begin{teaserfigure}
\includegraphics[width=\textwidth]{sampleteaser}
\caption{This is a teaser}
\label{fig:teaser}
\end{teaserfigure}
```

`\settopmatter` Some information in the top matter is printed for certain journals or proceedings and suppressed for others. You can override these defaults using the command `\settopmatter{<settings>}`. The settings and their meanings are listed in Table 4. For example,

```
\settopmatter{printacmref=false, printccs=true, printfolios=true}
```

The parameter `authorsperrow` requires some explanation. In conference proceedings authors' information is typeset in boxes, several boxes per row (see `sample-sigconf.pdf`, `sample-sigplan.pdf`, etc.). The number of boxes per row is determined automatically. If you want to override this, you can do it using this parameter, for example,

```
\settopmatter{authorsperrow=4}
```

However, in most cases you should *not* do this and should use the default settings. Setting `authorsperrow` to 0 will revert it to the default settings.

The parameter `printacmref` specifies whether to print the ACM bibliographic entry (default), or not. Note that this entry is required for all articles over one page in length, and is optional for one-page articles (abstracts).

`\received` The command `\received[<stage>]{<date>}` sets the history of the publication.

Table 4: Settings for the `\settopmatter` command

Parameter	Values	Meaning
<code>printccs</code>	true/false	Whether to print CCS categories
<code>printacmref</code>	true/false	Whether to print the ACM bibliographic entry
<code>printfolios</code>	true/false	Whether to print page numbers (folios)
<code>authorsperrow</code>	numeric	Number of authors per row for the title page in conference proceedings formats

The `[\stage]` argument is optional; the default is Received for the first date and revised for the subsequent ones. For example,

```
\received{20 February 2007}
\received[revised]{12 March 2009}
\received[accepted]{5 June 2009}
```

`\maketitle` The macro `\maketitle` must be the last command in the top-matter group. That is it must follow the commands defined in this section.

`\shortauthors` After the command `\maketitle`, the macro `\shortauthors` stores the names of the authors for the running head. You can redefine it if the list of author's name is too long, e.g.,

```
\maketitle
\renewcommand{\shortauthors}{Zhou et al.}
```

2.4 Top matter of ACM Engage materials

ACM Engage materials resemble conference proceedings, but have some special features. First, as a rule, they are released under a Creative Commons license. By default CC-BY is used. However, if you want to use another variant of CC license, use `\setcctype` command, for example, `\setcctype{by-nc}`. Second, abstract is called *synopsis*. Third, there are special top matter items used for the materials, such as *Course*, *Resource Type*, *Programming Language*, *CS Topics*.

`\setengagemetadata` These items are set with the command `\setengagemetadata{<name>}{<value>}`, for example,

```
\setengagemetadata{Course}{CS1}
\setengagemetadata{Programming Language}{Python}
\setengagemetadata{Knowledge Unit}{Programming Concepts}
\setengagemetadata{CS Topics}{Functions, Data Types, Expressions,
Mathematical Reasoning}
```

Note that the type of Creative Commons license, if such license is used, is automatically added to the metadata.

2.5 ACM cover page

ACM cover pages are forms of extended abstracts that are added to journals at the late stage. Authors prepare them as separate `.tex` files using `acmcp` format. At present only JDS uses them, but in the future this may change.

There are several top matter commands specific for this format.

`\acmArticleType` There are five article types accepted by JDS: *Research* (the default), *Review*, *Discussion*, *Invited*, and *Position*. The command `\acmArticleType{<type>}` sets the article type, for example

```
\acmArticleType{Review}
```

`\acmCodeLink` The commands `\acmCodeDataLink{<link>}` and `\acmDataLink{<link>}` set the links
`\acmDataLink` to the data and code accompanying the paper, for example,

```
\acmCodeLink{https://github.com/repository/code}
\acmDataLink{https://datadryad.org/stash/dataset/doi:DOI}
```

You may repeat these commands if you have several repositories.

`\acmContributions` The command `\acmContributions{<contributions>}` sets the contributions of the authors, for example,

```
\acmContributions{AW designed the study, CD performed it, all
authors contributed to the writing.}
```

ACM cover page should have the following obligatory sections:

- Problem statement,
- Methods,
- Results,
- Significance.

Sometimes the addresses extracted from the authors' data are too long to fit on the page. In this case the command `\authorsaddresses` can be use to override them, for example,

```
\authorsaddresses{Corresponding author: Ben Trovato,
\href{mailto:trovato@corporation.com}{trovato@corporation.com};
Institute for Clarity in Documentation, P.O. Box 1212, Dublin,
Ohio, USA, 43017-6221}
```

The design of the cover page may require additional runs of latex to make the elements of the page align.

2.6 Internationalization

ACM accepts publications in languages other than English, as well as papers in English with translations of titles, subtitles, keywords and abstracts into other languages. Papers in languages other than English usually have titles, subtitles (if applicable), keywords and abstracts in English. Note that CCS concepts are always typeset in English.

To submit these papers you need to set the option `language` in the `\documentclass` command. This option can be repeated, for example,

```
\documentclass[sigconf, language=french, language=english]{acmart}
```

The last language in the list is the main language of the paper, i.e. the one for the main title, abstract, body, etc. The other languages are *secondary*, and used for translated titles, keywords, abstracts. Thus the paper above is written in English, and has a secondary abstract and a secondary title in French. On the other hand, a paper in French with secondary titles and abstracts in English and German should use, for example

```
\documentclass[sigconf,
               language=german,
               language=english,
               language=french]{acmart}
```

This key can use any language defined in *babel* package [3] (currently the package is tested with English, French, German and Spanish languages. If `\keywords` are used, then other languages may require a translation of `\keywordsname` macro). Actually *acmart* loads *babel* internally, so you can use the facilities provided by this package.

If this key is set, you have access to several additional top matter commands.

`\translatedtitle` The commands `\translatedtitle`, `\translatedsubtitle`, and (if keywords are
`\translatedsubtitle` used) `\translatedkeywords` have two mandatory arguments each: `{<language>}` and
`\translatedkeywords` `{<keywords>}`. They are used to set title, subtitle and keywords in the secondary language. For example, a paper in English with French title and abstract may set

```
\title{A note on computational complexity}
\translatedtitle{french}{Remarque sur la complexit\'e de calcul}
```

while a paper in French should set

```
\title{Remarque sur la complexit\'e de calcul}
\translatedtitle{english}{A note on computational complexity}
```

`translatedabstract (env)` Similarly, `translatedabstract` environment has a mandatory language argument, for example,

```
\begin{translatedabstract}{english}
  This is the English version of the abstract
\end{translatedabstract}
```

You can repeat these commands if a paper has more than one secondary language.

Use the standard commands (`\title`, `\subtitle`, `\keywords`, `abstract`) for the main language of the paper.

2.7 Tagging

ACM is firmly committed to produce fully tagged PDFs compliant with the accessibility standards.

The $\LaTeX$ tagging project, <https://latex3.github.io/tagging-project/> is aimed to create accessible and standards compliant PDFs. To create a tagged PDF, start your document with the `\DocumentMetadata` command, for example,

```
\DocumentMetadata{
  lang          = en,
  pdfstandard   = ua-2,
```

```
pdfstandard = a-4f,
tagging=on,
tagging-setup={math/setup=mathml-SE}
}
```

You must use `lualatex` to make the document fully tagged. For some features you may need `lualatex-dev`.

At present this code is still experimental. If you use it, please *do not ask ACM for support*. On the other hand, bug reports at <https://github.com/borisveytsman/acmart/issues> will be appreciated.

2.8 Algorithms

There are now several good packages for typesetting algorithms [4, 5, 6], and the authors are free to choose their favorite one.

2.9 Figures and tables

ACM styles use the standard $\LaTeX$ interface for figures and tables. There are some important items to be aware of, however.

1. The captions for figures must be entered *after* the figure bodies and for tables *before* the table bodies.
2. The ACM uses the standard types for figures and tables and adds several new ones. In total there are the following types:

figure, table: a standard figure or table taking a full text width in one-column formats and one column width in two-column formats.

figure*, table* in two-column formats, a special figure or table taking a full text width.

teaserfigure: a special figure before `\maketitle`.

3. Accordingly, when scaling images, one should use the following sizes:
 - (a) For `teaserfigure`, `figure` in one-column mode or `figure*` in two-column mode, use `\textwidth`. In one-column mode, you can also use `\columnwidth`, which coincides with `\textwidth` in this case.
 - (b) For `figure` in two-column mode, use `\columnwidth`.

It is strongly recommended to use the package `booktabs` [7] and follow its main principles of typography with respect to tables:

1. Never, ever use vertical rules.
2. Never use double rules.

It is also a good idea not to overuse horizontal rules.

For table *footnotes* you have several options described in the TeX FAQ [1]. The simplest one is to use a `\minipage` environment:

```
\begin{table}
```

```

\caption{Simulation Configuration}
\label{tab:conf}
\begin{minipage}{\columnwidth}
\begin{center}
\begin{tabular}{ll}
\toprule
TERRAIN\footnote{This is a table footnote. This is a
    table footnote. This is a table footnote.} &
    (200\,m$\times$200\,m) Square\\
Node Number      & 289\\
Node Placement   & Uniform\\
Application      & Many-to-Many/Gossip CBR Streams\\
Payload Size     & 32 bytes\\
Routing Layer    & GF\\
MAC Layer        & CSMA/MMSN\\
Radio Layer      & RADIO-ACCNOISE\\
Radio Bandwidth  & 250Kbps\\
Radio Range      & 20m--45m\\
\bottomrule
\end{tabular}
\end{center}
\bigskip
\footnotesize\emph{Source:} This is a table
sourcenote. This is a table sourcenote. This is a table
sourcenote.

\emph{Note:} This is a table footnote.
\end{minipage}
\end{table}

```

Tables and figures are by default centered. However, in some cases (for example, when you use several subimages per figure) you may need to override this. A good way to do so is to put the contents into a `\minipage` of the width `\columnwidth`.

If tagging is enabled (see Section 2.7), it assumes that the first row of table contains headers. To change this assumption, use `tagpdfsetup`, for example, if you have two rows of headers,

```
\tagpdfsetup{table/header-rows={1,2}}
```

2.10 Descriptions of images

Some readers of ACM publications might be visually challenged. These readers might use a voice-over software to read aloud the papers. It is important to provide them a description of each image used in the paper.

There are two ways to include descriptions in the figures.

`\Description` The old way, that predates L^AT_EX tagging project is based on the `\Description` command. The command `\Description[<short description>]{<long description>}` should be placed inside every figure, `teaserfigure` or `marginfigure` environment to provide a description of the image(s) used in the figure. Unlike `\caption`, which is used alongside the image, `\Description` is intended to be used instead of the image, for example,

```
\begin{figure}
```

```

\centering
\includegraphics{voltage}
\Description{A bell-like histogram centered at $0.5\sim V$ with most
measurements between $0.2\sim V$ and $0.8\sim V$}
\caption{Histogram of the measurements of voltage}
\label{fig:voltage}
\end{figure}

```

The new way, which is switched on by the `\DocumentMetada` command is based on the `alt` keyword for the `\includegraphics` command, for example,

```

\begin{figure}
\centering
\includegraphics[alt={A bell-like histogram
centered at $0.5\sim V$ with most
measurements between $0.2\sim V$ and $0.8\sim V$}]{voltage}
\caption{Histogram of the measurements of voltage}
\label{fig:voltage}
\end{figure}

```

At present the lack of descriptions generates a warning at compilation.

2.11 Theorems

The ACM classes define two theorem styles and several pre-defined theorem environments:

acmplain: this is the style used for theorem, conjecture, proposition, lemma and corollary, and

acmdefinition: this is the style used for example and definition.

These environments are defined by default. In the unusual circumstance that a user does not wish to have these environments defined, the option `acmthm=false` in the preamble will suppress them.

Sometimes authors want to define new theorem-like constructs that use theorem counters. These constructs must be defined either after `\begin{document}`, or delayed using `\AtEndPreamble` macro, for example,

```

\AtEndPreamble{%
\theoremstyle{acmdefinition}
\newtheorem{remark}[theorem]{Remark}}

```

If you use *cleveref* and share counters between the theorem-like constructs, you need to explicitly tell *cleveref* the kind of the construct you use, for example,

```

\newtheorem{theorem}{Theorem}[section]
\newtheorem{lemma}[theorem]{Lemma}
% Theorems and lemmas share counters
...
\begin{theorem}\label{thm:test} % This is a theorem
...
\begin{lemma}\label[lemma]{test-lemma} % This is a lemma

```

2.12 Online-only and offline-only material

`printonly` (*env.*) Some supplementary material in ACM publications is put online but not in the printed version. The text inside the environment `screenonly` will be typeset only when the option `screen` (see Section 2.2) is set to true. Conversely, the text inside the environment `printonly` is typeset only when this option is set to false. For example,

```
\section{Supplementary materials}

\begin{printonly}
  Supplementary materials are available in the online version of this paper.
\end{printonly}

\begin{screenonly}
  (The actual supplementary materials.)
\end{screenonly}
```

We use the `comment` package for typesetting this code, so `\begin` and `\end` should start on a line of their own with no leading or trailing spaces.

2.13 Note about anonymous mode

`anonsuppress` (*env.*) When the option `anonymous` is selected, $\text{\TeX}$ suppresses author information (including the number of authors) for an anonymous review. However, sometimes the information identifying the authors may be present in the body of the paper. For example,

```
\begin{anonsuppress}
  This is the continuation of the previous work by the author
  \cite{prev1, prev2}.
\end{anonsuppress}
```

As for the `printonly` and `screenonly` environments, `\begin{anonsuppress}` and `\end{anonsuppress}` should start on a line of their own with no leading or trailing spaces.

`\anon` To suppress short snippets of information, use the command `\anon[<substitute>]{<suppressed-text>}`. By default [*<substitute>*] is the word ANONYMOUS. Examples:

```
This work was performed at \anon{NSA}.
This work was performed at \anon[No Such Agency]{NSA}.
```

2.14 Acknowledgments

The traditional “Acknowledgments” section is conventionally used to thank persons and granting agencies for their help and support. However, there are several important considerations about this section.

First, in anonymous mode this section must be omitted: it gives too much information to reviewers. Second, data about grants is extracted and stored separately by the postprocessing software. ACM classes provide facilities for both these tasks.

`acks` (*env.*) The environment `acks` starts an unnumbered section “Acknowledgments” unless the anonymous mode is chosen. Put all thanks inside this environment.

As for the `printonly` and `screenonly` environments, `\begin{acks}` and `\end{acks}` should start on a line of their own with no leading or trailing spaces or comments.

`\grantsponsor` All financial support *must* be listed using the commands `\grantsponsor` and `\grantnum`. These commands tell the postprocessing software about the granting organization and grant. The format of these commands is the following:

```
\grantsponsor{\sponsorID}{\name}{\url}
\grantnum[\url]{\sponsorID}{\number}.
```

Here `{\sponsorID}` is the unique ID used to match grants to sponsors, `{\name}` is the name of the sponsor, `{\url}` is its URL, and `{\number}` is the grant number. The `{\sponsorID}` of the `\grantnum` command must correspond to the `{\sponsorID}` of a `\grantsponsor` command. Some awards have their own web pages, which you can include using the optional argument of the `\grantnum` command.

At present `{\sponsorID}` is chosen by the authors and can be an arbitrary key in the same way the label of a `\cite` is arbitrarily chosen. There might be a change to this policy if the ACM decides to create a global database of sponsoring organizations.

Example:

```
\begin{acks}
The authors would like to thank Dr. Yuhua Li for providing the
matlab code of the \textit{BEPS} method.

The authors would also like to thank the anonymous referees for
their valuable comments and helpful suggestions. This work is
supported by the \grantsponsor{my-grant-GS501100001809}{National Natural
Science Foundation of
China}{https://doi.org/10.13039/501100001809} under Grant
No.: ~\grantnum{GS501100001809}{61273304}
and ~\grantnum[http://www.nnsf.cn/youngscientists]{GS501100001809}{Young
Scientists' Support Program}.
\end{acks}
```

2.15 Bibliography

The ACM lets you use either Bib $\text{\TeX}$ or Bib $\text{\LaTeX}$ to process your references: they require slightly different setup of your $\text{\LaTeX}$ file, as detailed in the following subsections.

2.15.1 Processing using Bib $\text{\TeX}$

This uses the `natbib` package for formatting references and the Bib $\text{\TeX}$ style file `ACM-Reference-Format.bst` for Bib $\text{\TeX}$ processing. You can disable loading of `natbib` using the option `natbib=false` in `\documentclass`. However, it is not recommended, as well as the use of Bib $\text{\TeX}$ styles other than `ACM-Reference-Format.bst`, and may delay the processing of the manuscript.

`\citestyle` If you use `natbib`, you can select one of two predefined citation styles using the command `\citestyle`: the author-year format `acmauthoryear` or the numeric format `acmnumeric`. For example,

```
\citestyle{acmauthoryear}
```

Note that numeric citations are the default mode for most formats.

`\setcitestyle` You can further customize natbib using the `\setcitestyle` command, for example,
`\setcitestyle{numbers,sort&compress}`

One of the more common versions is

`\setcitestyle{nosort}`

It is useful if you do not like the way natbib sorts citation lists.

If you use natbib, then commands like `\citep` and `\citeauthor` are automatically supported. The command `\shortcite` is the same as `\cite` in numerical mode and cites the year in author-date mode.

Note that before version 1.48 the command `\citeyear` put the year in parentheses. In version 1.48 and later it produces just the year; the command `\citeyearpar` can be used to emulate its old behavior. You can use these commands to produce in-text citations, for example,

So we see that `\citeauthor{Burano2026}`
proved this conjecture in `\citeyear{Burano2026}`~`\cite{Burano2026}`.

There are several customized B_BT_EX entry types and fields in the ACM style file `ACM-Reference-Format.bst` that you may want to be aware of. You can find many examples on the Web page <https://www.acm.org/publications/authors/bibtex-formatting>.

2.15.2 URL, DOI, eprint

The style supports the fields `doi` and `url`, for example,

```
doi = "10.1145/1188913.1188915",
url = "http://ccrma.stanford.edu/~jos/bayes/bayes.pdf",
```

Normally the printing of URL is suppressed if DOI is present. However, there is a special field `distinctURL`. If it is present and has any value, URL is printed even if DOI is present, for example,

```
distinctURL = 'true',
```

The dynamic nature of Internet makes URLs not permanent: sometimes they disappear or change. The problem of “stale links” exists from the early days of WWW. That is why ACM prefers DOIs to URLs: unlike the latter, the former are supposed to work forever. However, if you really need to use URLs, there are two facilities that help to mitigate some problems with the URLs. The field `lastaccessed` gives the date when the given URL was visited and had the information you saw. Presumably, some service like Internet Wayback Machine (<https://web.archive.org/>) allows the reader to retrieve this link at the given date. Moreover, sometimes a page is explicitly archived by this or other service. In this case the field `archived` will give the reference of the archival version. Examples:

```
lastaccessed = {March 2, 2005},
archived = {https://web.archive.org/web/20240505055615}
```

The style supports the arXiv-recommended fields `eprint` and (optionally) `primaryclass`, for example,

```
eprint = "960935712",
primaryclass = "cs",
```

2.15.3 Special entry types

There are several special entry types. Types `online` and `game` are used for Web pages and games, for example,

```
@online{Thornburg01,
  author = "Harry Thornburg",
  year = "2001",
  title = "Introduction to Bayesian Statistics",
  url = "http://ccrma.stanford.edu/~jos/bayes/bayes.html",
  month = mar,
  lastaccessed = "March 2, 2005",
}
```

Entry types `artifactsoftware`, `artifactdataset` (with synonyms `software` and `dataset`) can be used to cite software artifacts and datasets, for example,

```
@ArtifactSoftware{R,
  title = {R: A Language and Environment for Statistical Computing},
  author = {{R Core Team}},
  organization = {R Foundation for Statistical Computing},
  address = {Vienna, Austria},
  year = {2019},
  url = {https://www.R-project.org/},
}
@ArtifactDataset{UMassCitations,
  author = {Sam Anzaroot and Andrew McCallum},
  title = {{UMass} Citation Field Extraction Dataset},
  year = 2013,
  url =
    {http://www.iesl.cs.umass.edu/data/data-umasscitationfield},
  lastaccessed = {May 27, 2019}
}
```

For these entry types you can use the `lastaccessed` field to add the access date for the URL and `archived` field to add the archival link.

There are two ways to enter video or audio sources in the bibliography corresponding to two different possibilities. For standalone sources available online, you can use an `online` entry and set its `howpublished` field. For example,

```
@online{Obama08,
  author = "Barack Obama",
  year = "2008",
  title = "A more perfect union",
  howpublished = "Video",
  day = "5",
```

```

url      = "http://video.google.com/videoplay?docid=6528042696351994555",
month    = mar,
lastaccessed = "March 21, 2008",
}

```

For sources available as supplemental material attached to conference proceedings and similar documents, you can use the usual `inproceedings` entry type and set its `howpublished` field:

```

@Inproceedings{Novak03,
  author = "Dave Novak",
  title = "Solder man",
  booktitle = "ACM SIGGRAPH 2003 Video Review on Animation theater Program",
  year = "2003",
  publisher = "ACM Press",
  address = "New York, NY",
  pages = "4",
  month = "March 21, 2008",
  doi = "10.9999/woot07-S422",
  howpublished = "Video",
}

```

Sometimes you need to cite a complete issue of a journal. The periodical entry type is intended for this:

```

@periodical{JCohen96,
  key = "Cohen",
  editor = "Jacques Cohen",
  title = "Special issue: Digital Libraries",
  journal = "Communications of the {ACM}",
  volume = "39",
  number = "11",
  month = nov,
  year = "1996",
}

```

Sometimes a work is presented at meeting, but is not (yet) published in the proceedings. The entry type `presentation` is used for this purpose. The field `venue` is used for the place of the meeting:

```

@Presentation{Reiser2014,
  author = {Brian J. Reiser},
  year = 2014,
  title = {Designing coherent storylines aligned with NGSS for the
           K-12 classroom},
  venue = {National Science Education Leadership Association
           Meeting, Boston, MA, USA},
  url = {https://www.academia.edu/6884962/}
}

```

The URL, if present, may be accompanied by `lastaccessed` and `archived` links.

A preprint can be cited using the entry `preprint`, for example,

```
@preprint{AnzarootPBM14,
  author    = {Sam Anzaroot and
               Alexandre Passos and
               David Belanger and
               Andrew McCallum},
  title     = {Learning Soft Linear Constraints with Application to
               Citation Field Extraction},
  year      = {2014},
  archivePrefix = {arXiv},
  eprint    = {1403.1349},
  doi       = {10.48550/arXiv.1403.1349}
}
```

Sometimes you need to cite an article under review. ACM prefers the authors to use preprints for this purpose, but sometimes it is not feasible. In this case you can use the entry `underreview`, for example,

```
@underreview{Baggett2025,
  author = {R. Baggett and M. Simecek and C. Chambellan
            and K. Tsui and M. Fraune},
  year = 2025,
  title = {Fluidity in the Phased Framework of Technology
            Acceptance: Case Study to Gain a Holistic Understanding
            of (Older Adult) Participant Advancement Through
            Acceptance Phases with Mobile Telepresence Robots},
  journal = {Robotics Aut. Systems}
}
```

2.15.4 Dates and sorting

If you do not know the year of publication, the style will add “[n. d.]” (for “no date”) to the entry.

If you do not know the author (this is often the case for online entries), use the key field to add a key for sorting and citations, for example,

```
@online{TUGInstmem,
  key = {TUG},
  year = 2017,
  title = "Institutional members of the {\TeX} Users Group",
  url = "http://wwtug.org/instmem.html",
  lastaccessed = "May 27, 2017",
}
```

A note about sorting. The current ACM bibliography styles always sort the entries according to authors names and publication year. There is a controversy about sorting names with “von” or “van” part: should Ludwig van Beethoven be sorted under “V” or under “B”? The American practice is to use “van” in sorting, i.e. to file van Beethoven under “V”. However, some authorities recommend to sort Dutch persons according to their last names (see e.g. <https://www.ifla.org/files/assets/cataloguing/pubs/names-of-persons.1996.pdf>). While I do not want to take a part in this dispute, I

would like to point to the old “noopsort” trick by Oren Patashnik. Add to the .bib file the line

```
@PREAMBLE{"\providecommand{\noopsort}[1]{}"}
```

and then encode the author as

```
author = {Ludwig {\noopsort{Beethoven}}van Beethoven},
```

This will make the author to be sorted as “Beethoven” rather than “van Beethoven”.

The current bst style defines a number of macros for common journal names. In particular, all journals listed in Table 5 are included, so you can use strings like `journal = taccess` for *ACM Transactions on Accessible Computing*.

2.15.5 Processing using Bib_{La}T_EX

You will find in this package two sets of style files for Bib_{La}T_EX, `acmnumeric` and `acmauthoryear`, that mimic the behavior of the `ACM-Reference-Format.bst` Bib_TE_X style. They provide you access to all the power of Bib_{La}T_EX and already include support for advanced citation of software artifacts from the `biblatex-software` package, also separately available on CTAN. Look at the `biblatex-software` documentation to learn more about what it offers. This support is highly experimental and currently is *not* supported by TAPS!

There are a few key differences in how the _{La}T_EX sources are set up when using Bib_{La}T_EX instead of Bib_TE_X, that we summarize briefly here (please refer to the official Bib_{La}T_EX documentation for more details).

First of all, you need to pass the `natbib=false` option to the document class, and remove the `\citetstyle{acmauthoryear}` command from the sources (if present).

Then, in the preamble of your document you need to load the Bib_{La}T_EX package and select the appropriate bibliography style, as follows

```
\RequirePackage[
  datamodel=acmdatamodel,
  style=acmnumeric, % use style=acmauthoryear for publications that require it
]{biblatex}
```

Also in the preamble, you need to declare the bibliography sources files using the `\addbibresource` command (one `\addbibresource` command per source file), e.g.:

```
\addbibresource{software.bib}
\addbibresource{sample-base.bib}
```

At the end of the document, where you want the bibliography to appear, you need to place the command `\printbibliography`.

Look at the `sample-*-biblatex.tex` files that can be found in the `samples` directory after running `make` for templates showcasing these Bib_{La}T_EX styles.

2.15.6 Manual bibliography

Some people create bibliographies manually, writing down `\bibitem` commands explicitly. This approach is *not* recommended for ACM styles. The reason is, ACM submissions, besides being typeset, are also processed by special programs that extract metadata and references. Bibliographies created automatically with ACM styles contain customized macros for these programs, for example,

```
\bibitem[Ablamowicz and Fauser(2007)]%
    {Ablamowicz07}
\bibfield{author}{\bibinfo{person}{Rafal Ablamowicz} {and}
    \bibinfo{person}{Bertfried Fauser}.} \bibinfo{year}{2007}\natexlab{}.
\newblock \bibinfo{booktitle}{\emph{CLIFFORD: a Maple 11 Package for Clifford
    Algebra Computations, version 11}}.
\newblock
\urldef\tempurl%
\url{http://math.tntech.edu/rafal/cliff11/index.html}
\showURL{%
Retrieved February 28, 2008 from \tempurl}
```

Manual bibliographies without these macros may slow down the publication process, and thus are not recommended for ACM submissions.

2.16 Colors

While printed ACM publications are usually black and white, screen mode allows the use of colors. The ACM classes pre-define several colors according to [8]: ACMBLue, ACMYellow, ACMOrange, ACMRed, ACMLightBlue, ACMGreen, ACMPurple and ACMDarkBlue. You can use them in color assignments.

The ACM provides the following recommendation on color use.

The most accessible approach would be to ensure that your article is still readable when printed in greyscale. The most notable reasons for this are:

1. The most common type of inherited Color Vision Deficiency (CVD) is red-green (in which similar-brightness colors that differ only in their amounts of red or green are often confused), and it affects up to 8% of males and 0.5% of females of Northern European descent.
2. The most common type of acquired Color Vision Deficiency (CVD) is blue-yellow (including mild cases for many older adults).
3. Most printing is in black and white.
4. Situational impairments (e.g., bright sunlight shining on a mobile screen) tend to reduce the entire color gamut, reducing color discriminability.

Note: It is *not* safe to encode information using only variations in color (i.e., only differences in hue and/or saturation) as there is bound to be someone affected!

To ensure that you are using the most accessible colors, the ACM recommends that you choose sets of colors to help ensure suitable variations in when printed in greyscale by using either of the following tools:

1. ColourBrewer: <http://colorbrewer2.org/>
2. ACE: The Accessible Colour Evaluator: <http://daprlab.com/ace/> for designing WCAG 2.0 compliant palettes.

2.17 Other notable packages and typographic remarks

Several other packages are recommended for specialized tasks.

The package `subcaption` [9] is recommended for complex figures with several subplots or subfigures that require separate subcaptioning. The packages `nomencl` [10] and `glossaries` [11] can be used for the automatic creation of the lists of symbols and concepts used.

By default `acmart` prevents all widows and orphans (i.e., lonely lines at the beginning or end of the page) and hyphenation at the end of the page. This is done by the rather strict settings

```
\widowpenalty=10000
\clubpenalty=10000
\brokenpenalty=10000
```

However, this may lead to frustrating results when the authors must obey a page limit. Setting these penalties to smaller values may help if you absolutely need to.

Another problem might be the too strict line breaking rules. Again, a strategically placed `\sloppy` command or putting the problematic paragraph inside `sloppypar` environment might help—but beware, the results might be, well, sloppy.

Note that the uppercasing in section titles is done using the `textcase` package [12], so the command `\NoCaseChange` inside the title may help to prevent extraneous uppercasing.

2.18 Counting words

Some ACM conferences use word count limits for papers. The calculation of word number for a paper with math, tables and figures is not a trivial task. Currently the authoritative word count is done by translating the PDF to text and using `wc -w` on the output. Authors can use the package `texcount` (used by Overleaf) to get an estimate of the word count. To facilitate this one adds to the beginning of the package metacomments

```
%TC:macro \cite [option:text,text]
%TC:macro \citep [option:text,text]
%TC:macro \citet [option:text,text]
%TC:envir table 0 1
%TC:envir table* 0 1
%TC:envir tabular [ignore] word
%TC:envir displaymath 0 word
%TC:envir math 0 word
%TC:envir comment 0 0
```

and uses `\begin{math}...\end{math}` instead of dollar signs for math. Note that the count is in any case approximate, and the final decision of editors is based on PDF count.

The script `texcount` provides a report of word count in the document.

2.19 Disabled or forbidden commands

The goal of `acmart` package is to provide a uniform look and feel for ACM publications. Accordingly, a number of commands is forbidden or disabled in `acmart`.

You may *not* put several authors or several e-mails into a `\author` or `\email` command. This may lead to errors or warning.

You cannot change `\baselinestretch` in your document: this produces an error.

You should not abuse the command `\vspace`: this command may disturb the type-setting of ACM papers.

You should not load `amssymb` package since the package `acmart` defines the corresponding symbols itself.

2.20 Notes for wizards

Sometimes you need to change the behavior of `acmart`. The usual way to do this is to redefine commands in the preamble. However, these definitions are executed *after* `acmart` is loaded and certain decisions are made. This presents a number of problems.

For example, one may want to use the `titletoc` package with `acmart`. This package should be loaded before `hyperref`. However, since `acmart` loads `hyperref` itself, the line `\usepackage{titletoc}` in the preamble will lead to grief (see <http://tex.stackexchange.com/questions/357265/using-titletoc-with-acm-acmart-style>).

Another example is passing options to a package. Suppose you want to use the `dvipsnames` option of the `xcolor` package. Normally you cannot do this because `acmart` loads this package itself without options.

The file `acmart-preload-hook.tex` can be used to solve these problems. If this file exists, it will be processed before any other package. You can use this file to load packages or pass options to them. For example, if you put in this file

```
\let\LoadClassOrig\LoadClass
\renewcommand\LoadClass[2][\LoadClassOrig[#1]{#2}%
\usepackage{titletoc}
```

then `titletoc` will be loaded before `hyperref`. If you put in this file

```
\PassOptionsToPackage{dvipsnames}{xcolor}
```

you will pass `dvipsnames` to `xcolor`.

Important note. This hook makes it too easy to create a manuscript that is not acceptable by the ACM. It is even easier to create a file that cannot be compiled. So please do not use it *unless you know what you are doing*. And if you use it, *do not ask for support*. If you decide to use this hook, you are on your own.

`\AtBeginMaketitle` Another hook is `\AtBeginMaketitle`. The commands in this hook are executed before `\maketitle`.

2.21 Currently supported publications

Table 5: ACM publications and arguments of the `\acmJournal` command

Abbreviation	Publication
ACMJCSS	ACM Journal on Computing and Sustainable Societies
ACMJDS	ACM Journal of Data Science
AILET	ACM AI Letters

Table 5: ACM publications and arguments of the \acmJournal command (continued)

Abbreviation	Publication
CIE	ACM Computers in Entertainment
CSUR	ACM Computing Surveys
DLT	Distributed Ledger Technologies: Research and Practice
DGOV	Digital Government: Research and Practice
DTRAP	Digital Threats: Research and Practice
FAC	Formal Aspects of Computing
GAMES	ACM Games: Research and Practice
HEALTH	ACM Transactions on Computing for Healthcare
IMWUT	PACM on Interactive, Mobile, Wearable and Ubiquitous Technologies
JACM	Journal of the ACM
JATS	ACM Journal on Autonomous Transportation Systems
JDIQ	ACM Journal of Data and Information Quality
JDS	ACM Journal of Data Science
JEA	ACM Journal of Experimental Algorithmics
JERIC	ACM Journal of Educational Resources in Computing
JETC	ACM Journal on Emerging Technologies in Computing Systems
JOCCH	ACM Journal on Computing and Cultural Heritage
JRC	ACM Journal on Responsible Computing
PACMCGIT	Proceedings of the ACM on Computer Graphics and Interactive Techniques
PACMHCI	PACM on Human-Computer Interaction
PACMOD	PACM on Management of Data
PACMNET	PACM on Networking
PACMPL	PACM on Programming Languages
PACMSE	PACM on Software Engineering
POMACS	PACM on Measurement and Analysis of Computing Systems
TAAS	ACM Transactions on Autonomous and Adaptive Systems
TACCESS	ACM Transactions on Accessible Computing
TACO	ACM Transactions on Architecture and Code Optimization
TAIS	ACM Transactions on AI for Science
TAISAP	ACM Transactions on AI Security and Privacy
TALG	ACM Transactions on Algorithms
TALLIP	ACM Transactions on Asian and Low-Resource Language Information Processing
TAP	ACM Transactions on Applied Perception
TCPS	ACM Transactions on Cyber-Physical Systems
TDS	ACM Transactions on Data Science
TEAC	ACM Transactions on Economics and Computation
TECS	ACM Transactions on Embedded Computing Systems
TELO	ACM Transactions on Evolutionary Learning and Optimization
THRI	ACM Transactions on Human-Robot Interaction
TIIS	ACM Transactions on Interactive Intelligent Systems
TIOT	ACM Transactions on Internet of Things
TISSEC	ACM Transactions on Information and System Security
TIST	ACM Transactions on Intelligent Systems and Technology
TKDD	ACM Transactions on Knowledge Discovery from Data

Table 5: ACM publications and arguments of the \acmJournal command (continued)

Abbreviation	Publication
TMIS	ACM Transactions on Management Information Systems
TOCE	ACM Transactions on Computing Education
TOCHI	ACM Transactions on Computer-Human Interaction
TOCL	ACM Transactions on Computational Logic
TOCS	ACM Transactions on Computer Systems
TOCT	ACM Transactions on Computation Theory
TODAES	ACM Transactions on Design Automation of Electronic Systems
TODS	ACM Transactions on Database Systems
TOG	ACM Transactions on Graphics
TOIS	ACM Transactions on Information Systems
TOIT	ACM Transactions on Internet Technology
TOMACS	ACM Transactions on Modeling and Computer Simulation
TOMM	ACM Transactions on Multimedia Computing, Communications and Applications
TOMPECS	ACM Transactions on Modeling and Performance Evaluation of Computing Systems
TOMS	ACM Transactions on Mathematical Software
TOPC	ACM Transactions on Parallel Computing
TOPLAS	ACM Transactions on Programming Languages and Systems
TOPML	ACM Transactions on Probabilistic Machine Learning
TOPS	ACM Transactions on Privacy and Security
TORS	ACM Transactions on Recommender Systems
TOS	ACM Transactions on Storage
TOSEM	ACM Transactions on Software Engineering and Methodology
TOSN	ACM Transactions on Sensor Networks
TQC	ACM Transactions on Quantum Computing
TRETS	ACM Transactions on Reconfigurable Technology and Systems
TSAS	ACM Transactions on Spatial Algorithms and Systems
TSC	ACM Transactions on Social Computing
TSLP	ACM Transactions on Speech and Language Processing
TWEB	ACM Transactions on the Web

Besides the publications listed in Table 5, there is a special “publication” type FACMP, a forthcoming ACM publication, reserved for new journals which are not assigned an ISSN yet.

2.22 Samples

There are several samples of acmart documents, that can serve as templates for the authors. Among them sample-sig... are templates for conference papers, while sample-acm... are templates for journal articles and special materials.

Here is the list of the samples:

acmmanuscript: A proceedings paper in the manuscript format.

acmsmall: A journal paper in the acmsmall format.

acmsmall-biblatex: A journal paper using experimental *biblatex* bibliography processing.

acmlarge: A journal paper in the acmlarge format.

acmtog: A journal paper in the acmtog format.

sigconf: A proceedings paper in the standard sigconf format.

sigconf-biblatex: A proceedings paper in the sigconf format using experimental *biblatex* bibliography processing.

sigconf-authordraft: A proceedings paper in the sigconf format with the authordraft option selected.

sigconf-i13n: A proceedings paper in the sigconf format with multilanguage titles and abstract.

sigconf-lualatex: A proceedings paper in the sigconf format, Lua $\text{\LaTeX}$ version.

sigplan: A proceedings paper in the sigplan format.

acmsmall-conf: A proceedings paper in the acmsmall format for the publication in a journal using this option.

acmtog-conf: A proceedings paper in the acmtog format for the publication in a journal using this option.

acmcp: A sample of ACM Cover Page used by JDS.

acmengage: A sample of ACM Engage publication.

acmsmall-tagged: A journal paper in the acmsmall format with the tagging code, see Section 2.7.

sigconf-tagged: A proceedings paper in the standard sigconf format with the tagging code, see Section 2.7.

2.23 A note about sigchi-a format

Starting in Spring 2020 ACM retired SIGCHI Extended Abstract format (sigchi-a). ACM will not, under any circumstances, accept documents in this format for publication and will not offer technical support to the authors who use this template.

You may use this format in the nonacm mode only, as in

```
\documentclass[sigchi-a, nonacm]{acmart}
```

sidebar (*env.*) This format has large margin uses for special figures and tables. This package provides three environments for this with optional captions:
marginfigure (*env.*)
margintable (*env.*)

sidebar: textual information in the margin,

marginfigure: a figure in the margin,

margintable: a table in the margin.

The environments `figure` and `table` produce figures and tables with the width of the text column. The environments `figure*` and `table*` produce “wide” figures and tables, which take a large part of the margin.

The horizontal sizes of figures are:

1. `figure: \columnwidth`,
2. `marginfigure: \marginparwidth`,
3. `figure*: \fulltextwidth`.

3 Implementation

3.1 Identification

We start with a declaration of who we are. Most .dtx files put driver code in a separate .drv driver file. We roll this code into the main file and use the pseudo-guard <gobble> for it.

```
1 <class>\NeedsTeXFormat{LaTeX2e}
2 <*gobble>
3 \ProvidesFile{acmart.dtx}
4 </gobble>
5 <class&!tagged>\ProvidesClass{acmart}
6 <tagged>\ProvidesClass{acmart-tagged}
7 [2026/08/30 v2.21 Typesetting articles for the Association for Computing Machinery]
```

And the driver code:

```
8 <*gobble>
9 \documentclass{ltxdoc}
10 \usepackage{array,booktabs,amsmath,graphicx,fancyvrb,tabularx, longtable}
11 \usepackage[tt=false, type1=true]{libertine}
12 \usepackage[varqu]{zi4}
13 \usepackage[libertine]{newtxmath}
14 \usepackage[tableposition=top]{caption}
15 \usepackage{hypdoc}
16 \PageIndex
17 \CodelineIndex
18 \RecordChanges
19 \EnableCrossrefs
20 \begin{document}
21   \DocInput{acmart.dtx}
22 \end{document}
23 </gobble>
24 <*class>
25 \def\@classname{acmart}
```

3.2 Preload hook

We preload acmart-preload-hook:

```
26 \InputIfFileExists{acmart-preload-hook.tex}{%
27   \ClassWarning{\@classname}{%
28     I am loading acmart-preload-hook.tex. You are fully responsible
29     for any problems from now on.}}{}
```

3.3 Tagging and old kernels

We want to use \DocumentMetadata command, but what with the old kernels which some authors might use? Here we just switch off the tagging code if this is not defined

```
\IfDocumentMetadata
20 \providecommand{\IfDocumentMetadataTF}{\@secondoftwo}
31 \providecommand{\IfDocumentMetadataT}{\@gobble}
32 \providecommand{\IfDocumentMetadataF}{\@firstofone}
```

3.4 Options

We need xkeyval since some of our options may have values:

```
33 \RequirePackage{xkeyval}
```

We use xstring to check whether user input is valid

```
34 \RequirePackage{xstring}
```

We need iftex to check the engine

```
35 \RequirePackage{iftex}
```

format The possible formats

```
36 \define@choicekey*+{\@classname.cls}{format}[\ACM@format\ACM@format@nr]{%
37 manuscript, acmsmall, acmlarge, acmtog, sigconf, siggraph,
38 sigplan, sigchi, sigchi-a, acmengage, acmcp}[manuscript]{}%
39 \ClassError{\@classname}{The option format must be manuscript,
40 acmsmall, acmlarge, acmtog, sigconf, siggraph,
41 sigplan, sigchi or sigchi-a}}
42 \def\@DeclareACMFormat#1{\DeclareOptionX{#1}{\setkeys{\@classname.cls}{format=#1}}}
43 \@DeclareACMFormat{manuscript}
44 \@DeclareACMFormat{acmsmall}
45 \@DeclareACMFormat{acmlarge}
46 \@DeclareACMFormat{acmtog}
47 \@DeclareACMFormat{sigconf}
48 \@DeclareACMFormat{siggraph}
49 \@DeclareACMFormat{sigplan}
50 \@DeclareACMFormat{sigchi}
51 \@DeclareACMFormat{sigchi-a}
52 \@DeclareACMFormat{acmengage}
53 \@DeclareACMFormat{acmcp}
54 \ExecuteOptionsX{format}
```

\if@ACM@screen Whether we use screen mode

```
55 \define@boolkey+{\@classname.cls}[@ACM@]{screen}[true]{%
56 \if@ACM@screen
57 \PackageInfo{\@classname}{Using screen mode}%
58 \else
59 \PackageInfo{\@classname}{Not using screen mode}%
60 \fi}{\PackageError{\@classname}{The option screen can be either true or
61 false}}
62 \ExecuteOptionsX{screen=false}
```

\if@ACM@urlbreakonhyphens

```
63 \define@boolkey+{\@classname.cls}[@ACM@]{urlbreakonhyphens}[true]{%
64 \if@ACM@urlbreakonhyphens
65 \PackageInfo{\@classname}{Using breaking urls on hyphens}%
66 \else
67 \PackageInfo{\@classname}{Not breaking urls on hyphens}%
68 \fi}{\PackageError{\@classname}{The option urlbreakonhyphens can be either true or
69 false}}
70 \ExecuteOptionsX{urlbreakonhyphens=true}
```

\if@ACM@acmthm Whether we define theorem-like environments.

```
71 \define@boolkey+{\@classname.cls}[@ACM@]{acmthm}[true]{%
```

```

72 \if@ACM@acmthm
73   \PackageInfo{\@classname}{Requiring acmthm}%
74 \else
75   \PackageInfo{\@classname}{Suppressing acmthm}%
76 \fi}{\PackageError{\@classname}{The option acmthm can be either true or
77   false}}
78 \ExecuteOptionsX{acmthm=true}

\if@ACM@review Whether we use review mode
79 \define@boolkey+{\@classname.cls}[@ACM@]{review}[true]{%
80   \if@ACM@review
81     \PackageInfo{\@classname}{Using review mode}%
82     \AtBeginDocument{\@ACM@printfoliostrue}%
83   \else
84     \PackageInfo{\@classname}{Not using review mode}%
85   \fi}{\PackageError{\@classname}{The option review can be either true or
86     false}}
87 \ExecuteOptionsX{review=false}

\if@ACM@authorversion Whether we use author's-version mode
88 \define@boolkey+{\@classname.cls}[@ACM@]{authorversion}[true]{%
89   \if@ACM@authorversion
90     \PackageInfo{\@classname}{Using authorversion mode}%
91   \else
92     \PackageInfo{\@classname}{Not using authorversion mode}%
93   \fi}{\PackageError{\@classname}{The option authorversion can be either true or
94     false}}
95 \ExecuteOptionsX{authorversion=false}

\if@ACM@nonacm Special option for non-ACM publications using the ACM typesetting options.
96 \define@boolkey+{\@classname.cls}[@ACM@]{nonacm}[true]{%
97   \if@ACM@nonacm
98     \PackageInfo{\@classname}{Using nonacm mode}%
99     \AtBeginDocument{\@ACM@printacmreffalse}%
100    % in 'nonacm' mode we disable the "ACM Reference Format"
101    % printing by default, but this can be re-enabled by the
102    % user using \settopmatter{printacmref=true}
103   \else
104     \PackageInfo{\@classname}{Not using nonacm mode}%
105   \fi}{\PackageError{\@classname}{The option nonacm can be either true or
106     false}}
107 \ExecuteOptionsX{nonacm=false}

\if@ACM@balance Whether to balance the last page
108 \define@boolkey+{\@classname.cls}[@ACM@]{balance}[true]{%
109   \PackageError{\@classname}{The option balance can be either true or
110     false}}
111 \ExecuteOptionsX{balance}

\if@ACM@pbalance Whether to balance the last page
112 \define@boolkey+{\@classname.cls}[@ACM@]{pbalance}[true]{%
113   \PackageError{\@classname}{The option pbalance can be either true or
114     false}}
115 \ExecuteOptionsX{pbalance=false}

```


`\if@ACM@natbib@override` This macro is no longer used.

`\if@ACM@natbib` Whether we use natbib mode

```
116 \define@boolkey+{\@classname.cls}[@ACM@]{natbib}[true]{%
117   \if@ACM@natbib
118     \PackageInfo{\@classname}{Explicitly selecting natbib mode}%
119   \else
120     \PackageInfo{\@classname}{Explicitly deselecting natbib mode}%
121   \fi}{\PackageError{\@classname}{The option natbib can be either true or
122     false}}
123 \ExecuteOptionsX{natbib=true}
```

`\if@ACM@anonymous` Whether we use anonymous mode

```
124 \define@boolkey+{\@classname.cls}[@ACM@]{anonymous}[true]{%
125   \if@ACM@anonymous
126     \PackageInfo{\@classname}{Using anonymous mode}%
127   \else
128     \PackageInfo{\@classname}{Not using anonymous mode}%
129   \fi}{\PackageError{\@classname}{The option anonymous can be either true or
130     false}}
131 \ExecuteOptionsX{anonymous=false}
```

`\if@ACM@timestamp` Whether we use timestamp mode

```
132 \define@boolkey+{\@classname.cls}[@ACM@]{timestamp}[true]{%
133   \if@ACM@timestamp
134     \PackageInfo{\@classname}{Using timestamp mode}%
135   \else
136     \PackageInfo{\@classname}{Not using timestamp mode}%
137   \fi}{\PackageError{\@classname}{The option timestamp can be either true or
138     false}}
139 \ExecuteOptionsX{timestamp=false}
```

`\if@ACM@authordraft` Whether we use author-draft mode

```
140 \define@boolkey+{\@classname.cls}[@ACM@]{authordraft}[true]{%
141   \if@ACM@authordraft
142     \PackageInfo{\@classname}{Using authordraft mode}%
143     \@ACM@timestamptrue
144     \@ACM@reviewtrue
145   \else
146     \PackageInfo{\@classname}{Not using authordraft mode}%
147   \fi}{\PackageError{\@classname}{The option authordraft can be either true or
148     false}}
149 \ExecuteOptionsX{authordraft=false}
```

`\ACM@fontsize` The font size to pass to the base class

```
150 % \changes{v1.87}{2022/08/27}{Added fontsize 8pt}
151 \def\ACM@fontsize{}
152 \DeclareOptionX{8pt}{\edef\ACM@fontsize{\CurrentOption}}
153 \DeclareOptionX{9pt}{\edef\ACM@fontsize{\CurrentOption}}
154 \DeclareOptionX{10pt}{\edef\ACM@fontsize{\CurrentOption}}
155 \DeclareOptionX{11pt}{\edef\ACM@fontsize{\CurrentOption}}
156 \DeclareOptionX{12pt}{\edef\ACM@fontsize{\CurrentOption}}
```

\ACM@languages The languages of the document

```
157 \def\ACM@languages{}
158 \DeclareOptionX{language}{%
159   \ifx\ACM@languages\@empty
160     \gdef\ACM@languages{english}\fi
161   \g@addto@macro\ACM@languages{, #1}}

162 \DeclareOptionX{draft}{\PassOptionsToClass{\CurrentOption}{amsart}}
163 \DeclareOptionX{*}{\PassOptionsToClass{\CurrentOption}{amsart}}
164 \ProcessOptionsX
165 \ClassInfo{\@classname}{Using format \ACM@format, number \ACM@format@nr}
```

3.5 Setting switches

\if@ACM@manuscript Whether we use manuscript mode

```
166 \newif\if@ACM@manuscript
```

\if@ACM@journal There are two kinds of publications: journals and books

```
167 \newif\if@ACM@journal
```

\if@ACM@journal@bibstrip Sometimes ACM wants a journal-like publication to have conference information in the bibstrip and vice versa, so we have an additional switch.

```
168 \newif\if@ACM@journal@bibstrip
```

\if@ACM@journal@bibstrip@or@tog ACM TOG sometimes uses journal-like bibstrip even for conference proceedings, so we need to set it separately

```
169 \newif\if@ACM@journal@bibstrip@or@tog
```

\if@ACM@sigchiamode The formatting of SIGCHI extended abstracts is quite unusual. We have a special switch for them.

```
170 \newif\if@ACM@sigchiamode
```

\if@ACM@engage ACM Engage course materials have special formatting

```
171 \newif\if@ACM@engage
172 \@ACM@engagefalse
```

\if@ACM@acmcp ACM cover page formatting

```
173 \newif\if@ACM@acmcp
174 \@ACM@acmcpfalse
```

Setting up switches

```
175 \ifnum\ACM@format@nr=5\relax % siggraph
176 \ClassWarning{\@classname}{%
177   The format siggraph is now obsolete.\MessageBreak
178   I am switching to sigconf.}
179 \setkeys{\@classname.cls}{format=sigconf}
180 \fi
181 \ifnum\ACM@format@nr=7\relax % sigchi
182 \ClassWarning{\@classname}{%
183   The format sigchi is now obsolete.\MessageBreak
184   I am switching to sigconf.}
185 \setkeys{\@classname.cls}{format=sigconf}
```

```

186 \fi
187 \ifnum\ACM@format@nr=8\relax % sigchi
188 \ClassWarning{\@classname}{%
189   ACM SIGCHI has retired the SIGCHI-A template\MessageBreak
190   effective immediately. ACM is keeping this template\MessageBreak
191   option available to authors who are working on legacy\MessageBreak
192   documents only. ACM will not, under any circumstances,\MessageBreak
193   accept documents in this format for publication and\MessageBreak
194   will not offer technical support to the authors who use\MessageBreak
195   this template.\MessageBreak
196   ACM SIGCHI is directing Conference leaders and\MessageBreak
197   authors to publish their articles using the SIGCONF\MessageBreak
198   template call.}
199 \fi
200 \ifnum\ACM@format@nr=0\relax
201   \@ACM@manuscripttrue
202 \else
203   \@ACM@manuscriptfalse
204 \fi
205 \@ACM@sigchiamodefalse
206 \ifcase\ACM@format@nr
207 \relax % manuscript
208   \@ACM@journaltrue
209 \or % acsmall
210   \@ACM@journaltrue
211 \or % acmlarge
212   \@ACM@journaltrue
213 \or % acmtog
214   \@ACM@journaltrue
215   \@ACM@journal@bibstrip@or@togtrue
216 \or % sigconf
217   \@ACM@journalfalse
218 \or % siggraph
219   \@ACM@journalfalse
220 \or % sigplan
221   \@ACM@journalfalse
222 \or % sigchi
223   \@ACM@journalfalse
224 \or % sigchi-a
225   \@ACM@journalfalse
226   \@ACM@sigchiamodetrue
227 \or % acmengage
228   \@ACM@journalfalse
229   \@ACM@engagetrue
230 \or % acmcp
231   \@ACM@journaltrue
232   \@ACM@acmcptrue
233   \AtBeginDocument{\@ACM@printacmreffalse}%
234 \fi
235 \if@ACM@journal
236   \@ACM@journal@bibstriptrue
237   \@ACM@journal@bibstrip@or@togtrue
238 \else
239   \@ACM@journal@bibstripfalse

```

240 \fi

3.6 Loading the base class and package

Before we load the base class we want to save the original definitions of sectioning commands since acmart redefines them, and we need the vanilla versions.

\@startsection@kernel

241 \let\@startsection@kernel\@startsection

\@xsect@kernel

242 \let\@xsect@kernel\@xsect

\@sect@kernel

243 \let\@sect@kernel\@sect

\@ssect@kernel

244 \let\@ssect@kernel\@ssect

At this point we either have \ACM@fontsize or use defaults

245 \ifx\ACM@fontsize\empty

246 \ifcase\ACM@format@nr

247 \relax % manuscript

248 \def\ACM@fontsize{9pt}%

249 \or % acmsmall

250 \def\ACM@fontsize{10pt}%

251 \or % acmlarge

252 \def\ACM@fontsize{10pt}%

253 \or % acmtog

254 \def\ACM@fontsize{9pt}%

255 \or % sigconf

256 \def\ACM@fontsize{9pt}%

257 \or % siggraph

258 \def\ACM@fontsize{9pt}%

259 \or % sigplan

260 \def\ACM@fontsize{10pt}%

261 \or % sigchi

262 \def\ACM@fontsize{9pt}%

263 \or % sigchi-a

264 \def\ACM@fontsize{10pt}%

265 \or % acmengage

266 \def\ACM@fontsize{10pt}%

267 \or % acmcp

268 \def\ACM@fontsize{9pt}%

269 \fi

270 \fi

271 \ClassInfo{\@classname}{Using fontsize \ACM@fontsize}

272 \LoadClass[\ACM@fontsize, reqno]{amsart}

273 \RequirePackage{microtype}

We need etoolbox for delayed code

274 \RequirePackage{etoolbox}

Booktabs is now the default

275 \RequirePackage{booktabs}

We need `totpages` to calculate the number of pages and `refcount` to use that number

```
276 \RequirePackage{refcount}
277 \RequirePackage{totpages}
```

The `\collect@body` macro in `amsmath` is defined using `\def`. We load `environ` to access the `\long` version of this command

```
278 \RequirePackage{environ}
```

We use `setspace` for double spacing

```
279 \if@ACM@manuscript
280 \RequirePackage{setspace}
281 \onehalfspacing
282 \fi
```

We no longer need `textcase` for better upcasing.

```
283 % \RequirePackage{textcase}

284 \if@ACM@acmcp
285 \RequirePackage{framed}
286 \RequirePackage{zref-savepos, zref-user}
287 \fi
```

`\@ACM@acmcp@delta` We need to store the `dimen` to store the insert length of `amcp` box

```
288 \newdimen\@ACM@acmcp@delta
289 \@ACM@acmcp@delta=0pt\relax
```

3.7 Citations

We patch `\setcitestyle` to allow, for example, `\setcitestyle{sort}` and `\setcitestyle{nosort}`. We patch `\citestyle` to warn about undefined citation styles.

```
290 \if@ACM@natbib
291   \RequirePackage{natbib}
292   \renewcommand{\bibsection}{%
293     \section*{\refname}%
294     \phantomsection\addcontentsline{toc}{section}{\refname}%
295   }
296   \renewcommand{\bibfont}{\bibliofont}
297   \renewcommand{\setcitestyle}[1]{
298     \@for\@tempa:=#1\do
299     {\def\@tempb{round}\ifx\@tempa\@tempb
300       \renewcommand\NAT@open{(\}\renewcommand\NAT@close{)}\fi
301       \def\@tempb{square}\ifx\@tempa\@tempb
302         \renewcommand\NAT@open{[\}\renewcommand\NAT@close{]}\fi
303       \def\@tempb{angle}\ifx\@tempa\@tempb
304         \renewcommand\NAT@open{$<$}\renewcommand\NAT@close{$>$}\fi
305       \def\@tempb{curly}\ifx\@tempa\@tempb
306         \renewcommand\NAT@open{\{\}\renewcommand\NAT@close{\}}\fi
307       \def\@tempb{semicolon}\ifx\@tempa\@tempb
308         \renewcommand\NAT@sep{;}\fi
309       \def\@tempb{colon}\ifx\@tempa\@tempb
310         \renewcommand\NAT@sep{;}\fi
311       \def\@tempb{comma}\ifx\@tempa\@tempb
312         \renewcommand\NAT@sep{,}\fi
```

```

313 \def\@tempb{authoryear}\ifx\@tempa\@tempb
314 \NAT@numbersfalse\fi
315 \def\@tempb{numbers}\ifx\@tempa\@tempb
316 \NAT@numberstrue\NAT@superfalse\fi
317 \def\@tempb{super}\ifx\@tempa\@tempb
318 \NAT@numberstrue\NAT@supertrue\fi
319 \def\@tempb{nobibstyle}\ifx\@tempa\@tempb
320 \let\bibstyle=\@gobble\fi
321 \def\@tempb{bibstyle}\ifx\@tempa\@tempb
322 \let\bibstyle=\@citestyle\fi
323 \def\@tempb{sort}\ifx\@tempa\@tempb
324 \def\NAT@sort{\@ne}\fi
325 \def\@tempb{nosort}\ifx\@tempa\@tempb
326 \def\NAT@sort{\z@}\fi
327 \def\@tempb{compress}\ifx\@tempa\@tempb
328 \def\NAT@cmprs{\@ne}\fi
329 \def\@tempb{nocompress}\ifx\@tempa\@tempb
330 \def\NAT@cmprs{\z@}\fi
331 \def\@tempb{sort&compress}\ifx\@tempa\@tempb
332 \def\NAT@sort{\@ne}\def\NAT@cmprs{\@ne}\fi
333 \def\@tempb{mcite}\ifx\@tempa\@tempb
334 \let\NAT@merge\@ne\fi
335 \def\@tempb{merge}\ifx\@tempa\@tempb
336 \@ifnum{\NAT@merge<\tw@}{\let\NAT@merge\tw@}{\fi
337 \def\@tempb{elide}\ifx\@tempa\@tempb
338 \@ifnum{\NAT@merge<\thr@}{\let\NAT@merge\thr@}{\fi
339 \def\@tempb{longnamesfirst}\ifx\@tempa\@tempb
340 \NAT@longnamestrue\fi
341 \def\@tempb{nonamebreak}\ifx\@tempa\@tempb
342 \def\NAT@nmfmt#1{\mbox{\NAT@up#1}}\fi
343 \expandafter\NAT@find@eq\@tempa=\relax\@nil
344 \if\@tempc\relax\else
345 \expandafter\NAT@rem@eq\@tempc
346 \def\@tempb{open}\ifx\@tempa\@tempb
347 \xdef\NAT@open{\@tempc}\fi
348 \def\@tempb{close}\ifx\@tempa\@tempb
349 \xdef\NAT@close{\@tempc}\fi
350 \def\@tempb{aysep}\ifx\@tempa\@tempb
351 \xdef\NAT@aysep{\@tempc}\fi
352 \def\@tempb{yysep}\ifx\@tempa\@tempb
353 \xdef\NAT@yrsep{\@tempc}\fi
354 \def\@tempb{notesep}\ifx\@tempa\@tempb
355 \xdef\NAT@cmt{\@tempc}\fi
356 \def\@tempb{citesep}\ifx\@tempa\@tempb
357 \xdef\NAT@sep{\@tempc}\fi
358 \fi
359 }%
360 \NAT@@setcites
361 }
362 \renewcommand\citestyle[1]{%
363 \ifcsname bibstyle@#1\endcsname%
364 \csname bibstyle@#1\endcsname\let\bibstyle\@gobble%
365 \else%
366 \@latex@error{Undefined `#1' citestyle}%

```

```

367   \fi
368 }%
369 \fi

```

`\bibstyle@acmauthoryear` The default author-year format:

```

370 \newcommand{\bibstyle@acmauthoryear}{%
371   \setcitestyle{%
372     authoryear,%
373     open={},close={},citesep={;},%
374     aysep={},yysep={},,%
375     notesep={, }}

```

`\bibstyle@acmnumeric` The default numeric format:

```

376 \newcommand{\bibstyle@acmnumeric}{%
377   \setcitestyle{%
378     numbers,sort&compress,%
379     open={},close={},citesep={},,%
380     notesep={, }}

```

The default is numeric:

```

381 \if@ACM@natbib
382 \citestyle{acmnumeric}
383 \fi

```

3.8 Internationalization

```

384 \if@ACM@journal
385   \renewcommand\keywordsname{Additional Key Words and Phrases}%
386 \else
387   \renewcommand\keywordsname{Keywords}%
388 \fi
389 \if@ACM@engage
390   \renewcommand\abstractname{Synopsis}%
391 \fi
392 \ifx\ACM@languages\@empty
393 \else
394   \RequirePackage[\ACM@languages]{babel}%
395   \addto\captionsenglish{%
396     \if@ACM@journal
397       \renewcommand\keywordsname{Additional Key Words and Phrases}%
398     \else
399       \renewcommand\keywordsname{Keywords}%
400     \fi
401     \renewcommand\acksname{Acknowledgements}%
402     \if@ACM@engage
403       \renewcommand\abstractname{Synopsis}%
404     \fi
405   }%
406   \addto\captionsfrench{%
407     \if@ACM@journal
408       \renewcommand\keywordsname{Mots Clés et Phrases Supplémentaires}%
409     \else
410       \renewcommand\keywordsname{Mots clés}%
411     \fi

```

```

412     \renewcommand\acksname{Remerciements}%
413 }%
414 \addto\captionsspanish{%
415     \if@ACM@journal
416         \renewcommand\keywordsname{Zusätzliche Schlagwörter und Phrasen}%
417     \else
418         \renewcommand\keywordsname{Schlagwörter}%
419     \fi
420     \renewcommand\acksname{Danksagungen}%
421 }%
422 \addto\captionsspanish{%
423     \if@ACM@journal
424         \renewcommand\keywordsname{Palabras y Frases Claves Adicionales}%
425     \else
426         \renewcommand\keywordsname{Palabras claves}%
427     \fi
428     \renewcommand\acksname{Expresiones de gratitud}%
429 }%
430 \fi

```

\ACM@lang@check Some commands should not be used if the document is monlingual

```

431 \newcommand\ACM@lang@check[1]{%
432     \ifx\ACM@languages\empty\relax
433     \ClassError{\@classname}{%
434         Command \string#1 \MessageBreak is used in monlingual document}{%
435         You used a command (\string#1) \MessageBreak
436         that does not have a meaning \MessageBreak
437         unless are languages are defined. \MessageBreak
438         Please choose the languages in \string\documentclass
439         \MessageBreak
440         (e.g. \string\documentclass[languages={french, english}]{acmart}),
441         \MessageBreak
442         or delete the command.}%
443     \fi}

```

\@translatedtitle

```
444 \def\@translatedtitle{}
```

\translatedtitle The title of the paper in a different language

```

445 \newcommand\translatedtitle[2]{\ACM@lang@check{\translatedtitle}%
446 \g@addto@macro\@translatedtitle{\par\foreignlanguage{#1}{#2}}}

```

\@translatedsubtitle

```
447 \def\@translatedsubtitle{}
```

\translatedsubtitle The subtitle of the paper in a different language

```

448 \newcommand\translatedsubtitle[2]{\ACM@lang@check{\translatedsubtitle}%
449 \g@addto@macro\@translatedsubtitle{\par\foreignlanguage{#1}{#2}}}

```

\@translatedkeywords

```
450 \def\@translatedkeywords{}
```

\translatedkeywords Add keywords to the paper in the language specified

```

451 \newcommand\translatedkeywords[2]{\ACM@lang@check{\translatedkeywords}%
452 \g@addto@macro\@translatedkeywords{\mktranslatedkeywords{#1}{#2}}}

```


`\@translatedabstracts` The container for translated abstracts.

```
453 \def\@translatedabstracts{}
```

`translatedabstract` We save translated abstracts into `\@translatedabstracts`

```
454 \newenvironment{translatedabstract}[1]{\Collect@Body
455   \@savetranslatedabstract\mktranslatedabstract{#1}}{}
```

`\@savetranslatedabstract` And saving the abstract

```
456 \long\def\@savetranslatedabstract#1{\if@ACM@maketitle@typeset
457   \ClassError{\@classname}{Abstract must be defined before maketitle
458     command. Please move it!}\fi
459   \ACM@lang@check{translatedabstract}%
460   \g@addto@macro\@translatedabstracts{\bgroup#1\egroup}}
```

3.9 Sectioning

`\@startsection` Before we call `hyperref`, we redefine `\startsection` commands to their $\text{\LaTeX}$ defaults since the `amsart` ones are too AMS-specific. We need to do this early since we want `hyperref` to have a chance to redefine them again:

```
461 \let\@startsection\@startsection@kernel
462 \def\@sect#1#2#3#4#5#6[#7]#8{%
463   \ifnum #2>\c@secnumdepth \else
464     \tochangmeasure{\csname the#1\endcsname}\fi
465   \@sect@kernel{#1}{#2}{#3}{#4}{#5}{#6}[#7]{#8}}
466 \let\@ssect\@ssect@kernel
467 \let\@xsect\@xsect@kernel
468 \def\@seccntformat#1{\csname the#1\endcsname\quad}
```

`\@starttoc` The `amsart` package redefines `\startsection`. Here we redefine it again to make the table of contents work.

```
469 \def\@starttoc#1#2{\begingroup\makeatletter
470   \setTrue{#1}%
471   \par\removelastskip\vskip\z@skip
472   \@startsection{section}\@M\z@{\linespacing\@plus\linespacing}%
473     {.5\linespacing}{\centering\contentsnamefont}{#2}%
474   \@input{\jobname.#1}%
475   \if@filesw
476     \xp@newwrite\csname tf@#1\endcsname
477     \immediate\xp@openout\csname tf@#1\endcsname \jobname.#1\relax
478   \fi
479   \global\@nobreakfalse \endgroup
480   \addvspace{32\p@\@plus14\p@}%
481 }
```

`\l@section` Section spacing is more generous than for `amsart`

```
482 \def\l@section{\@tocline{1}{0pt}{1pc}{2pc}{}}
```

`\l@subsection` The spacing in `amsart` is too large

```
483 \def\l@subsection{\@tocline{2}{0pt}{1pc}{3pc}{}}
```

`\l@subsubsection` The spacing in `amsart` is too large

```
484 \def\l@subsubsection{\@tocline{3}{0pt}{1pc}{5pc}{}}
```

We need to define footnote-related stuff before the call to hyperref (Ross Moore)

`\@makefnmark` We do not use indentation for footnotes

```
485 \def\@makefnmark{\noindent\@makefnmark}
```

`\@footnotetext` In sigchi-a mode our footnotes are in the margin!

```
486 \if@ACM@sigchiamode
487 \long\def\@footnotetext#1{\marginpar{%
488   \reset@font\small
489   \interlinepenalty\interfootnotelinepenalty
490   \protected@edef\@currentlabel{%
491     \csname p@footnote\endcsname\@thefnmark
492   }%
493   \color@begingroup
494     \@makefnmark{%
495       \rule{z@footnotesep}{\ignorespaces#1\@finalstrut\strutbox}%
496     \color@endgroup}}%
497 \fi
```

`\@mpfootnotetext` We want the footnotes in minipages centered:

```
498 \long\def\@mpfootnotetext#1{%
499   \global\setbox\@mpfootins\vbox{%
500     \unvbox\@mpfootins
501     \reset@font\footnotesize
502     \hsize\columnwidth
503     \@parboxrestore
504     \protected@edef\@currentlabel
505       {\csname p@mpfootnote\endcsname\@thefnmark}%
506     \color@begingroup\centering
507       \@makefnmark{%
508         \rule{z@footnotesep}{\ignorespaces#1\@finalstrut\strutbox}%
509       \color@endgroup}}
```

`\@makefnmark` AMS classes use a buggy definition of `\makefnmark`. We revert to the standard one.

```
510 \def\@makefnmark{\hbox{\@textsuperscript{\normalfont\@thefnmark}}}
```

3.10 Hyperxmp and hyperref

Adding hyperref

```
511 \let\@footnotemark@nolink\@footnotemark
512 \let\@footnotetext@nolink\@footnotetext
513 \RequirePackage[bookmarksnumbered,unicode]{hyperref}
514 \RequirePackage{hyperxmp}
515 \pdfstringdefDisableCommands{%
516   \def\addtocounter#1#2{%
517     \def\unskip{}%
518     \def\textbullet{- }%
519     \def\texttrightarrow{ -> }%
520     \def\footnotemark{}%
521   }
522 \urlstyle{rm}
523 \ifcase\ACM@format@nr
524 \relax % manuscript
```

```

525 \or % acmsmall
526 \or % acmlarge
527 \or % acmtog
528 \or % sigconf
529 \or % siggraph
530 \or % sigplan
531 \urlstyle{sf}
532 \or % sigchi
533 \or % sigchi-a
534 \urlstyle{sf}
535 \or % acmengage
536 \or % acmcp
537 \fi
538 \AtEndPreamble{%
539   \if@ACM@urlbreakonhyphens
540     \def\do@url@hyp{\do\~}%
541   \fi
542   \if@ACM@screen
543     \hypersetup{colorlinks,
544       linkcolor=ACMPurple,
545       citecolor=ACMPurple,
546       urlcolor=ACMDarkBlue,
547       filecolor=ACMDarkBlue}
548   \else
549     \hypersetup{hidelinks}
550   \fi
551   \hypersetup{pdflang={en},
552     pdfdisplaydoctitle}}
  Bibliography mangling.
553 \if@ACM@natbib
554   \let\cite\cite
555   \let\cite\citep
556   \let\citeANP\citeauthor
557   \let\citeNN\citeyearpar
558   \let\citeyearNP\citeyear
559   \let\citeNP\citealt
560   \DeclareRobustCommand\citeA
561     {\begingroup\NAT@swafalse
562       \let\NAT@ctype\@ne\NAT@partrue\NAT@fullfalse\NAT@open\NAT@citetp}%
563   \providecommand\newblock{}%
564 \else
565   \AtBeginDocument{%
566     \let\shortcite\cite%
567     \providecommand\citename[1]{#1}}
568 \fi
569 \newcommand\shortcite[2][]{%
570   \ifNAT@numbers\cite[#1]{#2}\else\citeyearpar[#1]{#2}\fi}

```

`\bibliographystyle` The `amsart` package redefines `\bibliographystyle` since it prefers the AMS bibliography style. We turn it back to the $\TeX$ definition:

```

571 \def\bibliographystyle#1{%
572   \ifx\@begindocumenthook\@undefined\else
573     \expandafter\AtBeginDocument

```

```

574 \fi
575 {\if@filesw
576 \immediate\write\auxout{\string\bibstyle{#1}}%
577 \fi}}

```

3.11 Other packages

Graphics and color.

```

578 \RequirePackage{graphicx}
579 \RequirePackage[prologue]{xcolor}

We define ACM colors according to [8]:
580 \definecolor[named]{ACMBlue}{cmyk}{1,0.1,0,0.1}
581 \definecolor[named]{ACMYellow}{cmyk}{0,0.16,1,0}
582 \definecolor[named]{ACMOrange}{cmyk}{0,0.42,1,0.01}
583 \definecolor[named]{ACMRed}{cmyk}{0,0.90,0.86,0}
584 \definecolor[named]{ACMLightBlue}{cmyk}{0.49,0.01,0,0}
585 \definecolor[named]{ACMGreen}{cmyk}{0.20,0,1,0.19}
586 \definecolor[named]{ACMPurple}{cmyk}{0.55,1,0,0.15}
587 \definecolor[named]{ACMDarkBlue}{cmyk}{1,0.58,0,0.21}

```

Author-draft mode or sigchi-a mode

```

588 \if@ACM@authordraft
589 \RequirePackage{draftwatermark}
590 \SetWatermarkFontSize{0.5in}
591 \SetWatermarkColor[gray]{.9}
592 \SetWatermarkText{\parbox{12em}{\centering
593     Unpublished working draft.\\
594     Not for distribution.}}
595 \else
596 \if@ACM@sigchiamode
597 \if@ACM@nonacm\else
598 \RequirePackage{draftwatermark}
599 \SetWatermarkFontSize{0.5in}
600 \SetWatermarkColor[gray]{.9}
601 \SetWatermarkText{\parbox{12em}{\centering
602     Legacy document. \\
603     Not for publication in an ACM venue}}
604 \fi
605 \fi
606 \fi

```

3.12 Paper size and paragraphing

We use geometry for dimensions. Note that the present margins do not depend on the font size option—we might need to change this. See <https://github.com/borisveytsman/acmart/issues/5#issuecomment-272881329>.

```

607 \RequirePackage{geometry}
608 \ifcase\ACM@format@nr
609 \relax % manuscript
610 \geometry{letterpaper,head=13pt,
611     marginparwidth=6pc,heightrounded}%
612 \or % acmsmall
613 \geometry{twoside=true,

```

```

614     includeheadfoot, head=13pt, foot=2pc,
615     paperwidth=6.75in, paperheight=10in,
616     top=58pt, bottom=44pt, inner=46pt, outer=46pt,
617     marginparwidth=2pc,heightrounded
618 }%
619 \or % acmlarge
620 \geometry{twoside=true, head=13pt, foot=2pc,
621     paperwidth=8.5in, paperheight=11in,
622     includeheadfoot,
623     top=78pt, bottom=114pt, inner=81pt, outer=81pt,
624     marginparwidth=4pc,heightrounded
625 }%
626 \or % acmtog
627 \geometry{twoside=true, head=13pt, foot=2pc,
628     paperwidth=8.5in, paperheight=11in,
629     includeheadfoot, columnsep=24pt,
630     top=52pt, bottom=75pt, inner=52pt, outer=52pt,
631     marginparwidth=2pc,heightrounded
632 }%
633 \or % sigconf
634 \geometry{twoside=true, head=13pt,
635     paperwidth=8.5in, paperheight=11in,
636     includeheadfoot, columnsep=2pc,
637     top=57pt, bottom=73pt, inner=54pt, outer=54pt,
638     marginparwidth=2pc,heightrounded
639 }%
640 \or % siggraph
641 \geometry{twoside=true, head=13pt,
642     paperwidth=8.5in, paperheight=11in,
643     includeheadfoot, columnsep=2pc,
644     top=57pt, bottom=73pt, inner=54pt, outer=54pt,
645     marginparwidth=2pc,heightrounded
646 }%
647 \or % sigplan
648 \geometry{twoside=true, head=13pt,
649     paperwidth=8.5in, paperheight=11in,
650     includeheadfoot=false, columnsep=2pc,
651     top=1in, bottom=1in, inner=0.75in, outer=0.75in,
652     marginparwidth=2pc,heightrounded
653 }%
654 \or % sigchi
655 \geometry{twoside=true, head=13pt,
656     paperwidth=8.5in, paperheight=11in,
657     includeheadfoot, columnsep=2pc,
658     top=66pt, bottom=73pt, inner=54pt, outer=54pt,
659     marginparwidth=2pc,heightrounded
660 }%
661 \or % sigchi-a
662 \geometry{twoside=false, head=13pt,
663     paperwidth=11in, paperheight=8.5in,
664     includeheadfoot, marginparsep=72pt,
665     marginparwidth=170pt, columnsep=20pt,
666     top=72pt, bottom=72pt, left=314pt, right=72pt
667 }%

```

```

668 \mparswitchfalse
669 \reversemarginpar
670 \or % acmengage
671 \geometry{twoside=true, head=13pt,
672   paperwidth=8.5in, paperheight=11in,
673   includeheadfoot, columnsep=2pc,
674   top=57pt, bottom=73pt, inner=54pt, outer=54pt,
675   marginparwidth=2pc,heightrounded
676   }%
677 \or % acmcp
678 \geometry{twoside=true,
679   includeheadfoot, head=13pt, foot=2pc,
680   paperwidth=6.75in, paperheight=10in,
681   top=58pt, bottom=44pt, inner=46pt, outer=46pt,
682   marginparwidth=2pc,heightrounded
683   }%
684 \fi

```

\parindent Paragraphing

```

\parskip 685 \setlength\parindent{10\p@}
686 \setlength\parskip{\z@}
687 \ifcase\ACM@format@nr
688 \relax % manuscript
689 \or % acmsmall
690 \or % acmlarge
691 \or % acmtog
692 \setlength\parindent{9\p@}%
693 \or % sigconf
694 \or % siggraph
695 \or % sigplan
696 \or % sigchi
697 \or % sigchi-a
698 \or % acmengage
699 \or % acmcp
700 \fi

```

\normalparindent The `amsart` package defines the `\normalparindent` length and initializes it to 12pt (the value of `\parindent` in `amsart`). It is later used to set the `\listparindent` length in the quotation environment and the `\parindent` length in the `\@footnotetext` command. We set `\normalparindent` to the value of `\parindent` as selected by `acmart` for consistent paragraph indents.

```
701 \setlength\normalparindent{\parindent}
```

Footnotes require some consideration. We have several layers of footnotes: front-matter footnotes, “regular” footnotes and the special insert for the bibstrip. In the old ACM classes, the bibstrip was a `\@float`. The problem with floats is that they tend to, well, float—and we want the guarantee they stay.

We use `manyfoot` for layered footnotes instead.

copyrightpermissionfootnoterule This is the footnote rule that separates the bibstrip from the rest of the paper. It is a full width rule.

```

702 \def\copyrightpermissionfootnoterule{\kern-3\p@
703   \hrule \@width \columnwidth \kern 2.6\p@}

```

```

704 \RequirePackage{manyfoot}
705 \SelectFootnoteRule[2]{copyrightpermission}
706 \DeclareNewFootnote{authorsaddresses}
707 \SelectFootnoteRule[2]{copyrightpermission}
708 \DeclareNewFootnote{copyrightpermission}

```

\footnoterule Tschichold's rules:

```

709 \def\footnoterule{\kern-3\p@
710   \hrule \@width 4pc \kern 2.6\p@}

```

\endminipage We do not use footnote rules in minipages

```

711 \def\endminipage{%
712   \par
713   \unskip
714   \ifvoid\@mpfootins\else
715     \vskip\skip\@mpfootins
716     \normalcolor
717     \unvbox\@mpfootins
718   \fi
719   \@minipagefalse
720   \color@endgroup
721   \egroup
722   \expandafter\@iiiparbox\@mpargs{\unvbox\@tempboxa}}

```

\@textbottom Add some stretch according to David Carlisle's advice at <http://tex.stackexchange.com/a/62318/5522>

```

723 \def\@textbottom{\vskip \z@ \@plus 1pt}
724 \let\@texttop\relax

```

In two-column layouts, force both columns to be the same height by inserting extra internal vertical space to fill out the page.

```

725 \ifcase\ACM@format@nr
726 \relax % manuscript
727 \or % acmsmall
728 \or % acmlarge
729 \or % acmtog
730   \flushbottom
731 \or % sigconf
732   \flushbottom
733 \or % siggraph
734   \flushbottom
735 \or % sigplan
736   \flushbottom
737 \or % sigchi
738   \flushbottom
739 \or % sigchi-a
740 \or % acmengage
741   \flushbottom
742 \or % acmcp
743   \flushbottom
744 \fi

```

3.13 Fonts

Somehow PDFTeX and XeTeX require different incantations to make a PDF compliant with the current Acrobat bugs. Xpdf is much better.

The code below is by Ross Moore.

```
745 \ifPDFTeX
746 \input{glyptounicode}
747 \pdfglyptounicode{f_f}{FB00}
748 \pdfglyptounicode{f_f_i}{FB03}
749 \pdfglyptounicode{f_f_l}{FB04}
750 \pdfglyptounicode{f_i}{FB01}
751 \pdfglyptounicode{t_t}{0074 0074}
752 \pdfglyptounicode{f_t}{0066 0074}
753 \pdfglyptounicode{T_h}{0054 0068}
754 \pdfgentounicode=1
755 \RequirePackage{cmap}
756 \fi
```

\if@ACM@newfonts Whether we load the new fonts

```
757 \newif\if@ACM@newfonts
758 \@ACM@newfontstrue
759 \IfFileExists{libertine.sty}{\ClassWarning{\@classname}{You do not
760   have the libertine package installed. Please upgrade your
761   TeX}\@ACM@newfontsfalse}
762 \IfFileExists{zi4.sty}{\ClassWarning{\@classname}{You do not
763   have the inconsolata (zi4.sty) package installed. Please upgrade your
764   TeX}\@ACM@newfontsfalse}
765 \IfFileExists{newtxmath.sty}{\ClassWarning{\@classname}{You do not
766   have the newtxmath package installed. Please upgrade your
767   TeX}\@ACM@newfontsfalse}
```

We use Libertine throughout. We use Libertine throughout.

```
768 \if@ACM@newfonts
769   \ifxetex
770     \RequirePackage{amssymb}
771     \RequirePackage{unicode-math}
772     \setmathfont[Scale=MatchUppercase]{LibertinusMath-Regular.otf}
773     \setmathfont{latinmodern-math.otf}[range={cal}]
774     \RequirePackage[tt=false]{libertine}
775     \setmonofont[StylisticSet=3]{inconsolata}
776   \else
777     \ifluatex
778       \RequirePackage{amssymb}
779       \RequirePackage{unicode-math}
780       \setmathfont[Scale=MatchUppercase]{LibertinusMath-Regular.otf}
781       \setmathfont{latinmodern-math.otf}[range={cal}]
782       \RequirePackage[tt=false]{libertine}
783       \setmonofont[StylisticSet=3]{inconsolata}
784     \else
785       \RequirePackage[T1]{fontenc}
786       \RequirePackage[tt=false, type1=true]{libertine}
787       \RequirePackage[varqu]{zi4}
788       \RequirePackage[libertine]{newtxmath}
789     \fi
```



```

790 \fi
791 \fi

```

`\liningnums` Libertine defines `\liningnums`, which makes fontspec unhappy. While we do not use fontspec, some users do.

```

792 \let\liningnums\@undefined
793 \AtEndPreamble{%
794   \DeclareTextFontCommand{\liningnums}{\libertineLF}}

```

The SIGCHI extended abstracts are sans serif:

```

795 \if@ACM@sigchiamode
796   \renewcommand{\familydefault}{\sfdefault}
797 \fi

```

3.14 Image descriptions

`\if@Description@present` A local switch: whether there is a description in the given figure

```

798 \newif\if@Description@present
799 \@Description@presenttrue

```

`\if@undescribed@images` A global switch whether some images were *not* described

```

800 \newif\if@undescribed@images
801 \@undescribed@imagesfalse

```

`\Description` The Description macro right now just sets switches

```

802 \newcommand\Description[2][\global\@Description@presenttrue\ignorespaces]

```

The global warning

```

803 \IfDocumentMetadataF{%
804 \AtEndDocument{\if@undescribed@images
805   \ClassWarningNoLine{\@classname}{Some images may lack
806     descriptions.\MessageBreak
807     ACM is committed to complying with the upcoming US ADA
808     https://accessiblyapp.com/accessibility-compliance/ada/ and European
809     Accessibility Act
810     (EAA) https://accessiblyapp.com/accessibility-compliance/eea/
811     regulations by actively working to ensure our publications and
812     application services are accessible to individuals with
813     disabilities, adhering to the WCAG guidelines to provide a seamless
814     experience for all users, and regularly reviewing our accessibility
815     practices to maintain compliance with evolving standards. \MessageBreak
816     To this end, we strongly encourage our authors to provide
817     alternative text and captions for images and multimedia content. It
818     is also important to optimize color contrast for the visually
819     impaired. Taking these important steps when creating your papers
820     will ensure that the widest possible audience can ingest your
821     work.}\fi}
822 }

```

Hooks for figures and tables. We also patch `algorithm` and `algorithm*` in case somebody uses them.

```

823 \AtBeginEnvironment{figure}{\@Description@presentfalse
824   \let\@vspace\@vspace@orig

```

```

825 \let\@vspacer\@vspacer@orig}
826 \AtBeginEnvironment{figure*}{\@Description@presentfalse
827 \let\@vspace\@vspace@orig
828 \let\@vspacer\@vspacer@orig}
829 \AtEndEnvironment{figure*}{\if@Description@present\else
830 \global\@undescribed@imagestrue
831 \ClassWarning{\@classname}{A possible image without description}\fi}
832 \AtEndEnvironment{figure*}{\if@Description@present\else
833 \global\@undescribed@imagestrue
834 \ClassWarning{\@classname}{A possible image without description}\fi}
835 \AtBeginEnvironment{table*}{\let\@vspace\@vspace@orig
836 \let\@vspacer\@vspacer@orig}
837 \AtBeginEnvironment{table*}{\let\@vspace\@vspace@orig
838 \let\@vspacer\@vspacer@orig}
839 \AtBeginEnvironment{algorithm*}{\let\@vspace\@vspace@orig
840 \let\@vspacer\@vspacer@orig}
841 \AtBeginEnvironment{algorithm*}{\let\@vspace\@vspace@orig
842 \let\@vspacer\@vspacer@orig}
843 \AtBeginEnvironment{lstlisting*}{\let\@vspace\@vspace@orig
844 \let\@vspacer\@vspacer@orig}
845 \AtBeginEnvironment{lstlisting*}{\let\@vspace\@vspace@orig
846 \let\@vspacer\@vspacer@orig}
847 \AtBeginEnvironment{minted*}{\let\@vspace\@vspace@orig
848 \let\@vspacer\@vspacer@orig}
849 \AtBeginEnvironment{minted*}{\let\@vspace\@vspace@orig
850 \let\@vspacer\@vspacer@orig}
851 \AtBeginEnvironment{listing*}{\let\@vspace\@vspace@orig
852 \let\@vspacer\@vspacer@orig}
853 \AtBeginEnvironment{listing*}{\let\@vspace\@vspace@orig
854 \let\@vspacer\@vspacer@orig}
855 \AtBeginEnvironment{lstinputlisting*}{\let\@vspace\@vspace@orig
856 \let\@vspacer\@vspacer@orig}
857 \AtBeginEnvironment{lstinputlisting*}{\let\@vspace\@vspace@orig
858 \let\@vspacer\@vspacer@orig}
859

```

3.15 Floats

We use caption package

```

860 \RequirePackage{caption, float}
861 \captionsetup[table]{position=top}
862 \if@ACM@journal
863 \captionsetup{labelfont={sf, small},
864 textfont={sf, small}, margin=\z@}
865 \captionsetup[figure]{name={Fig.}}
866 \else
867 \captionsetup{labelfont={bf},
868 textfont={bf}, labelsep=colon, margin=\z@}
869 \ifcase\ACM@format@nr
870 \relax % manuscript
871 \or % acmsmall
872 \or % acmlarge
873 \or % acmtog
874 \or % sigconf

```

```

875 \or % siggraph
876   \captionsetup{textfont={it}}
877 \or % sigplan
878   \captionsetup{labelfont={bf},
879     textfont={normalfont}, labelsep=period, margin=\z@}
880 \or % sigchi
881   \captionsetup[figure]{labelfont={bf, small},
882     textfont={bf, small}}
883   \captionsetup[table]{labelfont={bf, small},
884     textfont={bf, small}}
885 \or % sigchi-a
886   \captionsetup[figure]{labelfont={bf, small},
887     textfont={bf, small}}
888   \captionsetup[table]{labelfont={bf, small},
889     textfont={bf, small}}
890 \or % acmengage
891 \or % acmcp
892 \fi
893 \fi

```

Unfortunately caption package so far is not tagging compatible. So we need to recreate it if tagging is enabled. This is from <https://github.com/latex3/tagging-project/issues/720>

```

894 \ExplSyntaxOn
895 \IfDocumentMetadataT{%
896   \RemoveFromHook{begindocument}[latex-lab-testphase-float]
897   \socket_new_plug:nnn{tagsupport/parbox/before}{caption}
898   {
899     \tagpdfparaOn %restart para tagging
900     \tl_if_empty:NTF\@current@float@struct
901     {
902       \tag_struct_begin:n{tag=Caption,firstkid}
903     }
904     {
905       \tag_struct_begin:n{tag=Caption,parent=\@current@float@struct,firstkid}
906     }
907   }
908   \socket_new_plug:nnn{tagsupport/parbox/after}{caption}
909   {
910     \tag_struct_end:
911   }
912   \l@addto@macro\caption@beginex@hook{%
913     \tagpdfparaOff %leavevmode in parbox should not start paragraph structure
914     \AssignSocketPlug{tagsupport/parbox/before}{caption}
915     \AssignSocketPlug{tagsupport/parbox/after}{caption}}
916 }
917 \ExplSyntaxOff

```

By default the first row of a table contains headers

```

918 \IfDocumentMetadataT{%
919   \tagpdfsetup{table/header-rows={1}}
920 }

```

sidebar The sidebar environment:

```

921 \newfloat{sidebar}{}{sbar}

```

```

922 \floatname{sidebar}{Sidebar}
923 \renewenvironment{sidebar}{\Collect@Body\@sidebar}{}

\@sidebar The processing of the saved text
924 \long\def\@sidebar#1{\bgroup\let\@vspace\@vspace@orig
925 \let\@vspacer\@vspacer@orig\captionsetup{type=sidebar}%
926 \marginpar{\small#1}\egroup}

marginfigure The marginfigure environment:
927 \newenvironment{marginfigure}{\Collect@Body\@marginfigure}{}

\@marginfigure The processing of the saved text
928 \long\def\@marginfigure#1{\bgroup
929 \let\@vspace\@vspace@orig
930 \let\@vspacer\@vspacer@orig
931 \captionsetup{type=figure}%
932 \marginpar{\@Description@presentfalse\centering
933 \small#1\if@Description@present\else
934 \global\@undescrbed@imagestrue
935 \ClassWarning{\@classname}{A possible image without description}
936 \fi}%
937 \egroup}

margintable The margintable environment:
938 \newenvironment{margintable}{\Collect@Body\@margintable}{}

\@margintable The processing of the saved text
939 \long\def\@margintable#1{\bgroup\let\@vspace\@vspace@orig
940 \let\@vspacer\@vspacer@orig\captionsetup{type=table}%
941 \marginpar{\centering\small#1}\egroup}

SIGCHI extended abstracts provide an interesting possibility to push into the margin.
Here we use figure* and table* for this.

\fulltextwidth We define the width of the boxes as
942 \newdimen\fulltextwidth
943 \fulltextwidth=\dimexpr(\textwidth+\marginparwidth+\marginparsep)

\@dblfloat We redefine the double-float command. First, we make the size bigger. Second, our
default position is going to be tp (to give marginalia a chance)
944 \if@ACM@sigchiamode
945 \def\@dblfloat{\bgroup\let\@vspace\@vspace@orig
946 \let\@vspacer\@vspacer@orig\columnwidth=\fulltextwidth
947 \let\@endfloatbox\@endwidefloatbox
948 \def\@fpsadddefault{\def\@fps{tp}}%
949 \@float}
950 \fi

\end@dblfloat And the end. Just adding a \bgroup.
951 \if@ACM@sigchiamode
952 \def\end@dblfloat{%
953 \end@float\egroup}
954 \fi

```

`\@endwidefloatbox` This is the end of a wide box—we basically move everything to the left

```
955 \def\@endwidefloatbox{%
956   \par\vskip\z@skip
957   \@minipagefalse
958   \outer@nobreak
959   \egroup
960   \color@endbox
961   \global\setbox\@currbox=\vbox{\moveleft
962     \dimexpr(\fulltextwidth-\textwidth)\box\@currbox}%
963   \wd\@currbox=\textwidth
964 }
```

3.16 Lists

```
965 \ifcase\ACM@format@nr
966 \relax % manuscript
967 \or % acmsmall
968 \or % acmlarge
969 \or % acmtog
970 \or % sigconf
971 \or % siggraph
972 \or % sigplan
973 \def\labelenumi{\theenumi.}
974 \def\labelenumii{\theenumii.}
975 \def\labelenumiii{\theenumiii.}
976 \def\labelenumiv{\theenumiv.}
977 \or % sigchi
978 \or % sigchi-a
979 \or % acmengage
980 \or % acmcp
981 \fi
```

The AMS uses generous margins for lists. Note that `amsart` defines `\leftmargin` values for list levels at the beginning of the document, so we must redefine them in the same manner. Also, note that `amsart` redefines the `enumerate` and `itemize` environments with a `\makelabel` command that uses `\llap`, so the `\labelwidth` value is (effectively) irrelevant; nonetheless, we follow `amsart` and set `\labelwidth` to `\leftmargin` minus `\labelsep`.

```
982 \newdimen\@ACM@labelwidth
983 \AtBeginDocument{%
984   \setlength\labelsep{4pt}
985   \setlength{\@ACM@labelwidth}{6.5pt}
986
987   %% First-level list: when beginning after the first line of an
988   %% indented paragraph or ending before an indented paragraph, labels
989   %% should not hang to the left of the preceding/following text.
990   \setlength\leftmargini{\z@}
991   \addtolength\leftmargini{\parindent}
992   \addtolength\leftmargini{2\labelsep}
993   \addtolength\leftmargini{\@ACM@labelwidth}
994
995   %% Second-level and higher lists.
996   \setlength\leftmarginii{\z@}
997   \addtolength\leftmarginii{0.5\labelsep}
```

```

998 \addtolength\leftmarginii{\@ACM@labelwidth}
999 \setlength\leftmarginiii{\leftmarginii}
1000 \setlength\leftmarginiv{\leftmarginiii}
1001 \setlength\leftmarginv{\leftmarginiv}
1002 \setlength\leftmarginvi{\leftmarginv}
1003 \@listi}
1004 \newskip\listisep
1005 \listisep\smallskipamount
1006 \def\@listI{\leftmargin\leftmargini
1007 \labelwidth\leftmargini \advance\labelwidth-\labelsep
1008 \listparindent\z@
1009 \topsep\listisep}
1010 \let\@listi\@listI
1011 \def\@listii{\leftmargin\leftmarginii
1012 \labelwidth\leftmarginii \advance\labelwidth-\labelsep
1013 \topsep\z@skip}
1014 \def\@listiii{\leftmargin\leftmarginiii
1015 \labelwidth\leftmarginiii \advance\labelwidth-\labelsep}
1016 \def\@listiv{\leftmargin\leftmarginiv
1017 \labelwidth\leftmarginiv \advance\labelwidth-\labelsep}
1018 \def\@listv{\leftmargin\leftmarginv
1019 \labelwidth\leftmarginv \advance\labelwidth-\labelsep}
1020 \def\@listvi{\leftmargin\leftmarginvi
1021 \labelwidth\leftmarginvi \advance\labelwidth-\labelsep}

\descriptionlabel We do not use a colon by default like amsart does:
1022 \renewcommand{\descriptionlabel}[1]{\upshape\bfseries #1}

\description Make the description environment indentation consistent with that of the itemize
and enumerate environments.
1023 \renewenvironment{description}{\list{}{%
1024 \labelwidth\@ACM@labelwidth
1025 \let\makelabel\descriptionlabel}%
1026 }{
1027 \endlist
1028 }
1029 \let\enddescription=\endlist % for efficiency

```

3.17 Top-matter data

`\if@ACM@maketitle@typeset` The switch to check whether `\maketitle` is already typeset. It is initially false, and setting top matter when it is true is wrong.

```

1030 \newif\if@ACM@maketitle@typeset
1031 \@ACM@maketitle@typesetfalse

```

We use the `xkeyval` interface to define journal titles and the relevant information

```

1032 \define@choicekey*+{ACM}{acmJournal}[\@journalCode\@journalCode@nr]{%
1033 ACMJCSS,%
1034 ACMJDS,%
1035 AILET,%
1036 CIE,%
1037 CSUR,%
1038 DGOV,%
1039 DLT,%

```

1040 DTRAP,%
1041 FAC,%
1042 GAMES,%
1043 HEALTH,%
1044 IMWUT,%
1045 JACM,%
1046 JATS,%
1047 JDIQ,%
1048 JDS,%
1049 JEA,%
1050 JERIC,%
1051 JETC,%
1052 JOCCH,%
1053 JRC,%
1054 PACMCGIT,%
1055 PACMHCI,%
1056 PACMMOD,%
1057 PACMNET,%
1058 PACMPL,%
1059 PACMSE,%
1060 POMACS,%
1061 TAAS,%
1062 TACCESS,%
1063 TACO,%
1064 TAIS,%
1065 TAISAP,%
1066 TALG,%
1067 TALLIP,%
1068 TAP,%
1069 TCPS,%
1070 TDS,%
1071 TEAC,%
1072 TECS,%
1073 TELO,%
1074 THRI,%
1075 TIIS,%
1076 TIOT,%
1077 TISSEC,%
1078 TIST,%
1079 TKDD,%
1080 TMIS,%
1081 TOCE,%
1082 TOCHI,%
1083 TOCL,%
1084 TOCS,%
1085 TOCT,%
1086 TODAES,%
1087 TODS,%
1088 TOG,%
1089 TOIS,%
1090 TOIT,%
1091 TOMACS,%
1092 TOMM,%
1093 TOMPECS,%

```

1094 TOMS,%
1095 TOPC,%
1096 TOPLAS,%
1097 TOPML,%
1098 TOPS,%
1099 TORS,%
1100 TOS,%
1101 TOSEM,%
1102 TOSN,%
1103 TQC,%
1104 TRETS,%
1105 TSAS,%
1106 TSC,%
1107 TSLP,%
1108 TWEB,%
1109 FACMP%
1110 }{%
1111 \ifcase\@journalCode@nr
1112 \relax % ACMJCSS
1113 \def\@journalName{ACM Journal on Computing and Sustainable Societies}%
1114 \def\@journalNameShort{ACM J. Comput. Sustain. Soc.}%
1115 \def\@permissionCodeOne{2834-5533}%
1116 \def\@permissionCodeTwo{2834-5533}%
1117 \or % ACMJDS
1118 \def\@journalName{ACM Journal of Data Science}%
1119 \def\@journalNameShort{ACM J. Data Sci.}%
1120 \def\@permissionCodeOne{3069-3497}%
1121 \def\@permissionCodeTwo{3069-3497}%
1122 \or % AILET
1123 \def\@journalName{ACM AI Letters}%
1124 \def\@journalNameShort{ACM AI Lett.}%
1125 \def\@permissionCodeOne{3068-8590}%
1126 \def\@permissionCodeTwo{3068-8590}%
1127 \or % CIE
1128 \def\@journalName{ACM Computers in Entertainment}%
1129 \def\@journalNameShort{ACM Comput. Entertain.}%
1130 \def\@permissionCodeOne{1544-3574}%
1131 \or % CSUR
1132 \def\@journalName{ACM Computing Surveys}%
1133 \def\@journalNameShort{ACM Comput. Surv.}%
1134 \def\@permissionCodeOne{0360-0300}%
1135 \def\@permissionCodeTwo{1557-7341}%
1136 \or % DGOV
1137 \def\@journalName{Digital Government: Research and Practice}%
1138 \def\@journalNameShort{Digit. Gov. Res. Pract.}%
1139 \def\@permissionCodeOne{2639-0175}%
1140 \def\@permissionCodeTwo{2639-0175}%
1141 \or % DLT
1142 \def\@journalName{Distributed Ledger Technologies: Research and Practice}%
1143 \def\@journalNameShort{Distrib. Ledger Technol.}%
1144 \def\@permissionCodeOne{2769-6472}%
1145 \def\@permissionCodeTwo{2769-6480}%
1146 \or % DTRAP
1147 \def\@journalName{Digital Threats: Research and Practice}%

```



```

1148 \def\@journalNameShort{Digit. Threat. Res. Pract.}%
1149 \def\@permissionCodeOne{2576-5337}%
1150 \def\@permissionCodeTwo{2576-5337}%
1151 \or % FAC
1152 \def\@journalName{Formal Aspects of Computing}%
1153 \def\@journalNameShort{Form. Asp. Comput.}%
1154 \def\@permissionCodeOne{0934-5043}%
1155 \def\@permissionCodeTwo{1433-299X}%
1156 \or % GAMES
1157 \def\@journalName{ACM Games: Research and Practice}%
1158 \def\@journalNameShort{ACM Games}%
1159 \def\@permissionCodeOne{2832-5516}%
1160 \def\@permissionCodeTwo{2832-5516}%
1161 \or % HEALTH
1162 \def\@journalName{ACM Transactions on Computing for Healthcare}%
1163 \def\@journalNameShort{ACM Trans. Comput. Healthcare}%
1164 \def\@permissionCodeOne{2637-8051}%
1165 \def\@permissionCodeTwo{2637-8051}%
1166 \or % IMWUT
1167 \def\@journalName{Proceedings of the ACM on Interactive, Mobile,
1168     Wearable and Ubiquitous Technologies}%
1169 \def\@journalNameShort{Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.}%
1170 \def\@permissionCodeOne{2474-9567}%
1171 \def\@permissionCodeTwo{2474-9567}%
1172 \@ACM@screentruer
1173 \PackageInfo{\@classname}{Using screen mode due to \@journalCode}%
1174 \or % JACM
1175 \def\@journalName{Journal of the ACM}%
1176 \def\@journalNameShort{J. ACM}%
1177 \def\@permissionCodeOne{0004-5411}%
1178 \def\@permissionCodeTwo{1557-735X}%
1179 \or % JATS
1180 \def\@journalName{Journal on Autonomous Transportation Systems}%
1181 \def\@journalNameShort{ACM J. Auton. Transport. Syst.}%
1182 \def\@permissionCodeOne{2833-0528}%
1183 \def\@permissionCodeTwo{2833-0528}%
1184 \or % JDIQ
1185 \def\@journalName{ACM Journal of Data and Information Quality}%
1186 \def\@journalNameShort{ACM J. Data Inform. Quality}%
1187 \def\@permissionCodeOne{1936-1955}%
1188 \def\@permissionCodeTwo{1936-1963}%
1189 \or % JDS
1190 \def\@journalName{ACM Journal of Data Science}%
1191 \def\@journalNameShort{ACM J. Data Sci.}%
1192 \def\@permissionCodeOne{2831-3194}%
1193 \def\@permissionCodeTwo{2831-3194}%
1194 \or % JEA
1195 \def\@journalName{ACM Journal of Experimental Algorithmics}%
1196 \def\@journalNameShort{ACM J. Exp. Algor.}%
1197 \def\@permissionCodeOne{1084-6654}%
1198 \def\@permissionCodeTwo{1084-6654}%
1199 \or % JERIC
1200 \def\@journalName{ACM Journal of Educational Resources in Computing}%
1201 \def\@journalNameShort{ACM J. Edu. Resources in Comput.}%

```

```

1202 \def\@permissionCodeOne{1073-0516}%
1203 \or % JETC
1204 \def\@journalName{ACM Journal on Emerging Technologies in Computing Systems}%
1205 \def\@journalNameShort{ACM J. Emerg. Technol. Comput. Syst.}%
1206 \def\@permissionCodeOne{1550-4832}%
1207 \def\@permissionCodeTwo{1550-4840}%
1208 \or % JOCCH
1209 \def\@journalName{ACM Journal on Computing and Cultural Heritage}%
1210 \def\@journalNameShort{ACM J. Comput. Cult. Herit.}%
1211 \def\@permissionCodeOne{1556-4673}%
1212 \def\@permissionCodeTwo{1556-4711}%
1213 \or % JRC
1214 \def\@journalName{ACM Journal on Responsible Computing}%
1215 \def\@journalNameShort{ACM J. Responsib. Comput.}%
1216 \def\@permissionCodeOne{2832-0565}%
1217 \def\@permissionCodeTwo{2832-0565}%
1218 \or % PACMCGIT
1219 \def\@journalName{Proceedings of the ACM on Computer Graphics and Interactive Techniques}%
1220 \def\@journalNameShort{Proc. ACM Comput. Graph. Interact. Tech.}%
1221 \def\@permissionCodeOne{2577-6193}%
1222 \def\@permissionCodeTwo{2577-6193}%
1223 \@ACM@screentru
1224 \PackageInfo{\@classname}{Using screen mode due to \@journalCode}%
1225 \or % PACMHCI
1226 \def\@journalName{Proceedings of the ACM on Human-Computer Interaction}%
1227 \def\@journalNameShort{Proc. ACM Hum.-Comput. Interact.}%
1228 \def\@permissionCodeOne{2573-0142}%
1229 \def\@permissionCodeTwo{2573-0142}%
1230 \@ACM@screentru
1231 \PackageInfo{\@classname}{Using screen mode due to \@journalCode}%
1232 \or % PACMMOD
1233 \def\@journalName{Proceedings of the ACM on Management of Data}%
1234 \def\@journalNameShort{Proc. ACM Manag. Data}%
1235 \def\@permissionCodeOne{2836-6573}%
1236 \def\@permissionCodeTwo{2836-6573}%
1237 \or % PACMNET
1238 \def\@journalName{Proceedings of the ACM on Networkng}%
1239 \def\@journalNameShort{Proc. ACM Netw.}%
1240 \def\@permissionCodeOne{2834-5509}%
1241 \def\@permissionCodeTwo{2834-5509}%
1242 \or % PACMPL
1243 \def\@journalName{Proceedings of the ACM on Programming Languages}%
1244 \def\@journalNameShort{Proc. ACM Program. Lang.}%
1245 \def\@permissionCodeOne{2475-1421}%
1246 \def\@permissionCodeTwo{2475-1421}%
1247 \@ACM@screentru
1248 \PackageInfo{\@classname}{Using screen mode due to \@journalCode}%
1249 \or % PACMSE
1250 \def\@journalName{Proceedings of the ACM on Software Engineering}%
1251 \def\@journalNameShort{Proc. ACM Softw. Eng.}%
1252 \def\@permissionCodeOne{2994-970X}%
1253 \def\@permissionCodeTwo{2994-970X}%
1254 \@ACM@screentru
1255 \PackageInfo{\@classname}{Using screen mode due to \@journalCode}%

```

```

1256 \or % POMACS
1257 \def\@journalName{Proceedings of the ACM on Measurement and Analysis of Computing Systems}%
1258 \def\@journalNameShort{Proc. ACM Meas. Anal. Comput. Syst.}%
1259 \def\@permissionCodeOne{2476-1249}%
1260 \def\@permissionCodeTwo{2476-1249}%
1261 \@ACM@screentru
1262 \PackageInfo{\@classname}{Using screen mode due to \@journalCode}%
1263 \or % TAAS
1264 \def\@journalName{ACM Transactions on Autonomous and Adaptive Systems}%
1265 \def\@journalNameShort{ACM Trans. Autonom. Adapt. Syst.}%
1266 \def\@permissionCodeOne{1556-4665}%
1267 \def\@permissionCodeTwo{1556-4703}%
1268 \or % TACCESS
1269 \def\@journalName{ACM Transactions on Accessible Computing}%
1270 \def\@journalNameShort{ACM Trans. Access. Comput.}%
1271 \def\@permissionCodeOne{1936-7228}%
1272 \def\@permissionCodeTwo{1936-7236}%
1273 \or % TACO
1274 \def\@journalName{ACM Transactions on Architecture and Code Optimization}%
1275 \def\@journalNameShort{ACM Trans. Arch. Code Optim.}%
1276 \def\@permissionCodeOne{1544-3566}%
1277 \def\@permissionCodeTwo{1544-3973}%
1278 \or % TAIS
1279 \def\@journalName{ACM Transactions on AI for Science}%
1280 \def\@journalNameShort{ACM Trans. AI Sci.}%
1281 \def\@permissionCodeOne{3066-4438}%
1282 \def\@permissionCodeTwo{3066-4438}%
1283 \or % TAISAP
1284 \def\@journalName{ACM Transactions on AI Security and Privacy}%
1285 \def\@journalNameShort{ACM Trans. AI Secur. Priv.}%
1286 \def\@permissionCodeOne{3068-3564}%
1287 \def\@permissionCodeTwo{3068-3564}%
1288 \or % TALG
1289 \def\@journalName{ACM Transactions on Algorithms}%
1290 \def\@journalNameShort{ACM Trans. Algor.}%
1291 \def\@permissionCodeOne{1549-6325}%
1292 \def\@permissionCodeTwo{1549-6333}%
1293 \or % TALLIP
1294 \def\@journalName{ACM Transactions on Asian and Low-Resource Language Information Processing}%
1295 \def\@journalNameShort{ACM Trans. Asian Low-Resour. Lang. Inf. Process.}%
1296 \def\@permissionCodeOne{2375-4699}%
1297 \def\@permissionCodeTwo{2375-4702}%
1298 \or % TAP
1299 \def\@journalName{ACM Transactions on Applied Perception}%
1300 \def\@journalNameShort{ACM Trans. Appl. Percept.}%
1301 \def\@permissionCodeOne{1544-3558}%
1302 \def\@permissionCodeTwo{1544-3965}%
1303 \or % TCPS
1304 \def\@journalName{ACM Transactions on Cyber-Physical Systems}%
1305 \def\@journalNameShort{ACM Trans. Cyber-Phys. Syst.}%
1306 \def\@permissionCodeOne{2378-962X}%
1307 \def\@permissionCodeTwo{2378-9638}%
1308 \or % TDS
1309 \def\@journalName{ACM Transactions on Data Science}%

```

```

1310 \def\@journalNameShort{ACM Trans. Data Sci.}%
1311 \def\@permissionCodeOne{2577-3224}%
1312 \or % TEAC
1313 \def\@journalName{ACM Transactions on Economics and Computation}%
1314 \def\@journalNameShort{ACM Trans. Econ. Comput.}%
1315 \def\@permissionCodeOne{2167-8375}%
1316 \def\@permissionCodeTwo{2167-8383}%
1317 \or % TECS
1318 \def\@journalName{ACM Transactions on Embedded Computing Systems}%
1319 \def\@journalNameShort{ACM Trans. Embedd. Comput. Syst.}%
1320 \def\@permissionCodeOne{1539-9087}%
1321 \def\@permissionCodeTwo{1558-3465}%
1322 \or % TELO
1323 \def\@journalName{ACM Transactions on Evolutionary Learning and Optimization}%
1324 \def\@journalNameShort{ACM Trans. Evol. Learn. Optim.}%
1325 \def\@permissionCodeOne{2688-299X}%
1326 \def\@permissionCodeTwo{2688-3007}%
1327 \or % THRI
1328 \def\@journalName{ACM Transactions on Human-Robot Interaction}%
1329 \def\@journalNameShort{ACM Trans. Hum.-Robot Interact.}%
1330 \def\@permissionCodeOne{2573-9522}%
1331 \def\@permissionCodeTwo{2573-9522}%
1332 \or % TIIS
1333 \def\@journalName{ACM Transactions on Interactive Intelligent Systems}%
1334 \def\@journalNameShort{ACM Trans. Interact. Intell. Syst.}%
1335 \def\@permissionCodeOne{2160-6455}%
1336 \def\@permissionCodeTwo{2160-6463}%
1337 \or % TIOT
1338 \def\@journalName{ACM Transactions on Internet of Things}%
1339 \def\@journalNameShort{ACM Trans. Internet Things}%
1340 \def\@permissionCodeOne{2577-6207}%
1341 \def\@permissionCodeTwo{2577-6207}%
1342 \or % TISSEC
1343 \def\@journalName{ACM Transactions on Information and System Security}%
1344 \def\@journalNameShort{ACM Trans. Info. Syst. Sec.}%
1345 \def\@permissionCodeOne{1094-9224}%
1346 \or % TIST
1347 \def\@journalName{ACM Transactions on Intelligent Systems and Technology}%
1348 \def\@journalNameShort{ACM Trans. Intell. Syst. Technol.}%
1349 \def\@permissionCodeOne{2157-6904}%
1350 \def\@permissionCodeTwo{2157-6912}%
1351 \or % TKDD
1352 \def\@journalName{ACM Transactions on Knowledge Discovery from Data}%
1353 \def\@journalNameShort{ACM Trans. Knowl. Discov. Data.}%
1354 \def\@permissionCodeOne{1556-4681}%
1355 \def\@permissionCodeTwo{1556-472X}%
1356 \or % TMIS
1357 \def\@journalName{ACM Transactions on Management Information Systems}%
1358 \def\@journalNameShort{ACM Trans. Manag. Inform. Syst.}%
1359 \def\@permissionCodeOne{2158-656X}%
1360 \def\@permissionCodeTwo{2158-6578}%
1361 \or % TOCE
1362 \def\@journalName{ACM Transactions on Computing Education}%
1363 \def\@journalNameShort{ACM Trans. Comput. Educ.}%

```

```

1364 \def\@permissionCodeOne{1946-6226}%
1365 \def\@permissionCodeTwo{1946-6226}%
1366 \or % TOCHI
1367 \def\@journalName{ACM Transactions on Computer-Human Interaction}%
1368 \def\@journalNameShort{ACM Trans. Comput.-Hum. Interact.}%
1369 \def\@permissionCodeOne{1073-0516}%
1370 \def\@permissionCodeTwo{1557-7325}%
1371 \or % TOCL
1372 \def\@journalName{ACM Transactions on Computational Logic}%
1373 \def\@journalNameShort{ACM Trans. Comput. Logic}%
1374 \def\@permissionCodeOne{1529-3785}%
1375 \def\@permissionCodeTwo{1557-945X}%
1376 \or % TOCS
1377 \def\@journalName{ACM Transactions on Computer Systems}%
1378 \def\@journalNameShort{ACM Trans. Comput. Syst.}%
1379 \def\@permissionCodeOne{0734-2071}%
1380 \def\@permissionCodeTwo{1557-7333}%
1381 \or % TOCT
1382 \def\@journalName{ACM Transactions on Computation Theory}%
1383 \def\@journalNameShort{ACM Trans. Comput. Theory}%
1384 \def\@permissionCodeOne{1942-3454}%
1385 \def\@permissionCodeTwo{1942-3462}%
1386 \or % TODAES
1387 \def\@journalName{ACM Transactions on Design Automation of Electronic Systems}%
1388 \def\@journalNameShort{ACM Trans. Des. Autom. Electron. Syst.}%
1389 \def\@permissionCodeOne{1084-4309}%
1390 \def\@permissionCodeTwo{1557-7309}%
1391 \or % TODS
1392 \def\@journalName{ACM Transactions on Database Systems}%
1393 \def\@journalNameShort{ACM Trans. Datab. Syst.}%
1394 \def\@permissionCodeOne{0362-5915}%
1395 \def\@permissionCodeTwo{1557-4644}%
1396 \or % TOG
1397 \def\@journalName{ACM Transactions on Graphics}%
1398 \def\@journalNameShort{ACM Trans. Graph.}%
1399 \def\@permissionCodeOne{0730-0301}%
1400 \def\@permissionCodeTwo{1557-7368}%
1401 \or % TOIS
1402 \def\@journalName{ACM Transactions on Information Systems}%
1403 \def\@journalNameShort{ACM Trans. Inf. Syst.}%
1404 \def\@permissionCodeOne{1046-8188}%
1405 \def\@permissionCodeTwo{1558-2868}%
1406 \or % TOIT
1407 \def\@journalName{ACM Transactions on Internet Technology}%
1408 \def\@journalNameShort{ACM Trans. Internet Technol.}%
1409 \def\@permissionCodeOne{1533-5399}%
1410 \def\@permissionCodeTwo{1557-6051}%
1411 \or % TOMACS
1412 \def\@journalName{ACM Transactions on Modeling and Computer Simulation}%
1413 \def\@journalNameShort{ACM Trans. Model. Comput. Simul.}%
1414 \def\@permissionCodeOne{1049-3301}%
1415 \def\@permissionCodeTwo{1558-1195}%
1416 \or % TOMM
1417 \def\@journalName{ACM Transactions on Multimedia Computing, Communications and Applications}%

```

```

1418 \def\@journalNameShort{ACM Trans. Multimedia Comput. Commun. Appl.}%
1419 \def\@permissionCodeOne{1551-6857}%
1420 \def\@permissionCodeTwo{1551-6865}%
1421 \or % TOMPECS
1422 \def\@journalName{ACM Transactions on Modeling and Performance Evaluation of Computing Systems}%
1423 \def\@journalNameShort{ACM Trans. Model. Perform. Eval. Comput. Syst.}%
1424 \def\@permissionCodeOne{2376-3639}%
1425 \def\@permissionCodeTwo{2376-3647}%
1426 \or % TOMS
1427 \def\@journalName{ACM Transactions on Mathematical Software}%
1428 \def\@journalNameShort{ACM Trans. Math. Softw.}%
1429 \def\@permissionCodeOne{0098-3500}%
1430 \def\@permissionCodeTwo{1557-7295}%
1431 \or % TOPC
1432 \def\@journalName{ACM Transactions on Parallel Computing}%
1433 \def\@journalNameShort{ACM Trans. Parallel Comput.}%
1434 \def\@permissionCodeOne{2329-4949}%
1435 \def\@permissionCodeTwo{2329-4957}%
1436 \or % TOPLAS
1437 \def\@journalName{ACM Transactions on Programming Languages and Systems}%
1438 \def\@journalNameShort{ACM Trans. Program. Lang. Syst.}%
1439 \def\@permissionCodeOne{0164-0925}%
1440 \def\@permissionCodeTwo{1558-4593}%
1441 \or % TOPML
1442 \def\@journalName{ACM Transactions on Probabilistic Machine Learning}%
1443 \def\@journalNameShort{ACM Trans. Probab. Mach. Learn.}%
1444 \def\@permissionCodeOne{2836-8924}%
1445 \def\@permissionCodeTwo{2836-8924}%
1446 \or % TOPS
1447 \def\@journalName{ACM Transactions on Privacy and Security}%
1448 \def\@journalNameShort{ACM Trans. Priv. Sec.}%
1449 \def\@permissionCodeOne{2471-2566}%
1450 \def\@permissionCodeTwo{2471-2574}%
1451 \or % TORS
1452 \def\@journalName{ACM Transactions on Recommender Systems}%
1453 \def\@journalNameShort{ACM Trans. Recomm. Syst.}%
1454 \def\@permissionCodeOne{2770-6699}%
1455 \def\@permissionCodeTwo{2770-6699}%
1456 \or % TOS
1457 \def\@journalName{ACM Transactions on Storage}%
1458 \def\@journalNameShort{ACM Trans. Storage}%
1459 \def\@permissionCodeOne{1553-3077}%
1460 \def\@permissionCodeTwo{1553-3093}%
1461 \or % TOSEM
1462 \def\@journalName{ACM Transactions on Software Engineering and Methodology}%
1463 \def\@journalNameShort{ACM Trans. Softw. Eng. Methodol.}%
1464 \def\@permissionCodeOne{1049-331X}%
1465 \def\@permissionCodeTwo{1557-7392}%
1466 \or % TOSN
1467 \def\@journalName{ACM Transactions on Sensor Networks}%
1468 \def\@journalNameShort{ACM Trans. Sensor Netw.}%
1469 \def\@permissionCodeOne{1550-4859}%
1470 \def\@permissionCodeTwo{1550-4867}%
1471 \or % TQC

```

```

1472 \def\@journalName{ACM Transactions on Quantum Computing}%
1473 \def\@journalNameShort{ACM Trans. Quantum Comput.}%
1474 \def\@permissionCodeOne{2643-6817}%
1475 \def\@permissionCodeTwo{2643-6817}%
1476 \or % TRET
1477 \def\@journalName{ACM Transactions on Reconfigurable Technology and Systems}%
1478 \def\@journalNameShort{ACM Trans. Reconfig. Technol. Syst.}%
1479 \def\@permissionCodeOne{1936-7406}%
1480 \def\@permissionCodeTwo{1936-7414}%
1481 \or % TSAS
1482 \def\@journalName{ACM Transactions on Spatial Algorithms and Systems}%
1483 \def\@journalNameShort{ACM Trans. Spatial Algorithms Syst.}%
1484 \def\@permissionCodeOne{2374-0353}%
1485 \def\@permissionCodeTwo{2374-0361}%
1486 \or % TSC
1487 \def\@journalName{ACM Transactions on Social Computing}%
1488 \def\@journalNameShort{ACM Trans. Soc. Comput.}%
1489 \def\@permissionCodeOne{2469-7818}%
1490 \def\@permissionCodeTwo{2469-7826}%
1491 \or % TSLP
1492 \def\@journalName{ACM Transactions on Speech and Language Processing}%
1493 \def\@journalNameShort{ACM Trans. Speech Lang. Process.}%
1494 \def\@permissionCodeOne{1550-4875}%
1495 \def\@permissionCodeTwo{2329-9304}%
1496 \or % TWEB
1497 \def\@journalName{ACM Transactions on the Web}%
1498 \def\@journalNameShort{ACM Trans. Web}%
1499 \def\@permissionCodeOne{1559-1131}%
1500 \def\@permissionCodeTwo{1559-114X}%
1501 \else % FACMP, a dummy journal
1502 \def\@journalName{ACM Just Accepted}%
1503 \def\@journalNameShort{ACM Accepted}%
1504 \def\@permissionCodeOne{XXXX-XXXX}%
1505 \fi
1506 \ClassInfo{\@classname}{Using journal code \@journalCode}%
1507 }{%
1508 \ClassError{\@classname}{Incorrect journal #1}%
1509 }%

```

\acmJournal And the syntactic sugar around it

```

1510 \def\acmJournal#1{\setkeys{ACM}{acmJournal=#1}%
1511 \global\@ACM@journal@bibstriptrue
1512 \global\@ACM@journal@bibstrip@or@togtrue}

```

The defaults:

```

1513 \def\@journalCode@nr{0}
1514 \def\@journalName{}%
1515 \def\@journalNameShort{\@journalName}%
1516 \def\@permissionCodeOne{XXXX-XXXX}%
1517 \def\@permissionCodeTwo{\@permissionCodeOne}%

```

\acmConference This is the conference command. The acmsmall format is used both for journals and for conference proceedings published as books. In the latter case the paper has no journal name, volume, issue or article number, so we switch it to the headers and footers of the

proceedings formats, and suppress the folios like these formats do: the page numbers of such a paper are assigned by the proceedings rather than by the article. An author who wants the folios can say `\settopmatter{printfolios=true}` after `\acmConference`. The `acmtog` format keeps its own conference headers and footers: there a conference paper is published as a journal issue and does have an article number.

```

1518 \newcommand\acmConference[4][\%
1519   \gdef\acmConference@shortname{#1}%
1520   \gdef\acmConference@name{#2}%
1521   \gdef\acmConference@date{#3}%
1522   \gdef\acmConference@venue{#4}%
1523   \ifx\acmConference@shortname\@empty
1524     \gdef\acmConference@shortname{#2}%
1525   \fi
1526   \global\@ACM@journal@bibstripfalse
1527   \ifnum\ACM@format@nr=1\relax % acmsmall
1528     \global\@ACM@journal@bibstrip@or@togfalse
1529   \settopmatter{printfolios=false}%
1530   \fi
1531   \ifx\@acmBooktitle\@empty\relax
1532     \acmBooktitle{Proceedings of \acmConference@name
1533       \ifx\acmConference@name\acmConference@shortname\else
1534         \ (\acmConference@shortname)\fi}%
1535   \fi
1536 }
1537 \if@ACM@journal\else
1538 \acmConference[Conference'17]{ACM Conference}{July 2017}{Washington,
1539   DC, USA}%
1540 \fi

```

`\acmBooktitle` The book title of the conference:

```

\@acmBooktitle 1541 \def\acmBooktitle#1{\gdef\@acmBooktitle{#1}}
1542 \acmBooktitle{}

```

`\@editorsAbbrev` How to abbreviate editors

```

1543 \def\@editorsAbbrev{(Ed.)}

```

`\@acmEditors` The list of editors

```

1544 \def\@acmEditors{}

```

`\editor` Add a new editor to the list

```

1545 \def\editor#1{\ifx\@acmEditors\@empty
1546   \gdef\@acmEditors{#1}%
1547 \else
1548   \gdef\@editorsAbbrev{(Eds.)}%
1549   \g@addto@macro\@acmEditors{\and#1}%
1550 \fi}

```

`\subtitle` The subtitle macro

```

1551 \def\subtitle#1{\def\@subtitle{#1}}
1552 \subtitle{}

```

`\num@authorgroups` The total number of “groups”. Each group is several authors with the same affiliations(s)

```

1553 \newcount\num@authorgroups
1554 \num@authorgroups=0\relax

```


`\num@authors` The total number of authors

```
1555 \newcount\num@authors
```

```
1556 \num@authors=0\relax
```

`\if@insideauthorgroup` Whether we are continuing an author group

```
1557 \newif\if@insideauthorgroup
```

```
1558 \@insideauthorgroupfalse
```

`\author` Adding an author to the list of authors and addresses.

Note that we want to typeset the link to the author's orcid if known. The problem is, we know the orcid only after the author is entered. The trick is based on the idea that `\csname...\endcsname` is `\relax` unless defined. Therefore we typeset authors by the special macro `\csname typeset@author\the\num@authors\endcsname`, which is defined by `\orcid` command.

```
1559 \renewcommand\author[2][{}%
```

```
1560 \IfSubStr{\detokenize{#2}}{,}{\ClassWarning{\@classname}{Do not put several  
1561 authors in the same \string\author\space macro!}}}%
```

```
1562 \global\advance\num@authors by 1\relax
```

```
1563 \if@insideauthorgroup\else
```

```
1564 \global\advance\num@authorgroups by 1\relax
```

```
1565 \global\@insideauthorgrouptrue
```

```
1566 \fi
```

```
1567 \ifx\addresses\@empty
```

```
1568 \if@ACM@anonymous
```

```
1569 \gdef\addresses{\@author{Anonymous Author(s)}%
```

```
1570 \ifx\@acmSubmissionID\@empty\else\Submission Id:
```

```
1571 \@acmSubmissionID\fi}}%
```

```
1572 \gdef\authors{Anonymous Author(s)}%
```

```
1573 \else
```

```
1574 \expandafter\gdef\expandafter\addresses\expandafter{%
```

```
1575 \expandafter\@author\expandafter{%
```

```
1576 \csname typeset@author\the\num@authors\endcsname{#2}}}%
```

```
1577 \gdef\authors{#2}%
```

```
1578 \fi
```

```
1579 \else
```

```
1580 \if@ACM@anonymous\else
```

```
1581 \expandafter\g@addto@macro\expandafter\addresses\expandafter{%
```

```
1582 \expandafter\and\expandafter\@author\expandafter{%
```

```
1583 \csname typeset@author\the\num@authors\endcsname{#2}}}%
```

```
1584 \g@addto@macro\authors{\and#2}%
```

```
1585 \fi
```

```
1586 \fi
```

```
1587 \if@ACM@anonymous
```

```
1588 \ifx\shortauthors\@empty
```

```
1589 \gdef\shortauthors{Anon.
```

```
1590 \ifx\@acmSubmissionID\@empty\else Submission Id:
```

```
1591 \@acmSubmissionID\fi}}%
```

```
1592 \fi
```

```
1593 \else
```

```
1594 \def\@tempa{#1}%
```

```
1595 \ifx\@tempa\@empty
```

```
1596 \ifx\shortauthors\@empty
```

```
1597 \gdef\shortauthors{#2}%
```

```

1598     \else
1599         \g@addto@macro\shortauthors{\and#2}%
1600     \fi
1601 \else
1602     \ifx\shortauthors\@empty
1603         \gdef\shortauthors{#1}%
1604     \else
1605         \g@addto@macro\shortauthors{\and#1}%
1606     \fi
1607 \fi
1608 \fi}

```

`\affiliation` The macro `\affiliation` mimics `\address` from `amsart`. Note that it has an optional argument, which we use differently from `amsart`.

```

1609 \newcommand{\affiliation}[2][{}]{%
1610     \global\@insideauthorgroupfalse
1611     \if@ACM@anonymous\else
1612         \g@addto@macro\addresses{\affiliation{#1}{#2}}%
1613     \fi}

```

`@ACM@affiliation@obeypunctuation` Whether to use the author's punctuation (false by default, which adds American-style address punctuation)

```

1614 \define@boolkey+{@ACM@affiliation@}[@ACM@affiliation@]{obeypunctuation}%
1615 [true]{}{\ClassError{\@classname}{The option obeypunctuation can be either true or false}}

```

`\additionalaffiliation` Additional affiliations go to footnotes

```

1616 \def\additionalaffiliation#1{\authornote{\@additionalaffiliation{#1}}}

```

`@\additionalaffiliation` Process `\additionalaffiliation` inside `\authornote`

```

1617 \def\@additionalaffiliation#1{\bgroup
1618     \def\position##1{\ignorespaces}%
1619     \def\institution##1{##1\ignorespaces}%
1620     \def\department{\@ifnextchar[{\@department}{\@department[]}}%
1621     \def\@department[##1]##2{, ##2 \ignorespaces}%
1622     \let\city\position
1623     \let\state\position
1624     \let\country\position
1625     Also with #1\unskip.\egroup}

```

`\email` The macro `\email` mimics `\email` from `amsart`. Again, it has an optional argument that we do not currently need but keep for possible future use.

```

1626 \renewcommand{\email}[2][{}]{%
1627     \IfSubStr{#2}{,}{\ClassWarning{\@classname}{Do not put several
1628         addresses in the same \string\email\space macro!}}{}%
1629     \if@ACM@anonymous\else
1630         \g@addto@macro\addresses{\email{#1}{#2}}%
1631     \fi}

```

`\orcid` We redefine the command to typeset the current author

```

1632 \def\orcid#1{\unskip\ignorespaces%
1633     \protected\def\orcidsite{https://orcid.org/%
1634     \IfBeginWith{#1}{http}{%
1635         \expandafter\gdef\cename

```

```

1636         typeset@author\the\num@authors\endcsname##1{%
1637             \href{#1}{##1}}}%
1638     \expandafter\gdef\csname
1639         typeset@author\the\num@authors\endcsname##1{%
1640             \href{\orcid site#1}{##1}}}%

```

`\authorsaddresses` Setting up authors' addresses

```

1641 \def\authorsaddresses#1{\def\@authorsaddresses{#1}}
1642 \authorsaddresses{\mkauthorsaddresses}

```

`\mktranslatedkeywords` We have two argument: the language to typeset keywords and the keywords.

```

1643 \newcommand\mktranslatedkeywords[2]{\bgroup
1644     \selectlanguage{#1}%
1645     {\@specialsection{\keywordsname}%
1646         \noindent#2\par}\egroup}

```

`\@titlenotes` The title notes. We put here the note for the corresponding author, so it is always printed first.

```

1647 \def\@titlenotes{%
1648     \if@ACM@corresponding@present
1649         \footnotetext[1]{Corresponding author}%
1650     \fi
1651 }

```

`\titlenote` Adding a note to the title

```

1652 \def\titlenote#1{%
1653     \g@addto@macro\@title{\footnotemark}%
1654     \if@ACM@anonymous
1655         \g@addto@macro\@titlenotes{%
1656             \stepcounter{footnote}\footnotetext{Title note}}%
1657     \else
1658         \g@addto@macro\@titlenotes{\stepcounter{footnote}\footnotetext{#1}}%
1659     \fi}

```

`\@subtitlenotes` The subtitle notes

```

1660 \def\@subtitlenotes{}

```

`\subtitlenote` Adding a note to the subtitle

```

1661 \def\subtitlenote#1{%
1662     \g@addto@macro\@subtitle{\footnotemark}%
1663     \if@ACM@anonymous
1664         \g@addto@macro\@subtitlenotes{%
1665             \stepcounter{footnote}\footnotetext{Subtitle note}}%
1666     \else
1667         \g@addto@macro\@subtitlenotes{%
1668             \stepcounter{footnote}\footnotetext{#1}}%
1669     \fi}

```

`\@authornotes` The author notes.

```

1670 \def\@authornotes{}

```

`\authornote` Adding a note to the author

```
1671 \def\authornote#1{%
1672   \if@ACM@anonymous\else
1673     \g@addto@macro\addresses{\@authornotemark}%
1674     \g@addto@macro\authornotes{%
1675       \stepcounter{footnote}\footnotetext{#1}}%
1676   \fi}
```

`\if@ACM@corresponding@present` A switch to control whether there is a corresponding author already defined.

```
1677 \newif\if@ACM@corresponding@present
1678 \@ACM@corresponding@presentfalse
```

`\correspondingauthor` Mark the author as corresponding author

```
1679 \newcommand\correspondingauthor{%
1680   \if@ACM@anonymous\else
1681     \if@ACM@corresponding@present
1682       \ClassError{\@classname}{You can have no more than one
1683         corresponding author}
1684     \fi
1685     \@ACM@corresponding@presenttrue
1686     \g@addto@macro\addresses{\@correspondingauthormark}%
1687
1688   \fi}
```

`\@correspondingauthormark` Print the envelope

```
1689 \def\@correspondingauthormark{\g@addto@macro\@currentauthors{%
1690   \advance\hfuzz by 5pt\relax\textsuperscript{*}\relax}}
```

`\authornotemark` Adding a footnote mark to the authors

```
1691 \newcommand\authornotemark[1][\relax]{%
1692   \if@ACM@anonymous\else
1693     \ifx#1\relax\relax\relax
1694       \g@addto@macro\addresses{\@authornotemark}%
1695     \else
1696       \g@addto@macro\addresses{\@authornotemark{#1}}%
1697     \fi
1698   \fi}
```

`\acmVolume` The current volume

```
1699 \def\acmVolume#1{\def\@acmVolume{#1}}
1700 \acmVolume{1}
```

`\acmNumber` The current number

```
1701 \def\acmNumber#1{\def\@acmNumber{#1}}
1702 \acmNumber{1}
```

`\acmArticle` The current article

```
1703 \def\acmArticle#1{\def\@acmArticle{#1}}
1704 \acmArticle{}
```

`\acmArticleSeq` The sequence number

```
1705 \def\acmArticleSeq#1{\def\@acmArticleSeq{#1}}
1706 \acmArticleSeq{\@acmArticle}
```

`\acmYear` The current year
1707 `\def\acmYear#1{\def\@acmYear{#1}}`
1708 `\acmYear{\the\year}`

`\acmMonth` The current month
1709 `\def\acmMonth#1{\def\@acmMonth{#1}}`
1710 `\acmMonth{\the\month}`

`\@acmPubDate` The publication date
1711 `\def\@acmPubDate{\ifcase\@acmMonth\or`
1712 `January\or February\or March\or April\or May\or June\or`
1713 `July\or August\or September\or October\or November\or`
1714 `December\fi~\@acmYear}`

`\acmPrice` The price—obsolete
1715 `\def\acmPrice#1{\ClassWarning{\@classname}{The macro \string\acmPrice`
1716 `is obsolete. ACM no longer prints the price in bibstrip.}}`

`\acmSubmissionID` The submission ID
1717 `\def\acmSubmissionID#1{\def\@acmSubmissionID{#1}}`
1718 `\acmSubmissionID{}`

`\acmISBN` The book ISBN
1719 `\def\acmISBN#1{\def\@acmISBN{#1}}`
1720 `\acmISBN{978-x-xxxx-xxxx-x/YYYY/MM}`

`\acmDOI` The paper DOI
1721 `\def\acmDOI#1{\def\@acmDOI{#1}}`
1722 `\acmDOI{10.1145/nnnnnnn.nnnnnnn}`

`\if@ACM@badge`

`\@ACM@badge@width` The width of the badge
1723 `\newlength\@ACM@badge@width`
1724 `\setlength\@ACM@badge@width{3pc}`

`\@ACM@title@width` The width of the title
1725 `\newlength\@ACM@title@width`

`\@ACM@badge@skip` The spacing between badges
1726 `\newlength\@ACM@badge@skip`
1727 `\setlength\@ACM@badge@skip{1pt}`

`\@acmBadgeR`
1728 `\def\@acmBadgeR{}`

`\@acmBadgeL`
1729 `\def\@acmBadgeL{}`

```

\acmBadgeR Setting the right badge
1730 \newcommand\acmBadgeR[2][{}]{%
1731   \ifx\@acmBadgeR\@empty
1732     \gdef\@acmBadgeR{%
1733       \smash{%
1734         \raisebox{0.5\height}{%
1735           \href{#1}{\includegraphics[width=\@ACM@badge@width]{#2}}}}}%
1736   \else
1737     \g@addto@macro{\@acmBadgeR}{%
1738       \hspace{\@ACM@badge@skip}%
1739       \smash{%
1740         \raisebox{0.5\height}{%
1741           \href{#1}{\includegraphics[width=\@ACM@badge@width]{#2}}}}}%
1742   \fi}

\acmBadgeL Setting the left badge
1743 \newcommand\acmBadgeL[2][{}]{%
1744   \ifx\@acmBadgeL\@empty
1745     \gdef\@acmBadgeL{%
1746       \smash{%
1747         \raisebox{0.5\height}{%
1748           \href{#1}{\includegraphics[width=\@ACM@badge@width]{#2}}}}}%
1749   \else
1750     \g@addto@macro{\@acmBadgeL}{%
1751       \hspace{\@ACM@badge@skip}%
1752       \smash{%
1753         \raisebox{0.5\height}{%
1754           \href{#1}{\includegraphics[width=\@ACM@badge@width]{#2}}}}}%
1755   \fi}

\acmBadge Just a syntax sugar for \acmBadgeR
1756 \let\acmBadge=\acmBadgeR

\startPage The start page of the paper
1757 \def\startPage#1{\def\@startPage{#1}}
1758 \startPage{}

\terms Terms are obsolete. We use CCS now.
1759 \def\terms#1{\ClassWarning{\@classname}{The command \string\terms{} is
1760   obsolete. I am going to ignore it}}

\keywords Keywords are mostly obsolete. We use CCS now. Still it makes sense to keep them for
compatibility.
1761 \def\keywords#1{\def\@keywords{#1}}
1762 \let\@keywords\@empty

1763 \AtEndDocument{\if@ACM@nonacm\else\ifx\@keywords\@empty
1764   \ifnum\getrefnumber{TotPages}>2\relax
1765   \ClassWarningNoLine{\@classname}{ACM keywords are mandatory
1766     for papers over two pages}%
1767   \fi\fi\fi}

abstract The amsart package puts abstract in a box. Since we do not know whether we will
use two-column mode, we prefer to save the text
1768 \renewenvironment{abstract}{\Collect@Body\@saveabstract}{\fi}

```

\@saveabstract And saving the abstract

```
1769 \long\def\@saveabstract#1{\if@ACM@maketitle@typeset
1770   \ClassError{\@classname}{Abstract must be defined before maketitle
1771     command. Please move it!}\fi
1772   \long\gdef\@abstract{#1}}
1773 \@saveabstract{}
```

\@empty The long version of \@empty (to compare with \@abstract)

```
1774 \long\def\@empty{}
```

\if@ACM@printccs Whether to print CCS

```
1775 \define@boolkey+{@ACM@topmatter@}[@ACM@]{printccs}[true]{%
1776   \if@ACM@printccs
1777     \ClassInfo{\@classname}{Printing CCS}%
1778   \else
1779     \ClassInfo{\@classname}{Suppressing CCS}%
1780   \fi}{\ClassError{\@classname}{The option printccs can be either true or false}}
```

\if@ACM@printacmref Whether to print the ACM bibstrip

```
1781 \define@boolkey+{@ACM@topmatter@}[@ACM@]{printacmref}[true]{%
1782   \if@ACM@printacmref
1783     \ClassInfo{\@classname}{Printing bibformat}%
1784   \else
1785     \ClassInfo{\@classname}{Suppressing bibformat}%
1786   \fi}{\ClassError{\@classname}{The option printacmref can be either true or false}}

1787 \AtEndDocument{\if@ACM@nonacm\else\if@ACM@printacmref\else
1788   \ifnum\getrefnumber{TotPages}>1\relax
1789   \ClassWarningNoLine{\@classname}{%
1790     ACM reference format is mandatory \MessageBreak
1791     for papers over one page. \MessageBreak
1792     Please add printacmref=true to the \MessageBreak
1793     \string\settopmatter\space command.}%
1794   \fi\fi\fi}
```

\if@ACM@printfolios Whether to print folios

```
1795 \define@boolkey+{@ACM@topmatter@}[@ACM@]{printfolios}[true]{%
1796   \if@ACM@printfolios
1797     \ClassInfo{\@classname}{Printing folios}%
1798   \else
1799     \ClassInfo{\@classname}{Suppressing folios}%
1800   \fi}{\ClassError{\@classname}{The option printfolios can be either true or false}}
```

\@ACM@authorssperrow The number of authors per row. 0 means use the default algorithm.

```
1801 \define@cmdkey{@ACM@topmatter@}[@ACM@]{authorssperrow}[0]{%
1802   \IfInteger{#1}{\ClassInfo{\@classname}{Setting authorssperrow to
1803     #1}}{\ClassWarning{\@classname}{The parameter authorssperrow must be
1804     numerical. Ignoring the input #1}\gdef\@ACM@authorssperrow{0}}}
```

\settopmatter The usual syntactic sugar

```
1805 \def\settopmatter#1{\setkeys{@ACM@topmatter@}{#1}}
```

Now the settings

```
1806 \settopmatter{printccs=true, printacmref=true}
1807 \if@ACM@manuscript
1808   \settopmatter{printfolios=true}
1809 \else
1810   \if@ACM@journal
1811     \settopmatter{printfolios=true}
1812   \else
1813     \settopmatter{printfolios=false}
1814   \fi
1815 \fi
1816 \settopmatter{authorsperrow=0}
```

\@received The container for the paper history

```
1817 \def\@received{}
```

\received The \received command

```
1818 \newcommand\received[2][\def\@tempa{#1}%
1819   \ifx\@tempa\@empty
1820     \ifx\@received\@empty
1821       \gdef\@received{Received #2}%
1822     \else
1823       \g@addto@macro{\@received}{; revised #2}%
1824     \fi
1825   \else
1826     \ifx\@received\@empty
1827       \gdef\@received{#1 #2}%
1828     \else
1829       \g@addto@macro{\@received}{; #1 #2}%
1830     \fi
1831   \fi}
1832 \AtEndDocument{%
1833   \ifx\@received\@empty\else
1834     \par\bigskip\noindent{\small\normalfont\@received\par}%
1835   \fi}
```

\acmArticleType Article type. We set up the color.

```
1836 \define@choicekey+{ACM}{articletype}[\ACM@ArticleType\ACM@ArticleType@nr]{%
1837   Research,Review,Discussion,Invited,Position}[Review]{%
1838   \ifcase\ACM@ArticleType@nr
1839     \relax % Research
1840     \colorlet{@ACM@Article@color}{ACMBlue}%
1841   \or % Review
1842     \colorlet{@ACM@Article@color}{ACMOrange}%
1843   \or % Discussion
1844     \colorlet{@ACM@Article@color}{ACMGreen}%
1845   \or % Invited
1846     \colorlet{@ACM@Article@color}{ACMPurple}%
1847   \or % Position
1848     \colorlet{@ACM@Article@color}{ACMRed}%
1849   \fi
1850 }{%
1851   \ClassError{\@classname}{Article Type must be Research, Review,\MessageBreak
1852     Discussion, Invited, or Position}}
```



```

1853 \def\acmArticleType#1{\setkeys{ACM}{articletype=#1}}
1854 \if@ACM@acmcp
1855 \acmArticleType{Research}%
1856 \fi
1857

\@ACM@color@frame The colored frame for the acmcp
1858 \newenvironment{@ACM@color@frame}{%
1859   \def\FrameCommand{\hspace*{-6.5pc}%
1860     \colorbox{@ACM@Article@color!10!white}}%
1861   \MakeFramed {\advance\hsize
1862     -6.5pc\relax\FrameRestore}}{\zsavaposy{@ACM@acmcpframe@y}%
1863   \endMakeFramed}

\acmCodeDataLink The code link
1864 \def\acmCodeLink#1{%
1865   \ifx\@acmCodeDataLink\@empty
1866     \gdef\@acmCodeDataLink{\url{#1}}%
1867   \else
1868     \g@addto@macro{\@acmCodeDataLink}{\ \url{#1}}%
1869   \fi}
1870 \def\@acmCodeDataLink{}

\acmContributions The authors' contrubution statement
1871 \def\acmContributions#1{\gdef\@acmContributions{#1}}
1872 \acmContributions{}

\acmDataLink The code link
1873 \let\acmDataLink\acmCodeLink

3.18 Concepts system

We exclude CCSXML stuff generated by the ACM system:
1874 \RequirePackage{comment}
1875 \excludecomment{CCSXML}

\@concepts This is the storage macro and counter for concepts
1876 \let\@concepts\@empty
1877 \newcounter{@concepts}

\ccsdesc The first argument is the significance, the second is the concept(s)
1878 \newcommand\ccsdesc[2][100]{%
1879   \ccsdesc@parse#1~#2~~\ccsdesc@parse@end}

\ccsdesc@parse The parser of the expression Significance~General~Specific (we need textcomp for
\textrightarrow). Note that Specific can be empty!
1880 \def\textrightarrow{\rightarrow}
1881 \def\ccsdesc@parse#1~#2~#3~{%
1882   \stepcounter{@concepts}%
1883   \expandafter\ifx\csname CCS@General@#2\endcsname\relax
1884     \expandafter\gdef\csname CCS@General@#2\endcsname{\textbullet\
1885       \textbf{#2}}%
1886     \expandafter\gdef\csname CCS@Punctuation@#2\endcsname{;} %

```

```

1887 \expandafter\gdef\csname CCS@Specific@#2\endcsname{%
1888 \g@addto@macro{\@concepts}{\csname CCS@General@#2\endcsname
1889 \csname CCS@Punctuation@#2\endcsname
1890 \csname CCS@Specific@#2\endcsname}%
1891 \fi
1892 \ifx#3\relax\relax\else
1893 \expandafter\gdef\csname CCS@Punctuation@#2\endcsname{
1894 \textrightarrow\ }%
1895 \expandafter\g@addto@macro\expandafter{\csname CCS@Specific@#2\endcsname}{%
1896 \addtocounter{\@concepts}{-1}%
1897 \ifnum#1>499\textbf{#3}\else
1898 \ifnum#1>299\textit{#3}\else
1899 #3\fi\fi\ifnum\value{\@concepts}=0.\else; \fi}%
1900 \fi
1901 \ccsdesc@parse@finish}

1902 \AtEndDocument{\if@ACM@nonacm\else\ifx\@concepts\@empty\relax
1903 \ifnum\getrefnumber{TotPages}>2\relax
1904 \ClassWarningNoLine{\@classname}{CCS concepts are mandatory
1905 for papers over two pages}%
1906 \fi\fi\fi}

```

```

\ccdesc@parse@finish Gobble everything to \ccsdesc@parse@end
1907 \def\ccsdesc@parse@finish#1\ccsdesc@parse@end{}

```

3.19 Copyright system

This is from acmcopyright.sty

```

\if@printcopyright Whether to print a copyright symbol
1908 \newif\if@printcopyright
1909 \@printcopyrighttrue

```

```

\if@printpermission Whether to print the permission block
1910 \newif\if@printpermission
1911 \@printpermissiontrue

```

```

\if@acmowned Whether the ACM owns the rights to the paper
1912 \newif\if@acmowned
1913 \@acmownedtrue

```

Keys:

```

1914 \define@choicekey*{ACM@}{acmcopyrightmode}[%
1915 \acm@copyrightinput\acm@copyrightmode]{none,%
1916 acmcopyright,acmlicensed,rightsretained,%
1917 usgov,usgovmixed,cagov,cagovmixed,licensedusgovmixed,%
1918 licensedcagov,licensedcagovmixed,othergov,licensedothergov,%
1919 iw3c2w3,iw3c2w3g,cc}{%
1920 \@printpermissiontrue
1921 \@printcopyrighttrue
1922 \@acmownedtrue
1923 \ifnum\acm@copyrightmode=0\relax % none
1924 \@printpermissionfalse

```

```

1925 \@printcopyrightfalse
1926 \@acmownedfalse
1927 \fi
1928 \ifnum\acm@copyrightmode=2\relax % acmlicensed
1929 \@acmownedfalse
1930 \fi
1931 \ifnum\acm@copyrightmode=3\relax % rightsretained
1932 \@acmownedfalse
1933 \fi
1934 \ifnum\acm@copyrightmode=4\relax % usgov
1935 \@printpermissiontrue
1936 \@printcopyrightfalse
1937 \@acmownedfalse
1938 \fi
1939 \ifnum\acm@copyrightmode=6\relax % cagov
1940 \@acmownedfalse
1941 \fi
1942 \ifnum\acm@copyrightmode=8\relax % licensedusgovmixed
1943 \@acmownedfalse
1944 \fi
1945 \ifnum\acm@copyrightmode=9\relax % licensedcagov
1946 \@acmownedfalse
1947 \fi
1948 \ifnum\acm@copyrightmode=10\relax % licensedcagovmixed
1949 \@acmownedfalse
1950 \fi
1951 \ifnum\acm@copyrightmode=11\relax % othergov
1952 \@acmownedtrue
1953 \fi
1954 \ifnum\acm@copyrightmode=12\relax % licensedothergov
1955 \@acmownedfalse
1956 \fi
1957 \ifnum\acm@copyrightmode=13\relax % iw3c2w3
1958 \@acmownedfalse
1959 \fi
1960 \ifnum\acm@copyrightmode=14\relax % iw3c2w3g
1961 \@acmownedfalse
1962 \fi
1963 \ifnum\acm@copyrightmode=15\relax % cc
1964 \@acmownedfalse
1965 \fi}

```

`\setcopyright` This is the syntactic sugar around setting keys.

```

1966 \def\setcopyright#1{\setkeys{ACM@}{acmcopyrightmode=#1}}
1967 \setcopyright{acmlicensed}

```

`\setcctype` The type of Creative Commons license used

```

1968 \newcommand\setcctype[2][4.0]{%
1969 \def\ACM@cc@version{#1}%
1970 \def\ACM@cc@type{#2}}
1971 \setcctype{by}

```

`\@copyrightowner` Here is the owner of the copyright

```

1972 \def\@copyrightowner{%

```

```

1973 \ifcase\acm@copyrightmode\relax % none
1974 \or % acmcopyright
1975 ACM\@.
1976 \or % acmlicensed
1977 Copyright held by the owner/author(s). Publication rights licensed to
1978 ACM\@.
1979 \or % rightsretained
1980 Copyright held by the owner/author(s).
1981 \or % usgov
1982 \or % usgovmixed
1983 Copyright held by the owner/author(s).
1984 \or % cagov
1985 Copyright Crown in Right of Canada.
1986 \or %cagovmixed
1987 Copyright held by the owner/author(s).
1988 \or %licensedusgovmixed
1989 Copyright held by the owner/author(s). Publication rights licensed to
1990 ACM\@.
1991 \or % licensedcagov
1992 Copyright held by the owner/author(s).
1993 \or %licensedcagovmixed
1994 Copyright held by the owner/author(s). Publication rights licensed to
1995 ACM\@.
1996 \or % othergov
1997 Copyright held by the owner/author(s).
1998 \or % licensedothergov
1999 Copyright held by the owner/author(s). Publication rights licensed to
2000 ACM\@.
2001 \or % ic2w3www
2002 IW3C2 (International World Wide Web Conference Committee), published
2003 under Creative Commons CC-BY~4.0 License.
2004 \or % ic2w3wwwgoogle
2005 IW3C2 (International World Wide Web Conference Committee), published
2006 under Creative Commons CC-BY-NC-ND~4.0 License.
2007 \or % cc
2008 Copyright held by the owner/author(s).
2009 \fi}

```

\@formatdoi Print a clickable DOI

```
2010 \def\@formatdoi#1{\url{https://doi.org/#1}}
```

\@ACM@copyright@check@cc This macro is no longer needed due to the change in ACM policy.

\@copyrightpermission The canned permission block.

```

2011 \def\@copyrightpermission{%
2012 \ifcase\acm@copyrightmode\relax % none
2013 \or % acmcopyright
2014 Permission to make digital or hard copies of all or part of this
2015 work for personal or classroom use is granted without fee provided
2016 that copies are not made or distributed for profit or commercial
2017 advantage and that copies bear this notice and the full citation on
2018 the first page. Copyrights for components of this work owned by
2019 others than ACM must be honored. Abstracting with credit is
2020 permitted. To copy otherwise, or republish, to post on servers or

```

2021 to redistribute to lists, requires prior specific permission
 2022 and\hspace*{.5pt}/or
 2023 a fee. Request permissions from permissions@acm.org.
 2024 \or % acmlicensed
 2025 Permission to make digital or hard copies of all or part of this
 2026 work for personal or classroom use is granted without fee provided
 2027 that copies are not made or distributed for profit or commercial
 2028 advantage and that copies bear this notice and the full citation on
 2029 the first page. Copyrights for components of this work owned by
 2030 others than the author(s) must be honored. Abstracting with credit
 2031 is permitted. To copy otherwise, or republish, to post on servers
 2032 or to redistribute to lists, requires prior specific permission
 2033 and\hspace*{.5pt}/or a fee. Request permissions from
 2034 permissions@acm.org.
 2035 \or % rightsretained
 2036 Permission to make digital or hard copies of all or part of this
 2037 work for personal or classroom use is granted without fee provided
 2038 that copies are not made or distributed for profit or commercial
 2039 advantage and that copies bear this notice and the full citation on
 2040 the first page. Copyrights for third-party components of this work
 2041 must be honored. For all other uses, contact the
 2042 owner\hspace*{.5pt}/author(s).
 2043 \or % usgov
 2044 This paper is authored by an employee(s) of the United States
 2045 Government and is in the public domain. Non-exclusive copying or
 2046 redistribution is allowed, provided that the article citation is
 2047 given and the authors and agency are clearly identified as its
 2048 source. Request permissions from
 2049 owner\hspace*{.5pt}/author(s).
 2050 \or % usgovmixed
 2051 ACM acknowledges that this contribution was authored or co-authored
 2052 by an employee, contractor, or affiliate of the United States
 2053 government. As such, the United States government retains a
 2054 nonexclusive, royalty-free right to publish or reproduce this
 2055 article, or to allow others to do so, for government purposes
 2056 only. Request permissions from owner\hspace*{.5pt}/author(s).
 2057 \or % cagov
 2058 This article was authored by employees of the Government of
 2059 Canada. As such, the Canadian government retains all interest in
 2060 the copyright to this work and grants to ACM a nonexclusive,
 2061 royalty-free right to publish or reproduce this article, or to
 2062 allow others to do so, provided that clear attribution is given
 2063 both to the authors and the Canadian government agency employing
 2064 them. Permission to make digital or hard copies for personal or
 2065 classroom use is granted. Copies must bear this notice and the full
 2066 citation on the first page. Copyrights for components of this work
 2067 owned by others than the Canadian Government must be honored. To
 2068 copy otherwise, distribute, republish, or post, requires prior
 2069 specific permission and/or a fee. Request permissions from
 2070 owner\hspace*{.5pt}/author(s).
 2071 \or % cagovmixed
 2072 ACM acknowledges that this contribution was co-authored by an
 2073 affiliate of the national government of Canada. As such, the Crown
 2074 in Right of Canada retains an equal interest in the

2075 copyright. Reprints must include clear attribution to ACM and the
 2076 author's government agency affiliation. Permission to make digital
 2077 or hard copies for personal or classroom use is granted. Copies
 2078 must bear this notice and the full citation on the first
 2079 page. Copyrights for components of this work owned by others than
 2080 ACM must be honored. To copy otherwise, distribute, republish, or
 2081 post, requires prior specific permission and/or a fee. Request
 2082 permissions from owner\hspace*{.5pt}/author(s).
 2083 \or % licensedusgovmixed
 2084 Publication rights licensed to ACM\@. ACM acknowledges that this
 2085 contribution was authored or co-authored by an employee, contractor
 2086 or affiliate of the United States government. As such, the
 2087 Government retains a nonexclusive, royalty-free right to publish or
 2088 reproduce this article, or to allow others to do so, for Government
 2089 purposes only. Request permissions from
 2090 owner\hspace*{.5pt}/author(s).
 2091 \or % licensedcagov
 2092 This article was authored by employees of the Government of
 2093 Canada. As such, the Canadian government retains all interest in
 2094 the copyright to this work and grants to ACM a nonexclusive,
 2095 royalty-free right to publish or reproduce this article, or to
 2096 allow others to do so, provided that clear attribution is given
 2097 both to the authors and the Canadian government agency employing
 2098 them. Permission to make digital or hard copies for personal or
 2099 classroom use is granted. Copies must bear this notice and the full
 2100 citation on the first page. Copyrights for components of this work
 2101 owned by others than the Canadian Government must be honored. To
 2102 copy otherwise, distribute, republish, or post, requires prior
 2103 specific permission and/or a fee. Request permissions from
 2104 owner\hspace*{.5pt}/author(s).
 2105 \or % licensedcagovmixed
 2106 Publication rights licensed to ACM. ACM acknowledges that this
 2107 contribution was authored or co-authored by an employee, contractor
 2108 or affiliate of the national government of Canada. As such, the
 2109 Government retains a nonexclusive, royalty-free right to publish or
 2110 reproduce this article, or to allow others to do so, for Government
 2111 purposes only. Request permissions from
 2112 owner\hspace*{.5pt}/author(s).
 2113 \or % othergov
 2114 ACM acknowledges that this contribution was authored or co-authored
 2115 by an employee, contractor or affiliate of a national
 2116 government. As such, the Government retains a nonexclusive,
 2117 royalty-free right to publish or reproduce this article, or to
 2118 allow others to do so, for Government purposes only. Request
 2119 permissions from owner\hspace*{.5pt}/author(s).
 2120 \or % licensedothergov
 2121 Publication rights licensed to ACM\@. ACM acknowledges that this
 2122 contribution was authored or co-authored by an employee, contractor
 2123 or affiliate of a national government. As such, the Government
 2124 retains a nonexclusive, royalty-free right to publish or reproduce
 2125 this article, or to allow others to do so, for Government purposes
 2126 only. Request permissions from owner\hspace*{.5pt}/author(s).
 2127 \or % iw3c2w3
 2128 This paper is published under the Creative Commons Attribution~4.0

```

2129 International (CC-BY~4.0) license. Authors reserve their rights to
2130 disseminate the work on their personal and corporate Web sites with
2131 the appropriate attribution.
2132 \or % iw3c2w3g
2133 This paper is published under the Creative Commons
2134 Attribution-NonCommercial-NoDerivs~4.0 International
2135 (CC-BY-NC-ND~4.0) license. Authors reserve their rights to
2136 disseminate the work on their personal and corporate Web sites with
2137 the appropriate attribution.
2138 \or % CC
2139 \IfEq{\ACM@cc@type}{zero}{%
2140   \def\ACM@CC@Url{https://creativecommons.org/publicdomain/zero/1.0}}{%
2141   \edef\ACM@CC@Url{https://creativecommons.org/licenses/\ACM@cc@type/\ACM@cc@version}}%
2142   \href{\ACM@CC@Url}{\includegraphics[height=5ex]{doclicense-CC-\ACM@cc@type-88x31}}\
2143   \href{\ACM@CC@Url}{%
2144     This work is licensed under a Creative Commons
2145     \IfEq{\ACM@cc@type}{zero}{CC0 1.0 Universal}{%
2146       \IfEq{\ACM@cc@type}{by}{Attribution}{%
2147         \IfEq{\ACM@cc@type}{by-sa}{Attribution-ShareAlike}{%
2148           \IfEq{\ACM@cc@type}{by-nd}{Attribution-NoDerivatives}{%
2149             \IfEq{\ACM@cc@type}{by-nc}{Attribution-NonCommercial}{%
2150               \IfEq{\ACM@cc@type}{by-nc-sa}{Attribution-NonCommercial-ShareAlike}{%
2151                 \IfEq{\ACM@cc@type}{by-nc-nd}{Attribution-NonCommercial-NoDerivatives}{%
2152                   ~\IfEq{\ACM@cc@version}{4.0}{4.0 International}{3.0 Unported}%
2153                 }
2154               License.}%
2155             \fi}

```

\copyrightyear By default, the copyright year is the same as \acmYear, but one can override this:

```

2156 \def\copyrightyear#1{\def\@copyrightyear{#1}}
2157 \copyrightyear{\@acmYear}

```

\@teaserfigures The teaser figures container

```

2158 \def\@teaserfigures{}

```

teaserfigure The teaser figure

```

2159 \newenvironment{teaserfigure}{\Collect@Body\@saveteaser}{}

```

\@saveteaser Saving the teaser

```

2160 \long\def\@saveteaser#1{\g@addto@macro\@teaserfigures{\@teaser{#1}}}

```

\thanks We redefine amsart \thanks so the anonymous key works

```

2161 \renewcommand{\thanks}[1]{%
2162   \@ifnotempty{#1}{%
2163     \if@ACM@anonymous
2164       \g@addto@macro\thankses{\thanks{A note}}%
2165     \else
2166       \g@addto@macro\thankses{\thanks{#1}}%
2167     \fi}}

```

\anon We provide \anon command, which obscures parts of the text if the package option anonymous is set

```

2168 \newcommand{\anon}[2][ANONYMIZED]{%

```

```

2169 \if@ACM@anonymous%
2170   {\color{ACMOrange}#1}%
2171 \else%
2172   #2%
2173 \fi}

```

3.20 Maketitle hook

The current $\LaTeX$ provides a nice lthooks mechanism. However, since it is relatively new, we will use oldfashioned approach—at least for now.

`\@beginmaketitlehook` The hook

```

2174 \ifx\@beginmaketitlehook\@undefined
2175   \let\@beginmaketitlehook\@empty
2176 \fi

```

`\AtBeginMaketitle` Adding to the hook

```

2177 \def\AtBeginMaketitle{\g@addto@macro\@beginmaketitlehook}

```

3.21 ACM Engage top matter

`\@acmengagemetadadata` The special metadata for ACM engage course materials

```

2178 \def\@acmengagemetadadata{}

```

`\setengagemetadadata` Adding topic to engage metadata

```

2179 \def\setengagemetadadata#1#2{%
2180   \g@addto@macro{\@acmengagemetadadata}{%
2181     \@setengagemetadadata{#1}{#2}}}

```

3.22 Typesetting top matter

`\mktitle@bx` Some of our formats use a two-column design. Some use a one-column design. In all cases we use a wide title. Thus we typeset the top matter in a special box to be used in the construction `\@twocolumn[ $\langle box \rangle$ ]`.

```

2182 \newbox\mktitle@bx

```

`\maketitle` The (in)famous `\maketitle`. Note that in sigchi-a mode, authors are *not* in the title box.

Another note: there is a subtle difference between author notes, title notes and thanks. The latter two refer to the paper itself and therefore belong to the copyright/permission block. By the way, this was the default behavior of the old ACM classes.

We reserve asterisk for corresponding authors, so we start footnotes from 2.

```

2183 \def\maketitle{\@beginmaketitlehook
2184   \@ACM@maketitle@typesettrue
2185   \if@ACM@anonymous
2186     % Anonymize omission of \author-s
2187     \ifnum\num@authorgroups=0\author{}\fi
2188   \fi
2189   \begingroup
2190   \let\@vspace\@vspace@orig
2191   \let\@vspacer\@vspacer@orig
2192   \let\@footnotemark\@footnotemark@nolink

```



```

2193 \let\@footnotetext\@footnotetext@nolink
2194 \renewcommand\thefootnote{\@fnsymbol\c@footnote}%
2195 \setcounter{footnote}{1}%
2196 \hsize=\textwidth
2197 \def\@makefnmark{\hbox{\@textsuperscript{\@thefnmark}}}%
2198 \mktitle\if@ACM@sigchiamode\else\mkauthors\fi\mkteasers
2199 \@printtopmatter
2200 \if@ACM@sigchiamode\mkauthors\fi
2201 \def\@makefnmark{\hbox{\@textsuperscript{\normalfont\@thefnmark}}}%
2202 \setcounter{footnote}{1}%
2203 \@titlenotes
2204 \@subtitlenotes
2205 \@authornotes
2206 \let\@makefnmark\relax
2207 \let\@thefnmark\relax
2208 \let\@makefntext\noindent
2209 \ifx\@empty\thankses\else
2210   \footnotetextauthorsaddresses{%
2211     \def\par{\let\par\@par}\parindent\z@\@setthanks}%
2212   \fi
2213 \if@ACM@acmcp\else
2214   \ifx\@empty\@authorsaddresses\else
2215     \if@ACM@anonymous\else
2216       \if@ACM@journal@bibstrip@or@tog
2217         \footnotetextauthorsaddresses{%
2218           \def\par{\let\par\@par}\parindent\z@\@setauthorsaddresses}%
2219         \fi
2220       \fi
2221     \fi
2222   \fi
2223 \if@ACM@nonacm
2224   \ifnum\acm@copyrightmode=15\relax % cc
2225     \footnotetextcopyrightpermission{\@copyrightpermission}%
2226   \fi
2227 \else
2228   \if@ACM@acmcp\else
2229     \footnotetextcopyrightpermission{%
2230       \if@ACM@authordraft
2231         \raisebox{-2ex}{\z@\z@{\makebox[0pt][l]{\large\bfseries
2232           Unpublished working draft. Not for distribution.}}}%
2233         \color[gray]{0.9}%
2234       \fi
2235       \parindent\z@\parskip0.1\baselineskip
2236       \if@ACM@authorversion\else
2237         \if@printpermission\@copyrightpermission\par\fi
2238       \fi
2239       \if@ACM@manuscript\else
2240         \if@ACM@journal@bibstrip\else % Print the conference information
2241           \if@ACM@engage
2242             {\itshape \@acmBooktitle, \@acmYear.}\par
2243           \else
2244             {\itshape \acmConference@shortname, \acmConference@venue}\par
2245           \fi
2246         \fi

```

```

2247 \fi
2248 \if@printcopyright
2249 \copyright\ \@copyrightyear\ \@copyrightowner\
2250 \else
2251 \ifx\@copyrightyear\@empty\else
2252 \@copyrightyear.\
2253 \fi
2254 \fi
2255 \if@ACM@manuscript
2256 Manuscript submitted to ACM\
2257 \else
2258 \if@ACM@authorversion
2259 This is the author's version of the work. It is posted here for
2260 your personal use. Not for redistribution. The definitive Version
2261 of Record was published in
2262 \if@ACM@journal@bibstrip
2263 \emph{\@journalName}%
2264 \else
2265 \emph{\@acmBooktitle}%
2266 \fi
2267 \ifx\@acmDOI\@empty
2268 .
2269 \else
2270 , \@formatdoi{\@acmDOI}.
2271 \fi\
2272 \else
2273 \if@ACM@nonacm\else
2274 \if@ACM@journal@bibstrip
2275 ACM~\@permissionCodeTwo/\@acmYear/\@acmMonth-ART\@acmArticle\
2276 \@formatdoi{\@acmDOI}%
2277 \else % Conference
2278 \ifx\@acmISBN\@empty\else ACM~ISBN~\@acmISBN\\\fi
2279 \ifx\@acmDOI\@empty\else\@formatdoi{\@acmDOI}\fi%
2280 \fi
2281 \fi
2282 \fi
2283 \fi}%
2284 \fi
2285 \fi
2286 \endgroup
2287 \if@ACM@engage\@typesetengagemetadata\fi
2288 \setcounter{footnote}{0}%
2289 \@mkabstract
2290 \ifx\@translatedabstracts\@empty\else
2291 \@translatedabstracts\fi
2292 \if@ACM@printccs
2293 \ifx\@concepts\@empty\else\bgroup
2294 {\@specialsection{CCS Concepts}%
2295 \noindent\@concepts\par}\egroup
2296 \fi
2297 \fi
2298 \if@ACM@acmcp\else
2299 \ifx\@keywords\@empty\else\bgroup
2300 {\@specialsection{\keywordsname}%

```

```

2301         \noindent\@keywords\par}\egroup
2302     \fi
2303     \ifx\@translatedkeywords\@empty\else
2304         \@translatedkeywords
2305     \fi
2306 \fi
2307 \let\metadata@authors=\authors
2308 \nxandlist{, }{, }{\, }\metadata@authors
2309 \def\@ACM@checkaffil{}%
2310 \hypersetup{%
2311     pdfauthor={\metadata@authors},
2312     pdftitle={\@title},
2313     pdfsubject={\@concepts},
2314     pdfkeywords={\@keywords},
2315     pdfcreator={LaTeX with acmart
2316         \csname ver@\@classname.cls\endcsname\space
2317         and hyperref
2318         \csname ver@hyperref.sty\endcsname}}}%
2319 \andify\authors
2320 \andify\shortauthors
2321 \global\let\authors=\authors
2322 \global\let\shortauthors=\shortauthors
2323 \if@ACM@printacmref
2324     \@mkbibcitation
2325 \fi
2326 \global\@topnum\z@ % this prevents floats from falling
2327                     % at the top of page 1
2328 \global\@botnum\z@ % we do not want them to be on the bottom either
2329 \@printendtopmatter
2330 \@afterindentfalse
2331 \@afterheading
2332 \if@ACM@acmcp
2333     \set@ACM@acmcpbox
2334     \AtEndDocument{\end@ACM@color@frame}%
2335     \@ACM@color@frame
2336 \fi
2337 }

```

\set@ACM@acmcpbox Setting infobox for acmcp

```

2338 \newbox\@ACM@acmcpbox
2339 \def\set@ACM@acmcpbox{%
2340     \bgroup
2341     \hsize=5pc
2342     \global\setbox\@ACM@acmcpbox=\vbox{%
2343         \setlength{\parindent}{\z@}%
2344         {\includegraphics[width=\hsize]{acm-jdslogo}\par}%
2345         \scriptsize
2346         \ifnum\getrefnumber{TotPages}>1\else
2347             \zrefused{\@ACM@acmcpbox@y}%
2348             \zrefused{\@ACM@acmcpframe@y}%
2349             \@tempdima=\dimexpr\zposy{\@ACM@acmcpbox@y}sp -
2350             \zposy{\@ACM@acmcpframe@y}sp+0.3\FrameSep+
2351             \@ACM@acmcp@delta\relax
2352             \ifdim\@tempdima>0pt\relax

```

```

2353         \vspace*{\@tempdima}%
2354         \protected@write\@auxout{}%
2355         {\string\global\@ACM@acmcp@delta=\the\@tempdima\relax}%
2356     \fi
2357 \fi
2358 \ifx\@acmCodeDataLink\@empty\else\bigskip
2359 Code and data links:\\ \@acmCodeDataLink\par\bigskip
2360 \fi
2361 \ifx\@keywords\@empty\else\bigskip
2362 Keywords: \@keywords\par
2363 \fi
2364 \ifx\@acmContributions\@empty\else\bigskip
2365 \@acmContributions\par
2366 \fi
2367 \ifx\@empty\@authorsaddresses\else\bigskip\@setauthorsaddresses\fi
2368 \zsavaposy{\@ACM@acmcpbox@y}%
2369 \par
2370 }
2371 \egroup}

```

`\@specialsection` This macro starts sections for proceedings and uses `\small` for journals

```

2372 \def\@specialsection#1{%
2373   \let\@vspace\@vspace@orig
2374   \let\@vspacer\@vspacer@orig
2375   \ifcase\ACM@format@nr
2376   \relax % manuscript
2377   \par\medskip\small\noindent#1: %
2378   \or % acmsmall
2379   \par\medskip\small\noindent#1: %
2380   \or % acmlarge
2381   \par\medskip\small\noindent#1: %
2382   \or % acmtog
2383   \par\medskip\small\noindent#1: %
2384   \or % sigconf
2385   \section*{#1}%
2386   \or % siggraph
2387   \section*{#1}%
2388   \or % sigplan
2389   \noindentparagraph*{#1:~}%
2390   \or % sigchi
2391   \section*{#1}%
2392   \or % sigchi-a
2393   \section*{#1}%
2394   \or % acmengage
2395   \section*{#1}%
2396   \or % acmcp
2397   \section*{#1}%
2398   \fi
2399   \let\@vspace\@vspace@acm
2400   \let\@vspacer\@vspacer@acm
2401 }

```

`\@printtopmatter` The printing of top matter starts a new page and uses the given title box.

Note that if there are too many authors, `\mktitle@bx` might overflow the page.

Therefore we start with checking this and if this happens, we split the box and print it page by page.

```

2402 \def\@printtopmatter{%
2403   \ifx\@startPage\@empty
2404     \gdef\@startPage{1}%
2405   \else
2406     \setcounter{page}{\@startPage}%
2407   \fi
2408   \@tempdima=\ht\mktitle@bx
2409   \advance\@tempdima by \dp\mktitle@bx
2410   \ifdim\@tempdima>0.9\textheight
2411     \loop
2412       \setbox\@tempboxa=\vsplit \mktitle@bx to 0.9\textheight
2413       \thispagestyle{firstpagestyle}%
2414       \noindent\unvbox\@tempboxa
2415       \clearpage
2416       \@tempdima=\ht\mktitle@bx
2417       \advance\@tempdima by \dp\mktitle@bx
2418       \ifdim\@tempdima>0.9\textheight\repeat
2419   \fi
2420   \thispagestyle{firstpagestyle}%
2421   \noindent
2422   \ifcase\ACM@format@nr
2423     \relax % manuscript
2424     \box\mktitle@bx\par
2425   \or % acmsmall
2426     \box\mktitle@bx\par
2427   \or % acmlarge
2428     \box\mktitle@bx\par
2429   \or % acmtog
2430     \twocolumn[\box\mktitle@bx]%
2431   \or % sigconf
2432     \twocolumn[\box\mktitle@bx]%
2433   \or % siggraph
2434     \twocolumn[\box\mktitle@bx]%
2435   \or % sigplan
2436     \twocolumn[\box\mktitle@bx]%
2437   \or % sigchi
2438     \twocolumn[\box\mktitle@bx]%
2439   \or % sigchi-a
2440     \par\box\mktitle@bx\par\bigskip
2441   \or % acmengage
2442     \twocolumn[\box\mktitle@bx]%
2443   \or % acmcp
2444     \box\mktitle@bx\par
2445   \fi
2446 }

```

\mktitle The title of the article

```

2447 \def\mktitle{%
2448   \ifcase\ACM@format@nr
2449     \relax % manuscript
2450     \@mktitle@i
2451   \or % acmsmall

```

```

2452 \mktitle@i
2453 \or % acmlarge
2454 \mktitle@i
2455 \or % acmtog
2456 \mktitle@i
2457 \or % sigconf
2458 \mktitle@iii
2459 \or % siggraph
2460 \mktitle@iii
2461 \or % sigplan
2462 \mktitle@iii
2463 \or % sigchi
2464 \mktitle@iii
2465 \or % sigchi-a
2466 \mktitle@iv
2467 \or % acmengage
2468 \mktitle@iii
2469 \or % acmcp
2470 \mktitle@i
2471 \fi
2472 }

```

`\@titlefont` The font to typeset the title

```

2473 \def\@titlefont{%
2474 \ifcase\ACM@format@nr
2475 \relax % manuscript
2476 \LARGE\sffamily\bfseries
2477 \or % acmsmall
2478 \LARGE\sffamily\bfseries
2479 \or % acmlarge
2480 \LARGE\sffamily\bfseries
2481 \or % acmtog
2482 \Huge\sffamily
2483 \or % sigconf
2484 \Huge\sffamily\bfseries
2485 \or % siggraph
2486 \Huge\sffamily\bfseries
2487 \or % sigplan
2488 \Huge\bfseries
2489 \or % sigchi
2490 \Huge\sffamily\bfseries
2491 \or % sigchi-a
2492 \Huge\bfseries
2493 \or % acmengage
2494 \Huge\sffamily\bfseries
2495 \or % acmcp
2496 \LARGE\sffamily\bfseries
2497 \fi}

```

`\@subtitlefont` The font to typeset the subtitle

```

2498 \def\@subtitlefont{\normalsize
2499 \ifcase\ACM@format@nr
2500 \relax % manuscript
2501 \mdseries

```

```

2502 \or % acmsmall
2503 \mdseries
2504 \or % acmlarge
2505 \mdseries
2506 \or % acmtog
2507 \LARGE
2508 \or % sigconf
2509 \LARGE\mdseries
2510 \or % siggraph
2511 \LARGE\mdseries
2512 \or % sigplan
2513 \LARGE\mdseries
2514 \or % sigchi
2515 \LARGE\mdseries
2516 \or % sigchi-a
2517 \mdseries
2518 \or % acmengage
2519 \LARGE\mdseries
2520 \or % acmcp
2521 \mdseries
2522 \fi}

```

\mktitle@i The version of \mktitle for most journals

```

2523 \def\mktitle@i{\hsize=\textwidth
2524 \if@ACM@acmcp
2525 \advance\hsize by -6pc%
2526 \fi
2527 \@ACM@title@width=\hsize
2528 \setbox\mktitle@bx=\vbox{\noindent\@titlefont
2529 \parbox[t]{\@ACM@title@width}{\raggedright
2530 \@titlefont\noindent
2531 \@title\@translatedtitle%
2532 \ifx\@subtitle\@empty\else
2533 \par\noindent{\@subtitlefont\@subtitle\@translatedsubtitle}%
2534 \fi}%
2535 \par\bigskip}}%

```

\mktitle@ii The version of \mktitle for TOG. Since v1.06, this is subsumed by the \mktitle@i macro

\mktitle@iii The version of \mktitle for SIG proceedings.

```

2536 \def\mktitle@iii{\hsize=\textwidth
2537 \setbox\mktitle@bx=\vbox{\@titlefont\centering
2538 \@ACM@title@width=\hsize
2539 \parbox[t]{\@ACM@title@width}{\centering\@titlefont
2540 \@title\@translatedtitle%
2541 \ifx\@subtitle\@empty\else
2542 \par\noindent{\@subtitlefont\@subtitle\@translatedsubtitle}%
2543 \fi
2544 }%
2545 \par\bigskip}}%

```

\mktitle@iv The version of \mktitle for sigchi-a

```

2546 \def\mktitle@iv{\hsize=\textwidth

```

```

2547 \setbox\mktitle@bx=\vbox{\raggedright\leftskip5pc\@titlefont
2548 \noindent\leavevmode\leaders\hrule height 2pt\hfill\kern0pt\par
2549 \noindent\@title\@translatedtitle%
2550 \ifx\@subtitle\@empty\else
2551 \par\noindent\@subtitlefont\@subtitle\@translatedsubtitle%
2552 \fi
2553 \par\bigskip}}%

```

\@ACM@addtoaddress

```

2554 \newbox\@ACM@commabox
2555 \def\@ACM@addtoaddress#1{%
2556 \ifvmode\else
2557 \if@ACM@affiliation@obeypunctuation\else
2558 \unskip,~%
2559 \fi\fi
2560 #1}

```

\institution Theoretically we can define the macros for \affiliation inside the \@mkauthors-style
\position commands. However, this would lead to a strange error if an author uses them outside
\department \affiliation. Of course we can make them produce an error message, but...

```

\streetaddress 2561 \def\streetaddress#1{\ClassWarning{\@classname}{ACM no longer collects
\city 2562 authors' postal addresses. I am ignoring your street
\state 2563 address}\unskip\ignorespaces}
\postcode 2564 \def\postcode#1{\ClassWarning{\@classname}{ACM no longer collects
\country 2565 authors' postal addresses. I am ignoring your postal
2566 code}\unskip\ignorespaces}
2567 \if@ACM@journal
2568 \def\position#1{\unskip\ignorespaces}
2569 \def\institution#1{\global\@ACM@instpresenttrue
2570 \unskip~#1\ignorespaces}
2571 \def\city#1{\global\@ACM@citypresenttrue\unskip\ignorespaces}
2572 \def\state#1{\unskip\ignorespaces}
2573 \newcommand\department[2][0]{\unskip\ignorespaces}
2574 \def\country#1{\StrDel{#1}{ }[\@tempa]%
2575 \ifx\@tempa\@empty\else
2576 \global\@ACM@countrypresenttrue\fi
2577 \if@ACM@affiliation@obeypunctuation\else, \fi#1\ignorespaces}
2578 \else
2579 \def\position#1{\if@ACM@affiliation@obeypunctuation#1\else#1\par\fi}%
2580 \def\institution#1{\global\@ACM@instpresenttrue
2581 \if@ACM@affiliation@obeypunctuation#1\else#1\par\fi}%
2582 \newcommand\department[2][0]{\if@ACM@affiliation@obeypunctuation
2583 #2\else#2\par\fi}%
2584 \def\city#1{\global\@ACM@citypresenttrue\@ACM@addtoaddress{#1}}%
2585 \let\state\@ACM@addtoaddress
2586 \def\country#1{\global\@ACM@countrypresenttrue\@ACM@addtoaddress{#1}}%
2587 \fi

```

\@mkauthors Typesetting the authors

```

2588 \def\@mkauthors{\begingroup
2589 \hsize=\textwidth
2590 \ifcase\ACM@format@nr
2591 \relax % manuscript
2592 \@mkauthors@i

```



```

2593 \or % acmsmall
2594 \mkauthors@i
2595 \or % acmlarge
2596 \mkauthors@i
2597 \or % acmtog
2598 \mkauthors@i
2599 \or % sigconf
2600 \mkauthors@iii
2601 \or % siggraph
2602 \mkauthors@iii
2603 \or % sigplan
2604 \mkauthors@iii
2605 \or % sigchi
2606 \mkauthors@iii
2607 \or % sigchi-a
2608 \mkauthors@iv
2609 \or % acmengage
2610 \mkauthors@iii
2611 \or % acmcp
2612 \mkauthors@i
2613 \fi
2614 \endgroup
2615 }

```

\@authorfont Somehow different conferences use different fonts for author names. Why?

```
2616 \def\@authorfont{\Large\sffamily}
```

\@affiliationfont Font for affiliations

```
2617 \def\@affiliationfont{\normalsize\normalfont}
```

Adjusting fonts for different formats

```

2618 \ifcase\ACM@format@nr
2619 \relax % manuscript
2620 \or % acmsmall
2621 \def\@authorfont{\large\sffamily}
2622 \def\@affiliationfont{\small\normalfont}
2623 \or % acmlarge
2624 \or % acmtog
2625 \def\@authorfont{\LARGE\sffamily}
2626 \def\@affiliationfont{\large}
2627 \or % sigconf
2628 \def\@authorfont{\LARGE}
2629 \def\@affiliationfont{\large}
2630 \or % siggraph
2631 \def\@authorfont{\normalsize\normalfont}
2632 \def\@affiliationfont{\normalsize\normalfont}
2633 \or % sigplan
2634 \def\@authorfont{\Large\normalfont}
2635 \def\@affiliationfont{\normalsize\normalfont}
2636 \or % sigchi
2637 \def\@authorfont{\bfseries}
2638 \def\@affiliationfont{\mdseries}
2639 \or % sigchi-a
2640 \def\@authorfont{\bfseries}

```

```

2641 \def\@affiliationfont{\mdseries}
2642 \or % acmengage
2643 \def\@authorfont{\LARGE}
2644 \def\@affiliationfont{\large}
2645 \or % acmcp
2646 \def\@authorfont{\large\sffamily}
2647 \def\@affiliationfont{\small\normalfont}
2648 \fi

\@typeset@author@line At this point we have \@currentauthors and \@currentaffiliations. We typeset
them in the journal style
2649 \def\@typeset@author@line{%
2650 \andify\@currentauthors\par\noindent
2651 \@currentauthors\def\@currentauthors{}%
2652 \ifx\@currentaffiliations\empty\else
2653 \andify\@currentaffiliations
2654 \unskip, {\@currentaffiliations}\par
2655 \fi
2656 \def\@currentaffiliations{}}

\if@ACM@instpresent Whether the given affiliation has institution
2657 \newif\if@ACM@instpresent
2658 \@ACM@instpresenttrue

\if@ACM@citypresent Whether the given affiliation has city
2659 \newif\if@ACM@citypresent
2660 \@ACM@citypresenttrue

\if@ACM@countrypresent Whether the given affiliation has country
2661 \newif\if@ACM@countrypresent
2662 \@ACM@countrypresenttrue

\@ACM@resetaffil Reset affiliation flags
2663 \def\@ACM@resetaffil{%
2664 \global\@ACM@instpresentfalse
2665 \global\@ACM@citypresentfalse
2666 \global\@ACM@countrypresentfalse
2667 }

\@ACM@checkaffil Check affiliation flags
2668 \def\@ACM@checkaffil{%
2669 \if@ACM@instpresent\else
2670 \ClassWarningNoLine{\@classname}{No institution present for an affiliation}%
2671 \fi
2672 \if@ACM@citypresent\else
2673 \ClassWarningNoLine{\@classname}{No city present for an affiliation}%
2674 \fi
2675 \if@ACM@countrypresent\else
2676 \ClassError{\@classname}{No country present for an affiliation}{ACM
2677 requires each author to indicate their country using country macro.}%
2678 \fi
2679 }

```

`\mkauthors@i` This version is used in most journal formats. Note that `\and` between authors with the same affiliation becomes `_and_`:

```

2680 \def\mkauthors@i{%
2681   \def\@currentauthors{}%
2682   \def\@currentaffiliations{}%
2683   \global\let\and\@typeset@author@line
2684   \def\@author##1{%
2685     \ifx\@currentauthors\@empty
2686       \gdef\@currentauthors{\@authorfont\MakeUppercase{##1}}%
2687     \else
2688       \g@addto@macro{\@currentauthors}{\and\MakeUppercase{##1}}%
2689     \fi
2690     \gdef\and{}%
2691   \def\email##1##2{}%
2692   \def\affiliation##1##2{%
2693     \def\@tempa{##2}\ifx\@tempa\@empty\else
2694       \ifx\@currentaffiliations\@empty
2695         \gdef\@currentaffiliations{%
2696           \setkeys{@ACM@affiliation@}{obeypunctuation=false}%
2697           \setkeys{@ACM@affiliation@}{##1}%
2698           \@ACM@resetaffil
2699           \@affiliationfont##2\@ACM@checkaffil}%
2700       \else
2701         \g@addto@macro{\@currentaffiliations}{\and
2702           \setkeys{@ACM@affiliation@}{obeypunctuation=false}%
2703           \setkeys{@ACM@affiliation@}{##1}\@ACM@resetaffil
2704           ##2\@ACM@checkaffil}%
2705       \fi
2706     \fi
2707     \global\let\and\@typeset@author@line}%
2708   \if@ACM@acmcp
2709     \advance\hsize by -6pc%
2710   \fi
2711   \global\setbox\mktitle@bx=\vbox{\noindent\unvbox\mktitle@bx\par\medskip
2712     \noindent\addresses\@typeset@author@line
2713     \par\medskip}%
2714 }
```

`\mkauthors@ii` The `\mkauthors@ii` command was the version used in `acmtog`. It is no longer necessary.

`\author@bx` The box to put an individual author in

```
2715 \newbox\author@bx
```

`\author@bx@wd` The width of the author box

```
2716 \newdimen\author@bx@wd
```

`\author@bx@sep` The separation between author boxes

```
2717 \newskip\author@bx@sep
2718 \author@bx@sep=1pc\relax
```

`\@typeset@author@bx` Typesetting the box with authors. Note that in `sigchi-a` the box is not centered.

```
2719 \def\@typeset@author@bx{\bgroup\hsize=\author@bx@wd
```

```

2720 \def\and{\par}\normalbaselines
2721 \global\setbox\author@bx=\vtop{\if@ACM@sigchiamode\else\centering\fi
2722   \@authorfont\@currentauthors\par\@affiliationfont
2723   \@currentaffiliation\par}\egroup
2724 \box\author@bx\hspace{\author@bx@sep}%
2725 \gdef\@currentauthors{}%
2726 \gdef\@currentaffiliation{}%

```

\@mkauthors@iii The sigconf version. Here we use a centered design with each author in a separate box.

```

2727 \def\@mkauthors@iii{%

```

First, we need to determine the design of the author strip. The boxes are separated by \author@bx@sep plus two \author@bx@sep margins. This means that each box must be of width $(\text{textwidth} - \text{author@bx@sep})/N - \text{author@bx@sep}$, where N is the number of boxes per row.

```

2728 \author@bx@wd=\textwidth\relax
2729 \advance\author@bx@wd by -\author@bx@sep\relax
2730 \ifnum\@ACM@authorsperrrow>0\relax
2731   \divide\author@bx@wd by \@ACM@authorsperrrow\relax
2732 \else
2733   \ifcase\@num@authorgroups
2734     \relax % 0?
2735     \or % 1=one author per row
2736     \or % 2=two authors per row
2737       \divide\author@bx@wd by \@num@authorgroups\relax
2738     \or % 3=three authors per row
2739       \divide\author@bx@wd by \@num@authorgroups\relax
2740     \or % 4=two authors per row (!)
2741       \divide\author@bx@wd by 2\relax
2742     \else % three authors per row
2743       \divide\author@bx@wd by 3\relax
2744   \fi
2745 \fi
2746 \advance\author@bx@wd by -\author@bx@sep\relax

```

Now, parsing of \addresses:

```

2747 \gdef\@currentauthors{}%
2748 \gdef\@currentaffiliation{}%
2749 \def\@author##1{\ifx\@currentauthors\@empty
2750   \gdef\@currentauthors{\par##1}%
2751 \else
2752   \g@addto@macro\@currentauthors{\par##1}%
2753 \fi
2754 \gdef\and{}}%
2755 \def\email##1##2{\ifx\@currentaffiliation\@empty
2756   \gdef\@currentaffiliation{\bgroup
2757     \mathchardef\UrlBreakPenalty=10000\href{mailto:##2}{##2}\egroup}%
2758 \else
2759   \g@addto@macro\@currentaffiliation{\par\bgroup
2760     \mathchardef\UrlBreakPenalty=10000\href{mailto:##2}{##2}\egroup}%
2761 \fi}%
2762 \def\affiliation##1##2{\ifx\@currentaffiliation\@empty
2763   \gdef\@currentaffiliation{%
2764     \setkeys{@ACM@affiliation@}{obeypunctuation=false}%
2765     \setkeys{@ACM@affiliation@}{##1}\@ACM@resetaffil

```

```

2766     ##2\@ACM@checkaffil}%
2767 \else
2768   \g@addto@macro\@currentaffiliation{\par
2769     \setkeys{@ACM@affiliation@}{obeypunctuation=false}%
2770     \setkeys{@ACM@affiliation@}{##1}\@ACM@resetaffil
2771     ##2\@ACM@checkaffil}%
2772 \fi
2773 \global\let\and\@typeset@author@bx
2774 }%

Actual typesetting is done by the \and macro:

2775 \hsize=\textwidth
2776 \global\setbox\mktitle@bx=\vbox{\noindent
2777   \unvbox\mktitle@bx\par\medskip\leavevmode
2778   \lineskip=1pc\relax\centering\hspace*{-1em}%
2779   \addresses\let\and\@typeset@author@bx\and\par\bigskip}}

```

\@mkauthors@iv The sigchi-a version. We put authors in the main text with no more than 2 authors per line, unless overridden.

```

2780 \def\@mkauthors@iv{%

First, we need to determine the design of the author strip. The boxes are separated by
\author@bx@sep plus two \author@bx@sep margins. This means that each box must
be of width (\textwidth - \author@bx@sep)/N - \author@bx@sep, where N is the
number of boxes per row.

2781 \author@bx@wd=\columnwidth\relax
2782 \advance\author@bx@wd by -\author@bx@sep\relax
2783 \ifnum\@ACM@authorsprow>0\relax
2784   \divide\author@bx@wd by \@ACM@authorsprow\relax
2785 \else
2786   \ifcase\@num@authorgroups
2787     \relax % 0?
2788     \or % 1=one author per row
2789     \else % 2=two authors per row
2790       \divide\author@bx@wd by 2\relax
2791   \fi
2792 \fi
2793 \advance\author@bx@wd by -\author@bx@sep\relax

```

Now, parsing of \addresses:

```

2794 \gdef\@currentauthors{}%
2795 \gdef\@currentaffiliation{}%
2796 \def\author##1{\ifx\@currentauthors\@empty
2797   \gdef\@currentauthors{\par##1}%
2798 \else
2799   \g@addto@macro\@currentauthors{\par##1}%
2800 \fi
2801 \gdef\and{}}%
2802 \def\email##1##2{\ifx\@currentaffiliation\@empty
2803   \gdef\@currentaffiliation{\href{mailto:##2}{##2}}%
2804 \else
2805   \g@addto@macro\@currentaffiliation{\par\href{mailto:##2}{##2}}%
2806 \fi}%
2807 \def\affiliation##1##2{\ifx\@currentaffiliation\@empty
2808   \gdef\@currentaffiliation{

```

```

2809     \setkeys{@ACM@affiliation@}{obeypunctuation=false}%
2810     \setkeys{@ACM@affiliation@}{##1}\@ACM@resetaffil
2811     ##2\@ACM@checkaffil}%
2812 \else
2813     \g@addto@macro\@currentaffiliation{\par
2814     \setkeys{@ACM@affiliation@}{obeypunctuation=false}%
2815     \setkeys{@ACM@affiliation@}{##1}\@ACM@resetaffil
2816     ##2\@ACM@checkaffil}%
2817 \fi
2818 \global\let\and\@typeset@author@bx}%
2819 %
Actual typesetting is done by the \and macro
2820     \bgroup\hsize=\columnwidth
2821     \par\raggedright\leftskip=\z@
2822     \lineskip=1pc\noindent
2823     \addresses\let\and\@typeset@author@bx\and\par\bigskip\egroup}

```

\@mkauthorsaddresses Typesetting authors' addresses in the footnote style

```

2824 \def\@mkauthorsaddresses{%
2825     \ifnum\@num@authors>1\relax
2826     Authors' \else Author's \fi
2827     Contact Information:
2828     \bgroup
2829     \def\streetaddress##1{\ClassWarning{\@classname}{ACM no longer collects
2830     authors' postal addresses. I am ignoring your street
2831     address}\unskip\ignorespaces}%
2832     \def\postcode##1{\ClassWarning{\@classname}{ACM no longer collects
2833     authors' postal addresses. I am ignoring your postal
2834     code}\unskip\ignorespaces}%
2835     \def\position##1{\unskip\ignorespaces}%
2836     \gdef\@ACM@institution@separator{,}%
2837     \def\institution##1{\unskip\@ACM@institution@separator ##1\gdef\@ACM@institution@separator{ and }}
2838     \def\city##1{\unskip, ##1}%
2839     \def\state##1{\unskip, ##1}%
2840     \renewcommand\department[2][0]{\unskip\@addpunct, ##2}%
2841     \def\country##1{\unskip, ##1}%
2842     \def\and{\unskip; \gdef\@ACM@institution@separator{,}}%
2843     \def\@author##1{##1}%
2844     \def\email##1##2{\unskip, \href{mailto:##2}{##2}}%
2845     \addresses
2846     \egroup}

2847 \AtEndDocument{\if@ACM@nonacm\else\if@ACM@journal
2848     \ifx\@authorsaddresses\@empty
2849     \ClassWarningNoLine{\@classname}{Authors'
2850     addresses are mandatory for ACM journals}%
2851     \fi\fi\fi}

```

\@setaddresses This is an amsart macro that we do not need.

```

2852 \def\@setaddresses{}

```

\@authornotemark Adding a footnote mark to authors. This version adds a “normal” footnote mark. Note that since we typeset the author in a box, we get superfluous box overfull; to protect against it we increase \hfuzz

```

2853 \def\@authornotemark{\g@addto@macro\@currentauthors{%
2854     \advance\hfuzz by 5pt\relax\footnotemark\relax}}

\@authornotemark Adding a footnote mark to authors with a given number
2855 \def\@authornotemark#1{\g@addto@macro\@currentauthors{%
2856     \advance\hfuzz by 5pt\relax\footnotemark[#1]}}

\mkteasers Typesetting the teasers
2857 \def\mkteasers{%
2858     \ifx\@teaserfigures\empty\else
2859     \def\@teaser##1{\par\bigskip\bgroup
2860         \captionsetup{type=figure}##1\egroup\par}
2861     \global\setbox\mktitle@bx=\vbox{\noindent\unvbox\mktitle@bx\par
2862         \noindent\@Description@presentfalse
2863         \@teaserfigures\par\if@Description@present\else
2864             \global\@undescrbed@imagestrue
2865             \ClassWarning{\@classname}{A possible image without
2866                 description}\fi
2867         \medskip}%
2868     \fi}

\mkabstract Typesetting the abstract
2869 \def\mkabstract{\bgroup
2870     \ifx\@abstract\empty\else
2871     {\phantomsection\addcontentsline{toc}{section}{\abstractname}%
2872         \if@ACM@journal
2873             \everypar{\setbox\z@\lastbox\everypar{}}\small
2874         \else
2875             \section*{\abstractname}%
2876             \fi
2877         \ignorespaces\@abstract\par}%
2878     \fi\egroup}

\mktranslatedabstract Typesetting a translated abstract
2879 \def\mktranslatedabstract#1{\selectlanguage{#1}%
2880     \if@ACM@journal
2881         \everypar{\setbox\z@\lastbox\everypar{}}\small
2882     \else
2883         \section*{\abstractname}%
2884     \fi
2885     \ignorespaces}

\mkbibcitation Print the bibcitation format
2886 \def\mkbibcitation{\bgroup
2887     \let\@vspace\@vspace@orig
2888     \let\@vspacer\@vspacer@orig
2889     \def\@pages@word{\ifnum\getrefnumber{TotPages}=1\relax page\else pages\fi}%
2890     \def\footnotemark{}%
2891     \def\{\unskip\} \ignorespaces}%
2892     \def\footnote{\ClassError{\@classname}{Please do not use footnotes
2893         inside a \string\title{} or \string\author{} command! Use
2894         \string\titlenote{} or \string\authornote{} instead!}}%
2895     \def\@article@string{\ifx\@acmArticle\empty{\ } \else,

```

```

2896 Article~\@acmArticle\ \fi}%
2897 \par\medskip\small\noindent{\bfseries ACM Reference Format:}\par\nobreak
2898 \noindent\bgroup
2899 \def\{\{\unskip\}, \ignorespaces}\authors\egroup. \@acmYear. \@title
2900 \ifx\@subtitle\@empty. \else: \@subtitle. \fi
2901 \if@ACM@nonacm\else
2902 % The 'nonacm' option disables 'printacmref' by default,
2903 % and the present \mkbibcitation definition is never used
2904 % in this case. The conditional remains useful if the user
2905 % explicitly sets \settopmatter{printacmref=true}.
2906 \if@ACM@journal@bibstrip
2907 \textit{\@journalNameShort}
2908 \@acmVolume, \@acmNumber \@article@string (\@acmPubDate),
2909 \ref{TotPages}~\@pages@word.
2910 \else
2911 In \textit{\@acmBooktitle}%
2912 \ifx\@acmEditors\@empty\textit{.}\else
2913 \andify\@acmEditors\textit{, }\@acmEditors~\@editorsAbbrev.%
2914 \fi\
2915 ACM, New York, NY, USA%
2916 \@article@string\unskip, \ref{TotPages}~\@pages@word.
2917 \fi
2918 \fi
2919 \ifx\@acmDOI\@empty\else\@formatdoi{\@acmDOI}\fi
2920 \par\egroup}

```

\@printendtopmatter End the top matter

```

2921 \def\@printendtopmatter{%
2922 \let\@vspace\@vspace@orig
2923 \let\@vspacer\@vspacer@orig
2924 \par\bigskip
2925 \let\@vspace\@vspace@acm
2926 \let\@vspacer\@vspacer@acm
2927 }

```

\@setthanks We redefine \setthanks using \long

```

2928 \def\@setthanks{\long\def\thanks##1{\par##1\@addpunct.}\thankses}

```

\@setauthorsaddresses

```

2929 \def\@setauthorsaddresses{\@authorsaddresses\unskip\@addpunct.}

```

\@typesetengagemetadata Typesetting special metadata for ACM Engage

```

2930 \def\@typesetengagemetadata{%
2931 \def\@setengagemetadata##1##2{\par\noindent\textbf{##1} ##2\par}%
2932 \@acmengagemetadata}

```

3.23 Headers and Footers

We use fancyhdr for our headers and footers:

```

2933 \RequirePackage{fancyhdr}

```

\ACM@restore@pagestyle The following code by Ross Moore protects against changes by the totpages package:

```

2934 \let\ACM@ps@plain\ps@plain

```



```

2935 \let\ACM@ps@myheadings\ps@myheadings
2936 \let\ACM@ps@headings\ps@headings
2937 \def\ACM@restore@pagestyle{%
2938   \let\ps@plain\ACM@ps@plain
2939   \let\ps@myheadings\ACM@ps@myheadings
2940   \let\ps@headings\ACM@ps@headings}
2941 \AtBeginDocument{\ACM@restore@pagestyle}

```

\ACM@linecount@bx This is the box displayed in review mode

```

2942 \if@ACM@review
2943   \newsavebox{\ACM@linecount@bx}
2944   \newlength\ACM@linecount@bxht
2945   \newcount\ACM@linecount
2946   \ACM@linecount\@ne\relax
2947   \def\ACM@mk@linecount{%
2948     \savebox{\ACM@linecount@bx}[4em][t]{\parbox[t]{4em}{\normalfont
2949       \normalsize
2950       \setlength{\ACM@linecount@bxht}{0pt}%
2951       \loop{\color{red}\scriptsize\the\ACM@linecount}}\
2952       \global\advance\ACM@linecount by \@ne
2953       \addtolength{\ACM@linecount@bxht}{\baselineskip}%
2954       \ifdim\ACM@linecount@bxht<\textheight\repeat
2955       {\color{red}\scriptsize\the\ACM@linecount}\hfill
2956       \global\advance\ACM@linecount by \@ne}}
2957 \fi

```

\ACM@linecountL How to display the box on the left

```

2958 \def\ACM@linecountL{%
2959   \if@ACM@review
2960     \ACM@mk@linecount
2961     \begin{picture}(\@,0)%
2962       \put(-26,-22){\usebox{\ACM@linecount@bx}}%
2963     \end{picture}%
2964 \fi}

```

\ACM@linecountR How to display the box on the right. In one column formats we do not step the numbers.

```

2965 \def\ACM@linecountR{%
2966   \if@ACM@review
2967     \ifcase\ACM@format@nr
2968       \relax % manuscript
2969       \relax
2970       \or % acmsmall
2971       \relax
2972       \or % acmlarge
2973       \relax
2974       \or % acmtog
2975       \ACM@mk@linecount
2976       \or % sigconf
2977       \ACM@mk@linecount
2978       \or % siggraph
2979       \ACM@mk@linecount
2980       \or % sigplan
2981       \ACM@mk@linecount
2982       \or % sigchi

```

```

2983         \ACM@mk@linecount
2984     \or % sigchi-a
2985         \ACM@mk@linecount
2986     \or % acmengage
2987         \ACM@mk@linecount
2988     \or % acmcp
2989     \relax
2990 \fi
2991 \begin{picture}(0,0)%
2992     \put(20,-22){\usebox{\ACM@linecount@bx}}%
2993 \end{picture}%
2994 \fi}

```

`\ACM@timestamp` The timestamp system

```

2995 \if@ACM@timestamp
2996 % Subtracting 30 from \time gives us the effect of rounding down despite
2997 % \numexpr rounding to nearest
2998 \newcounter{ACM@time@hours}
2999 \setcounter{ACM@time@hours}{\numexpr (\time - 30) / 60 \relax}
3000 \newcounter{ACM@time@minutes}
3001 \setcounter{ACM@time@minutes}{\numexpr \time - \theACM@time@hours * 60 \relax}
3002 \newcommand\ACM@timestamp{%
3003     \footnotesize%
3004     \ifx\@acmSubmissionID\@empty\relax\else
3005     Submission ID: \@acmSubmissionID.{ }%
3006     \fi
3007     \the\year-\two@digits{\the\month}-\two@digits{\the\day}{ }%
3008     \two@digits{\theACM@time@hours}:\two@digits{\theACM@time@minutes}{. }%
3009     Page \thepage\ of \@startPage--\pageref*{TotPages}.%
3010 }
3011 \fi

```

`\@shortauthors` Even if the author redefined `\shortauthors`, we do not print it in the headers when in anonymous mode:

```

3012 \def\@shortauthors{%
3013     \if@ACM@anonymous
3014     Anon.
3015     \ifx\@acmSubmissionID\@empty\else Submission Id: \@acmSubmissionID\fi
3016     \else\shortauthors\fi}

```

`\@headfootfont` The font to typeset header and footer text.

```

3017 \def\@headfootfont{\sffamily\footnotesize}

```

`standardpagestyle` The page style for all pages but the first one The page style for all pages but the first one

```

3018 \AtBeginDocument{%
3019 \fancypagestyle{standardpagestyle}{%
3020     \fancyhf{}%
3021     \renewcommand{\headrulewidth}{\z@}%
3022     \renewcommand{\footrulewidth}{\z@}%
3023     \def\@acmArticlePage{%
3024         \ifx\@acmArticle\empty%
3025         \if@ACM@printfolios\thepage\fi%

```

```

3026 \else%
3027 \acmArticle\if@ACM@printfolios:\thepage\fi%
3028 \fi%
3029 }%
3030 \if@ACM@journal@bibstrip@or@tog
3031 \ifcase\ACM@format@nr
3032 \relax % manuscript
3033 \fancyhead[LE]{\ACM@linecountL\if@ACM@printfolios\thepage\fi}%
3034 \fancyhead[RO]{\if@ACM@printfolios\thepage\fi}%
3035 \fancyhead[RE]{\@shortauthors}%
3036 \fancyhead[LO]{\ACM@linecountL\shorttitle}%
3037 \if@ACM@nonacm\else%
3038 \fancyfoot[RO,LE]{\footnotesize Manuscript submitted to ACM}
3039 \fi%
3040 \or % acmsmall
3041 \fancyhead[LE]{\ACM@linecountL\@headfootfont\@acmArticlePage}%
3042 \fancyhead[RO]{\@headfootfont\@acmArticlePage}%
3043 \fancyhead[RE]{\@headfootfont\@shortauthors}%
3044 \fancyhead[LO]{\ACM@linecountL\@headfootfont\shorttitle}%
3045 \if@ACM@nonacm\else%
3046 \fancyfoot[RO,LE]{\footnotesize \@journalNameShort, Vol. \@acmVolume, No.
3047 \@acmNumber, Article \@acmArticle. Publication date: \@acmPubDate.}%
3048 \fi
3049 \or % acmlarge
3050 \fancyhead[LE]{\ACM@linecountL\@headfootfont
3051 \@acmArticlePage\quad\textbullet\quad\@shortauthors}%
3052 \fancyhead[LO]{\ACM@linecountL}%
3053 \fancyhead[RO]{\@headfootfont
3054 \shorttitle\quad\textbullet\quad\@acmArticlePage}%
3055 \if@ACM@nonacm\else%
3056 \fancyfoot[RO,LE]{\footnotesize \@journalNameShort, Vol. \@acmVolume, No.
3057 \@acmNumber, Article \@acmArticle. Publication date: \@acmPubDate.}%
3058 \fi
3059 \or % acmtog
3060 \fancyhead[LE]{\ACM@linecountL\@headfootfont
3061 \@acmArticlePage\quad\textbullet\quad\@shortauthors}%
3062 \fancyhead[LO]{\ACM@linecountL}%
3063 \fancyhead[RE]{\ACM@linecountR}%
3064 \fancyhead[RO]{\@headfootfont
3065 \shorttitle\quad\textbullet\quad\@acmArticlePage\ACM@linecountR}%
3066 \if@ACM@nonacm\else
3067 \if@ACM@journal@bibstrip
3068 \fancyfoot[RO,LE]{\footnotesize \@journalNameShort,
3069 Vol. \@acmVolume, No.\@acmNumber, Article \@acmArticle.
3070 Publication date: \@acmPubDate.}%
3071 \else
3072 \fancyfoot[RO,LE]{\footnotesize \acmConference@shortname,
3073 \acmConference@date, \acmConference@venue.}%
3074 \fi
3075 \fi
3076 \else % Proceedings
3077 \fancyfoot[C]{\if@ACM@printfolios\footnotesize\thepage\fi}%
3078 \fancyhead[LO]{\ACM@linecountL\@headfootfont\shorttitle}%
3079 \fancyhead[RE]{\@headfootfont\@shortauthors\ACM@linecountR}%

```

```

3080 \if@ACM@nonacm
3081 \fancyhead[LE]{\ACM@linecountL}%
3082 \fancyhead[RO]{\ACM@linecountR}%
3083 \else%
3084 \if@ACM@engage
3085 \fancyhead[LE]{\ACM@linecountL\@headfootfont\footnotesize
3086 EngageCSEdu. \ifx\@acmDOI\@empty\else\@formatdoi{\@acmDOI}\fi}%
3087 \fancyhead[RO]{\@headfootfont
3088 EngageCSEdu. \ifx\@acmDOI\@empty\else\@formatdoi{\@acmDOI}\fi
3089 \ACM@linecountR}%
3090 \else
3091 \fancyhead[LE]{\ACM@linecountL\@headfootfont\footnotesize
3092 \acmConference@shortname,
3093 \acmConference@date, \acmConference@venue}%
3094 \fancyhead[RO]{\@headfootfont
3095 \acmConference@shortname,
3096 \acmConference@date, \acmConference@venue\ACM@linecountR}%
3097 \fi
3098 \fi
3099 \fi
3100 \else % Proceedings
3101 \fancyfoot[C]{\if@ACM@printfolios\footnotesize\thepage\fi}%
3102 \fancyhead[LO]{\ACM@linecountL\@headfootfont\shorttitle}%
3103 \fancyhead[RE]{\@headfootfont\@shortauthors\ACM@linecountR}%
3104 \if@ACM@nonacm
3105 \fancyhead[LE]{\ACM@linecountL}%
3106 \fancyhead[RO]{\ACM@linecountR}%
3107 \else%
3108 \if@ACM@engage
3109 \fancyhead[LE]{\ACM@linecountL\@headfootfont
3110 EngageCSEdu. \ifx\@acmDOI\@empty\else\@formatdoi{\@acmDOI}\fi}%
3111 \fancyhead[RO]{\@headfootfont
3112 EngageCSEdu. \ifx\@acmDOI\@empty\else\@formatdoi{\@acmDOI}\fi
3113 \ACM@linecountR}%
3114 \else
3115 \fancyhead[LE]{\ACM@linecountL\@headfootfont
3116 \acmConference@shortname,
3117 \acmConference@date, \acmConference@venue}%
3118 \fancyhead[RO]{\@headfootfont
3119 \acmConference@shortname,
3120 \acmConference@date, \acmConference@venue\ACM@linecountR}%
3121 \fi
3122 \fi
3123 \fi
3124 \if@ACM@sigchiamode
3125 \fancyheadoffset[L]{\dimexpr(\marginparsep+\marginparwidth)}%
3126 \fi
3127 \if@ACM@timestamp
3128 \fancyfoot[LO,RE]{\ACM@timestamp}
3129 \fi
3130 \if@ACM@acmcp
3131 \renewcommand{\footrulewidth}{0.1\p@}%
3132 \fancyheadoffset[L]{46pt}%
3133 \fancyhead[L]{\makebox[\z@][l]{%

```

```

3134 \raisebox{-\dimexpr(0.2\textheight*(\ACM@ArticleType@nr-2))}{%
3135 \rotatebox{90}{\colorbox{@ACM@Article@color}{\color{white}%
3136 \strut\ACM@ArticleType~Article}}}%
3137 \ACM@linecountL}%
3138 \fancyhead[R]{\makebox[\z@][r]{\box\@ACM@acmcpbox}}%
3139 \fancyfoot[L,C]{}%
3140 \fancyfoot[R]{\footnotesize
3141 \@journalName, Volume~\@acmVolume, Issue~\@acmNumber,
3142 \ifx\@acmArticle\@empty\else Article~\@acmArticle\fi\space
3143 (\@acmPubDate)\ifx\@acmDOI\@empty\else\\@formatdoi{\@acmDOI}\fi}
3144 \fi
3145 }%
3146 \pagestyle{standardpagestyle}
3147 }

```

\@folio@wd Folio blob width, height, offsets and max number

\@folio@ht

\@folio@blob The macro to typeset the folio blob.

\@folio@max
firstpagestyle The page style for the first page only.

```

3148 \AtBeginDocument{%
3149 \fancypagestyle{firstpagestyle}{%
3150 \fancyhf{}%
3151 \renewcommand{\headrulewidth}{\z@}%
3152 \renewcommand{\footrulewidth}{\z@}%
3153 \if@ACM@journal@bibstrip@or@tog
3154 \ifcase\ACM@format@nr
3155 \relax % manuscript
3156 \fancyhead[L]{\ACM@linecountL\@acmBadgeL}%
3157 \fancyhead[R]{\@acmBadgeR}%
3158 \fancyfoot[RO,LE]{\if@ACM@printfolios\small\thepage\fi}%
3159 \if@ACM@nonacm\else%
3160 \fancyfoot[RE,LO]{\footnotesize Manuscript submitted to ACM}%
3161 \fi%
3162 \or % acmsmall
3163 \if@ACM@nonacm\else%
3164 \fancyfoot[RO,LE]{\footnotesize \@journalNameShort, Vol. \@acmVolume, No.
3165 \@acmNumber, Article \@acmArticle. Publication date:
3166 \@acmPubDate.}%
3167 \fi%
3168 \fancyhead[LE]{\ACM@linecountL\@acmBadgeL}%
3169 \fancyhead[LO]{\ACM@linecountL\@acmBadgeL}%
3170 \fancyhead[RO]{\@acmBadgeR}%
3171 \fancyhead[RE]{\@acmBadgeR}%
3172 \or % acmlarge
3173 \if@ACM@nonacm\else%
3174 \fancyfoot[RO,LE]{\footnotesize \@journalNameShort, Vol. \@acmVolume, No.
3175 \@acmNumber, Article \@acmArticle. Publication date:
3176 \@acmPubDate.}%
3177 \fi%
3178 \fancyhead[RO]{\@acmBadgeR}%
3179 \fancyhead[RE]{\@acmBadgeR}%
3180 \fancyhead[LE]{\ACM@linecountL\@acmBadgeL}%
3181 \fancyhead[LO]{\ACM@linecountL\@acmBadgeL}%

```

```

3182 \or % acmtog
3183 \if@ACM@nonacm\else%
3184 \if@ACM@journal@bibstrip
3185 \fancyfoot[RO,LE]{\footnotesize \@journalNameShort,
3186 Vol. \@acmVolume, No.\@acmNumber, Article \@acmArticle.
3187 Publication date: \@acmPubDate.}%
3188 \else
3189 \fancyfoot[RO,LE]{\footnotesize \acmConference@shortname,
3190 \acmConference@date, \acmConference@venue.}%
3191 \fi
3192 \fi%
3193 \fancyhead[L]{\ACM@linecountL\@acmBadgeL}%
3194 \fancyhead[R]{\@acmBadgeR\ACM@linecountR}%
3195 \else % Conference proceedings
3196 \fancyhead[L]{\ACM@linecountL\@acmBadgeL}%
3197 \fancyhead[R]{\@acmBadgeR\ACM@linecountR}%
3198 \fancyfoot[C]{\if@ACM@printfolios\footnotesize\thepage\fi}%
3199 \fi
3200 \else
3201 \fancyhead[L]{\ACM@linecountL\@acmBadgeL}%
3202 \fancyhead[R]{\@acmBadgeR\ACM@linecountR}%
3203 \fancyfoot[C]{\if@ACM@printfolios\footnotesize\thepage\fi}%
3204 \fi
3205 \if@ACM@timestamp
3206 \ifnum\ACM@format@nr=0\relax % Manuscript
3207 \fancyfoot[LO,RE]{\ACM@timestamp\quad
3208 \if@ACM@nonacm\else
3209 \footnotesize Manuscript submitted to ACM
3210 \fi}
3211 \else
3212 \fancyfoot[LO,RE]{\ACM@timestamp}
3213 \fi
3214 \fi
3215 \if@ACM@acmcp
3216 \renewcommand{\footrulewidth}{0.1\p@}%
3217 \fancyheadoffset[L]{46pt}%
3218 \fancyhead[L]{\makebox[\z@][l]{%
3219 \raisebox{-\dimexpr(0.2\textheight*(\ACM@ArticleType@nr-2))}{%
3220 \rotatebox{90}{\colorbox{@ACM@Article@color}{\color{white}%
3221 \strut\ACM@ArticleType~Article}}}%
3222 \ACM@linecountL\@acmBadgeL}%
3223 \fancyhead[R]{\@acmBadgeR\makebox[\z@][r]{\box\@ACM@acmcpbox}}%
3224 \fancyfoot[L,C]{}%
3225 \fancyfoot[R]{\footnotesize
3226 \@journalName, Volume~\@acmVolume, Issue~\@acmNumber,
3227 \ifx\@acmArticle\@empty\else Article~\@acmArticle\fi\space
3228 (\@acmPubDate)\ifx\@acmDOI\@empty\else\\@formatdoi{\@acmDOI}\fi}
3229 \fi
3230 }}

```

3.24 Sectioning

\ACM@Nadjust Ross Moore's macro.

Initially `\ACM@sect@format@` just passes through its #1 argument unchanged. When a sectioning macro such as `\section` occurs, the `\ACM@NRadjust` redefines `\ACM@sect@format@` to expand into the styling commands; e.g., `\@secfont`. Then `\Sectionformat` is temporarily redefined to produce `\realSectionformat{\ACM@sect@format{#1}}{#2}`, which ultimately becomes `\Sectionformat{\@secfont{#1}}{#2}`. This is done to allow `\Sectionformat` to be variable, as may be needed in particular circumstances. Its current expansion is saved as `\realSectionformat` which is used with the modified 1st argument, then `\Sectionformat` is reverted to its former expansion.

To style the section-number, the line `#1{}` is used at the end of `\ACM@NRadjust`. This imposes style-changing commands for fonts and sizes, but any `\MakeUppercase` acts on just the `{}`. It's a bit hacky, by essentially assuming that only numbers occur here, not letters to be case-changed. (This should be acceptable in a class-file, as it ought to be possible to change the case elsewhere, if that was ever desired.) Also, it can result in `\@adddotafter` being called too soon, so a slight adjustment is made, via `\ACM@adddotafter`, which now tests whether its argument is empty. But the coding has to allow for other packages to have also patched `\@adddotafter`.

```

3231 \def\ACM@NRadjust#1{%
3232   \begingroup
3233   \expandafter\ifx\csname Sectionformat\endcsname\relax
3234   % do nothing when \Sectionformat is unknown
3235   \def\next{\endgroup #1}%
3236   \else
3237   \def\next{\endgroup
3238     \let\realSectionformat\Sectionformat
3239     \def\ACM@sect@format@{#1}%
3240     \let\Sectionformat\ACM@NR@adjustedSectionformat
3241     %% next lines added 2018-06-17 to ensure section number is styled
3242     \let\real@adddotafter\@adddotafter
3243     \let\@adddotafter\ACM@adddotafter
3244     #1{ }% imposes the styles, but nullifies \MakeUppercase
3245     \let\@adddotafter\real@adddotafter
3246   }%
3247   \fi \next
3248 }
```

`\ACM@NR@adjustedSectionformat`

```

3249 \def\ACM@NR@adjustedSectionformat#1#2{%
3250   \realSectionformat{\ACM@sect@format{#1}}{#2}%
3251   \let\Sectionformat\realSectionformat
3252   \DeclareRobustCommand{\ACM@sect@format@}{\ACM@sect@format@}
3253   \def\ACM@sect@format@null#1{#1}
3254   \let\ACM@sect@format@\ACM@sect@format@null
3255   \AtBeginDocument{%
3256     \expandafter\ifx\csname LTX@adddotafter\endcsname\relax
3257     \let\LTX@adddotafter\@adddotafter
3258   \fi
3259 }
```

`\ACM@adddotafter`

```

3260 \def\ACM@adddotafter#1{\ifx\relax#1\relax\else\LTX@adddotafter{#1}\fi}
```

Sectioning is different for different levels

```

3261 \renewcommand\section{\def\@toclevel{1}%
3262 \@startsection{section}{1}{\z@}%
3263 {- .75\baselineskip \@plus -2\p@ \@minus -.2\p@}%
3264 {.25\baselineskip}%
3265 {\ACM@NRadjust\@secfont}}
3266 \renewcommand\subsection{\def\@toclevel{2}%
3267 \@startsection{subsection}{2}{\z@}%
3268 {- .75\baselineskip \@plus -2\p@ \@minus -.2\p@}%
3269 {.25\baselineskip}%
3270 {\ACM@NRadjust\@subsecfont}}
3271 \renewcommand\subsubsection{\def\@toclevel{3}%
3272 \@startsection{subsubsection}{3}{\z@}%
3273 {- .5\baselineskip \@plus -2\p@ \@minus -.2\p@}%
3274 {-3.5\p@}%
3275 {\ACM@NRadjust{\@subsubsecfont\@adddotafter}}}}
3276 \renewcommand\paragraph{\def\@toclevel{4}%
3277 \@startsection{paragraph}{4}{\parindent}%
3278 {- .5\baselineskip \@plus -2\p@ \@minus -.2\p@}%
3279 {-3.5\p@}%
3280 {\ACM@NRadjust{\@parfont\@adddotafter}}}}
3281 \newcommand\noindentparagraph{\def\@toclevel{4}%
3282 \@startsection{paragraph}{4}{\z@}%
3283 {- .5\baselineskip \@plus -2\p@ \@minus -.2\p@}%
3284 {-3.5\p@}%
3285 {\ACM@NRadjust{\@parfont}}}}
3286 \renewcommand\part{\def\@toclevel{9}%
3287 \@startsection{part}{9}{\z@}%
3288 {-10\p@ \@plus -4\p@ \@minus -2\p@}%
3289 {4\p@}%
3290 {\ACM@NRadjust\@parfont}}

```

\section@raggedright Special version of \raggedright compatible with \MakeUppercase

```

3291 \def\section@raggedright{\@rightskip\@flushglue
3292 \rightskip\@rightskip
3293 \leftskip\z@skip
3294 \parindent\z@}

```

\@secfont Fonts for sections etc. are different for different formats.

```

\@subsecfont 3295 \def\@secfont{\sffamily\bfseries\section@raggedright}
\@subsubsecfont 3296 \def\@subsecfont{\sffamily\bfseries\section@raggedright}
\@parfont 3297 \def\@subsubsecfont{\sffamily\itshape}
3298 \def\@parfont{\itshape}
3299 \setcounter{secnumdepth}{3}
3300 \ifcase\ACM@format@nr
3301 \relax % manuscript
3302 \or % acmsmall
3303 \or % acmlarge
3304 \def\@secfont{\sffamily\large\section@raggedright}
3305 \def\@subsecfont{\sffamily\large\section@raggedright}
3306 \or % acmtog
3307 \def\@secfont{\sffamily\large\section@raggedright}
3308 \def\@subsecfont{\sffamily\large\section@raggedright}
3309 \or % sigconf

```



```

3310 \def\@secfont{\bfseries\Large\section@raggedright}
3311 \def\@subsecfont{\bfseries\Large\section@raggedright}
3312 \or % siggraph
3313 \def\@secfont{\sffamily\bfseries\Large\section@raggedright}
3314 \def\@subsecfont{\sffamily\bfseries\Large\section@raggedright}
3315 \or % sigplan
3316 \def\@secfont{\bfseries\Large\section@raggedright}
3317 \def\@subsecfont{\bfseries\section@raggedright}
3318 \def\@subsubsecfont{\bfseries\section@raggedright}
3319 \def\@parfont{\bfseries\itshape}
3320 \def\@subparfont{\itshape}
3321 \or % sigchi
3322 \setcounter{secnumdepth}{1}
3323 \def\@secfont{\sffamily\bfseries\section@raggedright}
3324 \def\@subsecfont{\sffamily\bfseries\section@raggedright}
3325 \or % sigchi-a
3326 \setcounter{secnumdepth}{0}
3327 \def\@secfont{\sffamily\bfseries\section@raggedright}
3328 \def\@subsecfont{\sffamily\bfseries\section@raggedright}
3329 \or % acmengage
3330 \def\@secfont{\bfseries\Large\section@raggedright}
3331 \def\@subsecfont{\bfseries\Large\section@raggedright}
3332 \or %acmcp
3333 \fi

```

Amasat does not define tocindent for paragraphs and subparagraphs. Here we add them

```

3334 \AtBeginDocument{%
3335   \@for\@tempa:=-1,0,1,2,3,4,5\do{%
3336     \ifundefined{r@tocindent\@tempa}{%
3337       \exp\gdef\csname r@tocindent\@tempa\endcsname{0pt}}{}%
3338   }%
3339 }
3340 \def\@writetocindents{%
3341   \begingroup
3342   \@for\@tempa:=-1,0,1,2,3,4,5\do{%
3343     \immediate\write\@auxout{%
3344       \string\newlabel{tocindent\@tempa}{%
3345         \csname r@tocindent\@tempa\endcsname}}%
3346   }%
3347   \endgroup}

```

\@adddotafter Add punctuation after a sectioning command

```

3348 \def\@adddotafter#1{#1\@addpunct{.}}

```

\@addspaceafter Add space after a sectioning command

```

3349 \def\@addspaceafter#1{#1\@addpunct{\enspace}}

```

```

3350 \if@ACM@acmcp
3351   \setcounter{secnumdepth}{-1}%
3352 \fi

```

3.25 TOC lists

`\@dotsep` Related to the `\tableofcontents` are all the horizontal fillers. Base $\LaTeX$ defines `\@dottedtocline`, which we should not disable. Yet, this command expects `\@dotsep` to be defined but leaves this to the class implementation. Since `amsart` does not provide this, we copy the standard variant from `article` here.

```
3353 \providecommand*\@dotsep{4.5}
```

3.26 Theorems

`\@acmplainbodyfont` The font to typeset the body of the `acmplain` theorem style.

```
3354 \def\@acmplainbodyfont{\itshape}
```

`\@acmplainindent` The amount to indent the `acmplain` theorem style.

```
3355 \def\@acmplainindent{\parindent}
```

`\@acmplainheadfont` The font to typeset the head of the `acmplain` theorem style.

```
3356 \def\@acmplainheadfont{\scshape}
```

`\@acmplainnotefont` The font to typeset the note of the `acmplain` theorem style.

```
3357 \def\@acmplainnotefont{\@empty}
```

Customization of the `acmplain` theorem style:

```
3358 \ifcase\ACM@format@nr
3359 \relax % manuscript
3360 \or % acmsmall
3361 \or % acmlarge
3362 \or % acmtog
3363 \or % sigconf
3364 \or % siggraph
3365 \or % sigplan
3366 \def\@acmplainbodyfont{\itshape}
3367 \def\@acmplainindent{\z@}
3368 \def\@acmplainheadfont{\bfseries}
3369 \def\@acmplainnotefont{\normalfont}
3370 \or % sigchi
3371 \or % sigchi-a
3372 \or % acmengage
3373 \or % acmcp
3374 \fi
```

`acmplain` The `acmplain` theorem style

```
3375 \newtheoremstyle{acmplain}%
3376 { .5\baselineskip \@plus .2\baselineskip
3377 \@minus .2\baselineskip}% space above
3378 { .5\baselineskip \@plus .2\baselineskip
3379 \@minus .2\baselineskip}% space below
3380 {\@acmplainbodyfont}% body font
3381 {\@acmplainindent}% indent amount
3382 {\@acmplainheadfont}% head font
3383 { .}% punctuation after head
3384 { .5em}% spacing after head
3385 {\thmname{#1}\thmnumber{ #2}\thmnote{ {\@acmplainnotefont{#3}}}}% head spec
```

`\acmdefinitionbodyfont` The font to typeset the body of the `acmdefinition` theorem style.

```
3386 \def\acmdefinitionbodyfont{\normalfont}
```

`\acmdefinitionindent` The amount to indent the `acmdefinition` theorem style.

```
3387 \def\acmdefinitionindent{\parindent}
```

`\acmdefinitionheadfont` The font to typeset the head of the `acmdefinition` theorem style.

```
3388 \def\acmdefinitionheadfont{\itshape}
```

`\acmdefinitionnotefont` The font to typeset the note of the `acmdefinition` theorem style.

```
3389 \def\acmdefinitionnotefont{\@empty}
```

Customization of the `acmdefinition` theorem style:

```
3390 \ifcase\ACM@format@nr
```

```
3391 \relax % manuscript
```

```
3392 \or % acmsmall
```

```
3393 \or % acmlarge
```

```
3394 \or % acmtog
```

```
3395 \or % sigconf
```

```
3396 \or % siggraph
```

```
3397 \or % sigplan
```

```
3398 \def\acmdefinitionbodyfont{\normalfont}
```

```
3399 \def\acmdefinitionindent{\z@}
```

```
3400 \def\acmdefinitionheadfont{\bfseries}
```

```
3401 \def\acmdefinitionnotefont{\normalfont}
```

```
3402 \or % sigchi
```

```
3403 \or % sigchi-a
```

```
3404 \or % acmengage
```

```
3405 \or % acmcp
```

```
3406 \fi
```

`acmdefinition` The `acmdefinition` theorem style

```
3407 \newtheoremstyle{acmdefinition}%
```

```
3408 { .5\baselineskip \@plus.2\baselineskip
```

```
3409 \@minus.2\baselineskip}% space above
```

```
3410 { .5\baselineskip \@plus.2\baselineskip
```

```
3411 \@minus.2\baselineskip}% space below
```

```
3412 {\acmdefinitionbodyfont}% body font
```

```
3413 {\acmdefinitionindent}% indent amount
```

```
3414 {\acmdefinitionheadfont}% head font
```

```
3415 { .}% punctuation after head
```

```
3416 { .5em}% spacing after head
```

```
3417 {\thmname{#1}\thmnumber{ #2}\thmnote{ {\acmdefinitionnotefont{#3}}}}% head spec
```

Make `acmplain` the default theorem style.

```
3418 \theoremstyle{acmplain}
```

Delay defining the theorem environments until after other packages have been loaded. In particular, the `cleveref` package must be loaded before the theorem environments are defined in order to show the correct environment name (see <https://github.com/borisveytsman/acmart/issues/138>). The `acmthm` option is used to suppress the definition of any theorem environments. Also, to avoid obscure errors arising from these environment definitions conflicting with environments defined by

the user or by user-loaded packages, we only define environments that have not yet been defined.

```

3419 \AtEndPreamble{%
3420   \if@ACM@acmthm
3421   \theoremstyle{acmplain}
3422   \@ifundefined{theorem}{%
3423     \newtheorem{theorem}{Theorem}[section]
3424   }{}
3425   \@ifundefined{conjecture}{%
3426     \newtheorem{conjecture}[theorem]{Conjecture}
3427   }{}
3428   \@ifundefined{proposition}{%
3429     \newtheorem{proposition}[theorem]{Proposition}
3430   }{}
3431   \@ifundefined{lemma}{%
3432     \newtheorem{lemma}[theorem]{Lemma}
3433   }{}
3434   \@ifundefined{corollary}{%
3435     \newtheorem{corollary}[theorem]{Corollary}
3436   }{}
3437   \theoremstyle{acmdefinition}
3438   \@ifundefined{example}{%
3439     \newtheorem{example}[theorem]{Example}
3440   }{}
3441   \@ifundefined{definition}{%
3442     \newtheorem{definition}[theorem]{Definition}
3443   }{}
3444   \fi
3445   \theoremstyle{acmplain}
3446 }
```

\@proofnamefont The font to typeset the proof name.

```

3447 \def\@proofnamefont{\scshape}
```

\@proofindent Whether or not to indent proofs.

```

3448 \def\@proofindent{\indent}
```

Customization of the proof environment.

```

3449 \ifcase\ACM@format@nr
3450 \relax % manuscript
3451 \or % acmsmall
3452 \or % acmlarge
3453 \or % acmtog
3454 \or % sigconf
3455 \or % siggraph
3456 \or % sigplan
3457   \def\@proofnamefont{\itshape}
3458   \def\@proofindent{\noindent}
3459 \or % sigchi
3460 \or % sigchi-a
3461 \or % acmengage
3462 \or % acmcp
3463 \fi
```

proof We want some customization of the proof environment.

```

3464 \renewenvironment{proof}[1][\proofname]{\par
3465   \pushQED{\qed}%
3466   \normalfont \topsep6\p@\plus6\p@\relax
3467   \trivlist
3468   \item[\@proofindent\hskip\labelsep
3469     {\@proofnamefont #1\@addpunct{.}}]\ignorespaces
3470 }{%
3471   \popQED\endtrivlist\endpfalse
3472 }

```

3.27 Balancing columns

We need balancing only if the user did not disable it, and we use a two column format. Note that pbalance uses a different mechanism.

```

3473 \AtEndPreamble{%
3474   \if@ACM@pbalance
3475     \global\@ACM@balancefalse
3476     \ifcase\ACM@format@nr
3477       \relax % manuscript
3478       \or % acmsmall
3479       \or % acmlarge
3480       \or % acmtog
3481         \RequirePackage{pbalance}%
3482       \or % sigconf
3483         \RequirePackage{pbalance}%
3484       \or % siggraph
3485         \RequirePackage{pbalance}%
3486       \or % sigplan
3487         \RequirePackage{pbalance}%
3488       \or % sigchi
3489         \RequirePackage{pbalance}%
3490       \or % sigchi-a
3491       \or % acmengage
3492         \RequirePackage{pbalance}%
3493       \or % acmcp
3494     \fi
3495   \fi
3496   \if@ACM@balance
3497     \ifcase\ACM@format@nr
3498       \relax % manuscript
3499       \global\@ACM@balancefalse
3500       \or % acmsmall
3501       \global\@ACM@balancefalse
3502       \or % acmlarge
3503       \global\@ACM@balancefalse
3504       \or % acmtog
3505         \RequirePackage{balance}%
3506       \or % sigconf
3507         \RequirePackage{balance}%
3508       \or % siggraph
3509         \RequirePackage{balance}%
3510       \or % sigplan

```

```

3511         \RequirePackage{balance}%
3512     \or % sigchi
3513         \RequirePackage{balance}%
3514     \or % sigchi-a
3515         \global\@ACM@balancefalse
3516     \or % acmengage
3517         \RequirePackage{balance}%
3518     \or % acmcp
3519         \global\@ACM@balancefalse
3520 \fi
3521 \fi
3522 }
3523 \AtEndDocument{%
3524     \if@ACM@balance
3525     \if@twocolumn
3526     \balance
3527     \fi\fi}

```

3.28 Acknowledgments

\acksname

```

3528 \newcommand\acksname{Acknowledgments}

```

\acks This is a comment-like structure

```

3529 \specialcomment{acks}{%
3530     \begin{group}
3531     \section*{\acksname}
3532     \phantomsection\addcontentsline{toc}{section}{\acksname}
3533 }{%
3534     \end{group}
3535 }

```

\grantsponsor We just typeset the name of the sponsor

```

3536 \def\grantsponsor#1#2#3{#2}

```

\grantnum

```

3537 \newcommand\grantnum[3][{}]{#3%
3538     \def\@tempa{#1}\ifx\@tempa\@empty\else\space(\url{#1})\fi}

```

3.29 Conditional typesetting

We use the comment package for conditional typesetting:

```

3539 \AtEndPreamble{%
3540     \if@ACM@screen
3541     \includecomment{screenonly}
3542     \excludecomment{printonly}
3543 \else
3544     \excludecomment{screenonly}
3545     \includecomment{printonly}
3546 \fi
3547 \if@ACM@anonymous
3548     \excludecomment{anonsuppress}
3549     \excludecomment{acks}

```

```

3550 \else
3551   \includecomment{anonsuppress}
3552 \fi}

```

3.30 Additional bibliography commands

`\showeprint` The command `\showeprint` has two arguments: the (optional) prefix and the eprint number. Right now the only prefix we understand is the (lowercase) word ‘arxiv’.

```

3553 \newcommand\showeprint[2][arxiv]{%
3554   \def\@tempa{#1}%
3555   \ifx\@tempa\empty\def\@tempa{arxiv}\fi
3556   \def\@tempb{arxiv}%
3557   \ifx\@tempa\@tempb\relax
3558     arXiv:\href{https://arxiv.org/abs/#2}{#2}%
3559   \else
3560     \def\@tempb{arXiv}%
3561     \ifx\@tempa\@tempb\relax
3562       arXiv:\href{https://arxiv.org/abs/#2}{#2}%
3563     \else
3564       #1:#2%
3565     \fi
3566 \fi}

```

`\showISBNx` The bibliography style calls `\showISBNx` for a 10-digit ISBN and `\showISBNxiii` for a 13-digit one.

```

3567 \newcommand\showISBNx[1]{ISBN~#1.}
3568 \newcommand\showISBNxiii[1]{ISBN-13~#1.}

```

3.31 Index

`\theindex` Standard amsart uses plain page style in index pages. This clashes with line numbering for review option, so we redefine it.

```

3569 \def\theindex{\@restonecoltrue\if@twocolumn\@restonecolfalse\fi
3570   \columnseprule\z@ \columnsep 35\p@
3571   \@indextitlestyle
3572   \let\item\@idxitem
3573   \parindent\z@ \parskip\z@\@plus.3\p@\relax
3574   \raggedright
3575   \hyphenpenalty\@M
3576   \footnotesize}

```

3.32 End of Class

`\vspace` We do not disable `\vspace`, but add warning to it

```

3577 \let\vspace@orig=\vspace
3578 \let\vspacer@orig=\vspacer
3579 \apptocmd{\vspace}{\ClassWarning{\@classname}{\string\vspace\space should
3580   only be used to provide space above/below surrounding
3581   objects}}{}{}
3582 \apptocmd{\vspacer}{\ClassWarning{\@classname}{\string\vspace\space should
3583   only be used to provide space above/below surrounding
3584   objects}}{}{}

```

```

3585 \let\@vspace@acm=\@vspace
3586 \let\@vspacer@acm=\@vspacer

```

\ACM@origbaselinestretch We produce an error if the user tries to change \baselinestretch

```

3587 \let\ACM@origbaselinestretch\baselinestretch
3588 \AtEndDocument{\ifx\baselinestretch\ACM@origbaselinestretch\else
3589   \ClassError{\@classname}{An attempt to redefine
3590     \string\baselinestretch\space detected. Please do not do this for
3591     ACM submissions!}\fi}

```

\ACM@origsection We produce an error if the user tries to change \section

```

3592 \let\ACM@origsection\section
3593 \AtEndDocument{\ifx\section\ACM@origsection\else
3594   \ClassError{\@classname}{An attempt to redefine
3595     \string\section\space detected. Please do not do this for
3596     ACM submissions!}\fi}

```

\ACM@origsubsection We produce an error if the user tries to change \subsection

```

3597 \let\ACM@origsubsection\subsection
3598 \AtEndDocument{\ifx\subsection\ACM@origsubsection\else
3599   \ClassError{\@classname}{An attempt to redefine
3600     \string\subsection\space detected. Please do not do this for
3601     ACM submissions!}\fi}

```

\ACM@origsubsubsection We produce an error if the user tries to change \subsubsection

```

3602 \let\ACM@origsubsubsection\subsubsection
3603 \AtEndDocument{\ifx\subsubsection\ACM@origsubsubsection\else
3604   \ClassError{\@classname}{An attempt to redefine
3605     \string\subsubsection\space detected. Please do not do this for
3606     ACM submissions!}\fi}

```

\ACM@origparagraph We produce an error if the user tries to change \paragraph

```

3607 \let\ACM@origparagraph\paragraph
3608 \AtEndDocument{\ifx\paragraph\ACM@origparagraph\else
3609   \ClassError{\@classname}{An attempt to redefine
3610     \string\paragraph\space detected. Please do not do this for
3611     ACM submissions!}\fi}

```

```

3612 \normalsize\normalfont\frenchspacing
3613 </class>

```


References

- [1] UK $\text{\TeX}$ Users Group. UK list of $\text{\TeX}$ frequently asked questions. <https://texfaq.org>, 2019.
- [2] Michael Downes and Barbara Beeton. *The amsart, amsproc, and amsbook document classes*. American Mathematical Society, August 2004. <http://www.ctan.org/pkg/amslatex>.
- [3] Johannes L. Braams and Javier Bezos. *Babel*, 2022. <http://www.ctan.org/pkg/babel>.
- [4] Cristophe Fiorio. *algorithm2e.sty—package for algorithms*, October 2015. <http://www.ctan.org/pkg/algorithm2e>.
- [5] Rogério Brito. *The algorithms bundle*, August 2009. <http://www.ctan.org/pkg/algorithms>.
- [6] Carsten Heinz, Brooks Moses, and Jobst Hoffmann. *The Listings Package*, June 2015. <http://www.ctan.org/pkg/listings>.
- [7] Simon Fear. *Publication quality tables in $\text{\LaTeX}$* , April 2005. <http://www.ctan.org/pkg/booktabs>.
- [8] Association for Computing Machinery. *ACM Visual Identity Standards*, 2007. <http://identitystandards.acm.org>.
- [9] Axel Sommerfeldt. *The subcaption package*, April 2013. <http://www.ctan.org/pkg/subcaption>.
- [10] Boris Veytsman, Bern Schandl, Lee Netherton, and C. V. Radhakrishnan. *A package to create a nomenclature*, September 2005. <http://www.ctan.org/pkg/nomencl>.
- [11] Nicola L. C. Talbot. *User Manual for glossaries.sty v4.44*, December 2019. <http://www.ctan.org/pkg/glossaries>.
- [12] David Carlisle. *The textcase package*, October 2004. <http://www.ctan.org/pkg/textcase>.

Change History

1.48	General: Added prologue option to xcolor	52	\ccsdesc@parse: Added warning for missing concepts	82
	Bib code cleanup (Zack Weinberg)	38	\if@ACM@printacmref: Added warning for longer papers	79
	Documentation update (siggraph) .	38	\keywords: Added warning for missing keywords	78
1.49	General: New journal: DTRAP	38	1.69	\vspace: Added the changed command to avoid grouping . . .
1.53	General: New journals: PACMCGIT, TIOT, TDSCI	38	1.70	General: Name change for TDS
	Rearranged docs	38	1.71	General: Bibliography change: volume for @inproceedings is now in brackets together with series . . .
1.54	General: Moved footnote stuff before hyperref call (Ross Moore)	38		LuaTeX now uses the OTF versions of fonts
1.56	General: Documented \Description	38		Retired sigchi and sigchi-a
1.57	General: Booktabs package is now the default	38	1.75	General: Documentation update . . .
1.58	General: Changes in samples (Enrico Gregorio)	38	1.78	General: Documentation update: Word count
	New journal: HEALTH. TDS is renamed to TDSCI	38	1.79	General: Updated indormation for TAP, TCPS, TEAC
1.60	General: New option: urlbreakonhyphens	38	1.80	General: Added DLT and FAC
	\if@ACM@urlbreakonhyphens: introduced macro	39	1.82	General: Updated abbreviations and codes for several journals
1.62	General: New journal: TELO	38	1.84	General: New journals: JDS, GAMES
1.63	General: New journal: FACMP	38	1.85	General: Added CC licenses
	New journal: TQUANT	38		Added: acmengage
1.63a	General: Move: TQUANT to TQC . .	38		New journal: JRC
1.64	General: Putting abstract after \maketitle now causes an error .	38		\@mkauthors: Added: acmengage . .
1.65	General: New journal: DGOV	38		\@mktitle: Added: acmengage
1.66	General: ACM reference format is now mandatory for papers over one page; CCS concepts and keywords are now mandatory for papers over two pages	38		\@printtopmatter: Added: acmengage
	Authors' addresses are mandatory for journal articles	38		\@specialsection: Added: acmengage
	\@mkauthorsaddresses: Added warning for missing addresses .	102		\@subtitlefont: Added: acmengage .
				\@titlefont: Added: acmengage . .
				\ACM@linecountR: Added: acmengage
				format: New format: acmengage . . .
				\parskip: Added: acmengage

1.87	General: New journals: ACMJCSS, JATS, TORS	62	//github.com/borisveytsman/ acmart/issues/1)	38
	format: New format: acmcp	39	Defined ACM colors	38
1.88	General: Added ISSN for ACMJCSS	62	Explicitly put draft option (closes https: //github.com/borisveytsman/ acmart/issues/4)	42
	Added ISSN for GAMES	62	Set headheight to 1pc for all formats (closes https: //github.com/borisveytsman/ acmart/issues/5)	38
	Added PACMNET	62		
1.89	General: Added PACMMOD	62		
	Added TOPML	62		
	\@printtopmatter: Moved badges to top header	92	v1.02 General: Added TOPS and TSC (closes https: //github.com/borisveytsman/ acmart/issues/12)	38
1.90	General: Added TOPML eISSN	62	Documentation changes (closes https: //github.com/borisveytsman/ acmart/issues/13)	38
1.90a	General: Changes in the sample keywords and concepts	38		
1.92	General: Documentation update: use of 'anonymous' throughout	38	v1.03 General: Added anonsuppress environment	38
	Moved hyperxmp again due to kernel update	50	Added authorversion option (closes https: //github.com/borisveytsman/ acmart/issues/9)	38
2.04	General: Deleted loading ifdraft due to the new release of hyperxmp	50	\@formatdoi: Added macro	84
	Explicitly load ifdraft since hyperxmp tries to load it inside a group	50	\if@ACM@authorversion: Added macro	40
2.05	General: New data for TELO	62	\maketitle: Added special case of authorversion	88
2.06	General: Added eISSN for a number of journals	62	v1.04 General: Updated bibliography for siggraph	38
2.07	General: Another change for TELO	62	\acmArticleSeq: Name change by Matthew Fluet	76
2.13	\orcid: Protected orchidcite against uppercasing	74	\acmDOI: Name change by Matthew Fluet	77
2.15	General: Added AILET, TAIS, TAISAP, ACMJDS	62	\acmISBN: Name change by Matthew Fluet	77
2.16	General: Corrected handling of journals in the options	62	v1.05 General: Added processing doi numbers for acmsiggraph and doi numbers for sigproc.bib	38
2.17	General: Fixed bug with ACMJDS	62	Patched \setcitestyle command; closes https: //github.com/borisveytsman/ acmart/issues/19	38
v1.00	General: First released version	38	v1.06 \@ACM@badge@skip: Added macro	77
v1.01	General: Changed hyperref colors in screen mode (closes https: //github.com/borisveytsman/ acmart/issues/19)	77	\@ACM@badge@width: Added macro	77
			\@ACM@title@width: Added macro	77

\@mktitle@i: Added processing		\@parfont: Moved to	
badges	95	\section@raggedright	112
Made generic	95	\copyrightpermissionfootnoterule:	
\@mktitle@ii: Now this macro is		Added macro	54
obsolete	95	\descriptionlabel: Redefined	62
\@mktitle@iii: Added processing		\footnoterule: Made shorter	55
badges	95	\if@ACM@natbib@override: Added	
Made more generic	95	macro	41
\@mktitle@iv: Made more generic	95	\if@ACM@newfonts: Added macro	56
\@printtopmatter: Added processing		\maketitle: Moved thankses to	
badges	92	copyrightpermission box	88
\@subtitlefont: Added macro	94	\section@raggedright: Introduced	
\@titlefont: Added macro	94	macro	112
\acmBadgeL: Added macro	78	v1.13	
\acmBadgeR: Added macro	78	General: Font adjustments for	
\if@ACM@badge: Added macro	77	acmsmall	97
v1.07		Increased font size for ACM Large	44
\maketitle: Corrected a bug with		Print bibliographic information by	
abstract footnotes	88	default for the proceedings	80
Corrected a bug with permssion		\@mpfootnotetext: Made minipage	
and footnotes order	88	footnotes centered	50
v1.08		\bibliographystyle: Redefined	
General: SIGPLAN reformatting by		macro	51
Matthew Fluet	38	\bibstyle@acmauthoryear: Added	
Typos corrected (Tobias Pape)	38	macro	47
v1.09		\bibstyle@acmnumeric: Added	
General: Revert SIGPLAN caption		macro	47
rules	38	\if@ACM@printfolios: Added macro	79
v1.10		standardpagestyle: Added headers	
General: Changes of		for sigs	106
‘licensedothergov’ wording	82	Expanded headers for sigchi-a	106
\maketitle: Corrected a bug with doi		Suppressed folios if sig	106
in manuscript and author version,		v1.14	
https://github.com/borisveytsman/		General: \citestyle updates	
acmart/issues/36	88	(Matthew Fluet)	38
standardpagestyle: Reversed folios		Patched \citestyle	45
location	106	\bibstyle@acmauthoryear: Moved	
v1.11		def of \bibstyle@acmauthoryear	
General: Customization of ACM		before use	47
theorem styles and proof		\bibstyle@acmnumeric: Moved def of	
environment by Matthew Fluet	38	\bibstyle@numeric before use	47
v1.12		v1.15	
General: Added graceful behavior		\@ACM@addtoaddress: Added macro	96
when libertine fonts are absent	56	\@mkauthors@iii: New authors	
Documentation updates	38	system	100
\@marginfigure: Now centering by		\@shortauthors: Introduced macro	106
default	60	\@typeset@author@bx: Moved to	
\@marginable: Now centering by		separate macro	99
default	60	\acks: Added macro	118
\@mkauthors@i: Removed		\affiliation: Added code for author	
MakeTextUppercase due to a bug		groups	74
with new kernel	99	\author: Added code for author	
		groups	73
		\city: Added macro	96

\country: Added macro	96	v1.19	General: Include 'Abstract', 'Acknowledgements', and 'References' in PDF bookmarks (Matthew Fluet)	38
\department: Added macro	96		Include 'References' in PDF bookmarks (Matthew Fluet)	45
\grantnum: Added macro	118		Tagging code for tables	59
\grantsponsor: Added macro	118		\@mkabstract: Include 'Abstract' in PDF bookmarks (Matthew Fluet)	103
\if@insideauthorgroup: Introduced macro	73		\acks: Include 'Acknowledgements' in PDF bookmarks (Matthew Fluet)	118
\institution: Added macro	96	v1.20	General: Bug fixes for bst	38
\num@authorgroups: Renamed	72		\@mkabstract: Deleted spurious space	103
\orcid: Introduced macro	74	v1.21	\showeprint: Added macro	119
\position: Added macro	96	v1.22	General: More bibliography changes for Aptara	38
\postcode: Added macro	96	v1.23	General: Add PACMPL journal option	38
\state: Added macro	96	v1.24	General: Add IMWUT journal option	62
\streetaddress: Added macro	96	v1.25	General: Updated PACMPL	62
standardpagestyle: Better handling of anonymous mode	106	v1.26	General: Corrected \shortcite bug	38
v1.16			Documentation typos fixed (thanks to Stephen Spencer)	38
General: Formatting header/footer (Matthew Fluet)	38	v1.27	\if@ACM@printccs: Typo corrected	79
\@headfootfont: Added macro	106	v1.28	General: Corrected option natbib behavior	47
standardpagestyle: Customize header/footer text font	106		Got rid of warnings in pdf keywords	50
v1.17			\ccsdesc@parse: Change from \to to \textrightarrow (Matteo Riondato)	81
General: Slightly decreased margins for sigs	52	v1.29	General: Increased head to 13pt	52
\@makefnmark: Redefined	50		\@mkabstract: Removed spurious indentation (John Wickerson)	103
\@mkauthors: TOG now uses the same authors block as other journals	96		\ccsdesc@parse: Add spacing after bullet and around rightarrow; semicolon separators no longer in bold/italic (John Wickerson)	81
\@mkauthors@ii: Deleted	99		standardpagestyle: Corrected printfolios (Matthew Fluet)	106
\@mkbibcitation: Added \nobreak Changed format for sigs	103	v1.30	General: Added \frenchspacing	120
\acmArticle: The default is now numerical	76		Bibtex style now recognizes https:// in doi	38
\acmMonth: The default must be numerical. Closes #50.	77			
\acmNumber: The default is now numerical	76			
\acmVolume: The default is now numerical	76			
\acmYear: The default is now numerical	77			
\description: Decreased indent	62			
\grantnum: Added url	118			
\grantsponsor: Renamed	118			
\if@ACM@printacmref: Renamed from \if@ACM@printbib	79			
standardpagestyle: Added paper title to sigs	106			
v1.18				
General: Natbib is now the default for all formats	38			
\@mkauthors@i: Now we andify affiliations	99			
\@typeset@author@line: Added macro	98			

Switched to T1: looks like libertine has problems with \l in OT1 . . .	56	\ACM@timestamp: Added current page number	106
Updated IMWUT and PACMPL . .	62	Added macro (Michael D. Adams) .	106
\department: Added optional parameter	96	Added submission id	106
v1.31		\acmSubmissionID: Added macro . .	77
General: Documentation changes . .	38	\city: Added obeypunctuation code	96
\@additionalaffiliation: Added macro	74	\country: Added obeypunctuation code	96
\@mkbibcitation: Disabled linebreak	103	\department: Added obeypunctuation code	96
\@startsection: Added \tochangmeasure	49	\if@ACM@affiliation@obeypunctuation: Added macro	74
\@starttoc: Redefined macro	49	\if@ACM@authordraft: Added macro	41
\@textbottom: Redefined	55	\if@ACM@natbib@override: Deleted macro	41
\acmMonth: The default now is the current month (Matteo Riondato)	77	\if@ACM@timestamp: Added macro (Michael D. Adams)	41
\acmYear: The default now is the current year (Matteo Riondato) . .	77	\institution: Added obeypunctuation code	96
\additionalaffiliation: Added macro	74	firstpagestyle: Added right linecount for two-column formats	109
\l@section: Redefined macro . .	49	Added timestamp (Michael D. Adams)	109
\l@subsubsection: Redefined macro	49	\maketitle: Fixed the bug with figures on top and bottom of the title page, thanks to David Epstein	88
v1.32		\position: Added obeypunctuation code	96
General: Added POMACS journal option	38	\postcode: Added obeypunctuation code	96
Format siggraph is now obsolete .	38	\state: Added obeypunctuation code	96
\@formatdoi: New doi format	84	\streetaddress: Added obeypunctuation code	96
v1.33		standardpagestyle: Added right linecount for two-column formats	106
General: Added acmart-preload-hook	38	Added timestamp (Michael D. Adams)	106
Added cmap and glyptounicode .	56	v1.34	
BibTeX comma before articleno bug corrected	38	\@mkbibcitation: Deleted DOI from doi numbers	103
BibTeX crossref bug corrected . .	38	\ACM@linecount@bx: Rulers now are continuous	105
BibTeX numpages bug corrected .	38	\ACM@linecountL: Rulers now are continuous	105
Documentation updates	38	\ACM@linecountR: Rulers now are continuous	105
Moved \citename definition for non-natbib bibliography, so a package may redefine it	51	\maketitle: Added date to the bibstrip in conf proceedings	88
Switched to Type 1 fonts for libertine even if OTF-capable engine is used (Kai Mindermann) .	56	Deleted 'DOI' from doi numbers .	88
\@ACM@addtoaddress: Added obeypunctuation code	96	Rearranged bibstrip	88
\@mkauthors@i: Added obeypunctuation code	99	The \terms command is now obsolete	88
\@mkauthors@iii: Added obeypunctuation code	100	\terms: The \terms command now just produces a warning	78
\@mkauthors@iv: Added obeypunctuation code	101		
\@subtitlefont: Added \normalsize	94		
\ACM@linecountL: Renamed macro .	105		
\ACM@linecountR: Added macro . .	105		

v1.35	General: BibTeX bug fixed: et al.	38	\@parfont: Switched to MakeTextUppercase	112
	If the copyright is set to usgov or rigtsretained, the price is suppressed	82	\ACM@linecount@bx: Work around a bug in xcolor: looks like cmyk colors in boxes do not work . . .	105
	\bibstyle@acmauthoryear: Square brackets for author-year style . . .	47	\ccdesc@parse@finish: Added macro	82
v1.36	General: Added PACMHCI journal options	62	\city: We now do not print this even in SIG	96
	Added the possibility to adjust number of author boxes per row in conference formats	38	\country: Fixed bugs with extra spaces	96
	\@ACM@authorsperrrow: Added macro	79	\l@section: Redefined macro . .	49
	\@mkauthors@iii: Added authorsperrrow override	100	\postcode: We now do not print this even in SIG	96
	\@mkauthors@iv: Added authorsperrrow override	101	\state: We now do not print this even in SIG	96
	\if@ACM@authordraft: Corrected typo, thanks to bargteil	41	\streetaddress: We now do not print this even in SIG	96
v1.37	General: Reduce list indentation (Matthew Fluet)	61	v1.41 General: Added new badges	38
	Set \normalparindent; Reduce list indentation (Matthew Fluet)	38	v1.42 General: Deleted ACM badges	38
	\description: Reduce list indentation (Matthew Fluet)	62	v1.43 \@starttoc: Added \makeatletter	49
	\descriptionlabel: Reduce list indentation (Matthew Fluet)	62	\country: Added comma before country for journals	96
	\normalparindent: Set \normalparindent (Matthew Fluet)	54	v1.44 General: Added package refcount . . .	38
v1.38	General: Increase default font size for SIGPLAN	38, 44	Deleted package cleveref	38
v1.39	\@@authornotemark: Added macro . .	103	Put theorem defs in a separate style	38
	\authornotemark: Added macro . . .	76	\@acmBooktitle: Added macro	72
v1.40	General: Added 'textcase' package . .	45	\@acmEditors: Added macro	72
	Added heightrounded to geometry	52	\@editorsAbbrev: Added macro . . .	72
	Added new copyright version: licensedcagov	38	\@folioblob: Suppress the blob if acmArticleSeq is empty	109
	Added package cleveref	38	\@mkbibcitation: If the paper has just one page, use 'page' instead of 'pages'	103
	Added Ross Moore code for glyphtounicode	56	\ACM@restore@pagestyle: Added macro	104
	Bibliography changes	38	\acmArticle: The default is now empty	76
	\@copyrightowner: Added new copyright version: licensedcagov .	83	\acmArticleSeq: Now acmArticle might be empty	76
	\@copyrightpermission: Added new copyright version: licensedcagov .	84	\acmBooktitle: Added macro	72
	\@mkauthors@i: Switched to MakeTextUppercase	99	\editor: Added macro	72
			\if@ACM@acmthm: Added macro	39
			\maketitle: Empty DOI or ISBN are not printed (by Michael Ekstrand)	88
			v1.45 \@folioblob: Switched \bfseries\sffamily to \sffamily\bfseries	109

\@parfont: Switched		\@mkabstract: Removed spurious indentation if abstract is followed by an empty line	103
\bfseries\sffamily to		\ACM@timestamp: Formatting change (Michael D. Adams)	106
\sffamily\bfseries	112	\if@ACM@review: Review mode now switches on folios	40
\@titlefont: Switched		\maketitle: Added acmart and version info to pdfcreator tag	88
\bfseries\sffamily to		Added language and doctitle attributes to PDF (Andreas Reichinger)	88
\sffamily\bfseries	94	Suppressed empty DOI (Serguei Mokhov)	88
v1.46		v1.50	
General: Added etoolbox	38	\@copyrightpermission: New copyright statements for iw3c2w3[g]	84
Bst file bug fixes: label width is calculated correctly	38	v1.51	
Delayed hypersetup since journal options may change screen mode	50	General: Corrected the bug with price suppression	82
PACM now set screen to true	62	Suppress price if the copyright is set to iw3c2w3 or iw3c2w3g	82
Restore theorem defs to class file	38	XeTeX now uses OTF fonts (Joachim Breitner)	56
\@mkauthorsaddresses: Introduced macro	102	firstpagestyle: We now use journal abbreviation for footers	109
\@mkbibcitation: Added subtitle	103	standardpagestyle: We now use journal abbreviation for footers	106
\@printendtopmatter: Made it		v1.52	
\par\bigskip uniformly	104	\@copyrightowner: Rewording of licenses	83
\@printtopmatter: Deleted rule	92	\@copyrightpermission: Another rewording of copyright statements for iw3c2w3[g]	84
\@setauthorsaddresses: Introduced macro	104	v1.53	
\ACM@linecount@bx: Rearranged the code to get rid of spurious underfull messages (Benjamin Byholm)	105	General: Added PACMCGIT, TIOT, TDSCI	62
\author: Started counting authors	73	v1.54	
\authorsaddresses: Introduced macro	75	General: Added Ross's patch for sectioning	112
\country: Corrected spacing for institution	96	Deleted indent for subsubsection	112
\if@ACM@acmthm: Modified description	39	XeTeX now uses T1 encoding	56
\liningnums: Workaround for compatibility with fontspec	57	\@mkbibcitation: Bug corrected (Ross Moore)	103
\maketitle: Added authors' addresses	88	\@shortauthors: Added submission id in anonymous mode	106
Thankses go before authors' addresses	88	\ACM@adddotafter: Added macro	111
\num@authors: Introduced macro	73	\ACM@NR@adjustedSectionFormat: Added macro	111
v1.47		\ACM@NRadjust: Added macro	110
General: New journal: THRI	38	\author: Added submission id in anonymous mode	73
v1.48		\if@ACM@nonacm: Added macro	40
General: \citeyear no longer behaves like \citeyearpar	51		
Code prettying (Michael D. Adams)	38		
Initial support for Biblatex (Daniel Thomas)	38		
Misc entries in the bibliography no longer produce a separate date	38		
Typos fixed (Jamie Davis)	38		
\@headfootfont: Deleted unnecessary switch (Michael D. Adams)	106		

v1.55	General: Now we use purple color for links	50	\maketitle: Original \vspace inside \maketitle	88
	Table captions for SIGCHI now use the same font size as figure captions (Luis Leiva)	58	\vspace: Preserved old versions as \vspace@orig and \vspace@orig	119
	Typo corrected (Zack Weinberg)	45	v1.59	
	\@mkbibcitation: Bug corrected	103	\acmConference: Setting \ACM@journal@bibstrip	71
	\maketitle: Fixes topnum	88	\acmJournal: Setting \ACM@journal@bibstrip	71
v1.56	General: Looks like XeTeX requires explicit monotype font declaration	56	\if@ACM@journal@bibstrip: Introduced macro	42
	Make two-column layouts flush (Philip Quinn)	52	firstpagestyle: Added logic for conference papers using journal format	109
	\@marginfigure: Added checking for description	60	Moved page styles to \AtBeginDocument	109
	\@mkteasers: Added check for description	103	standardpagestyle: Added logic for conference papers using journal format	106
	\acksname: Added macro (Philip Quinn)	118	Moved page styles to \AtBeginDocument	106
	\ccsdesc@parse: End the concepts list with a period (Philip Quinn)	81	v1.60	
	\Description: Added macro	57	\@headfootfont: Moved to footnotesize	106
	\if@Description@present: Added macro	57	v1.61	
	\if@undescribed@images: Added macro	57	\@typeset@author@bx: Added \normalbaselines	99
v1.57	\ACM@origbaselinestretch: Added check whether the user changed \baselinestretch	120	v1.64	
	\author: Added warning if \author has comma	73	\ACM@linecount@bx: Bug fix: made the spacing on the left and the right size equal	105
	\email: Added warning if \author has comma	74	\if@ACM@maketitle@typeset: Introduced macro	62
	\if@ACM@balance: Added macro	40	\maketitle: Added a switch setting to show that \maketitle is typeset	88
	\vspace: Added warning	119	v1.65	
v1.58	General: Added HEALTH. TDSCI is renamed to TDS	62	\@mkauthorsaddresses: Deleted cs@addpunct	102
	Suppressed \addtocounter in pdf subject	50	v1.66	
	\@dblfloat: Deleted spurious \vspace warning	60	General: Patching algorithm and algorithm* to prevent spurious space warnings	57
	\@marginfigure: Deleted spurious \vspace warning	60	\@specialsection: Suppress warnings about \vspace	92
	\@margintable: Deleted spurious \vspace warning	60	v1.67	
	\@mkbibcitation: Deleted spurious \vspace warning	103	General: Patching lstlisting and lstlisting* to prevent spurious space warnings	57
	\@printendtopmatter: Deleted spurious \vspace warning	104	\@parfont: Deleted redefinitions of sections	112
	\@sidebar: Deleted spurious \vspace warning	60	\@specialsection: Paragraph inside group does not print its argument. Added a bugfix	92

v1.68	General: New command for unindent paras 112	v1.75	\@ACM@checkaffil: Changed warning to error for country 98
	\@specialsection: Deleted grouping altogether 92		\@beginmaketitlehook: Introduced macro 88
v1.69	General: No dots for unindented paragraphs 112		\AtBeginMaketitle: Introduced macro 88
	\@printendtopmatter: Deleted grouping 104		\maketitle: Added \@beginmaketitlehook 88
	\@specialsection: Changed dot to colon for sigplan 92	v1.76	General: Moved hyperxmp before hyperref, see https://github.com/borisveytsman/ acmart/issues/425 50
	\ACM@linecountR: Do not increase numbers in one column format . . 105		\author: Added code for orcid links . . 73
	\ccsdesc@parse: Use math right arrow in text 81		\ifACM@pbalance: Added macro . . . 40
v1.71	\l@subsubsection: Bug fixed (thanks to Andrew Black) 49		\maketitle: Put \par inside group for keywords 88
v1.72	\author: Added \detokenize (Kuldeep S. Meel) 73		\orcid: Added code for orcid links . . 74
	\maketitle: Do not andify authors for pdf metadata (Scott Pakin) . . . 88		standardpagestyle: Fixed bug with no line numbers for nonacm . . . 106
v1.73	General: Patching minted and minted* to prevent spurious space warnings 57	v1.77	\mkauthorsaddresses: New separator of institutions 102
	\@ACM@checkaffil: Added macro . . . 98	v1.78	\mkauthors@i: Unboxing \mktitle@bx 99
	\@ACM@resetaffil: Added macro . . . 98		\mkauthors@iii: Unboxing \mktitle@bx 100
	\mkauthors@i: Check the presence of required elements for affiliations 99		\mkteasers: Unboxing \mktitle@bx 103
	\mkauthors@iii: Check the presence of required elements for affiliations 100		\@printtopmatter: Added processing the overflowing title box 92
	\mkauthors@iv: Check the presence of required elements for affiliations 101		\@specialsection: Again suppressed \vspace warning 92
	\city: Check the presence in affil . . 96	v1.82	\anon: Introduced macro 87
	\country: Check the presence in affil . 96	v1.83	General: Use babel for multilanguage papers 47
	\ifACM@citypresent: Added macro . 98		\mkabstract: I13n 103
	\ifACM@countrypresent: Added macro 98		\mktitle@i: Added \translatedtitle and \translatedsubtitle 95
	\ifACM@instpresent: Added macro . 98		\mktitle@iii: Added \translatedtitle and \translatedsubtitle 95
	\institution: Check the presence in affil 96		\mktitle@iv: Added \translatedtitle and \translatedsubtitle 95
	\maketitle: Do not check again the presense of address fields 88		\mktranslatedabstract: Added macro 103
v1.74	General: Changed the order of font loading 56		\mktranslatedkeywords: Added macro 75

\@translatedabstracts: Added macro	49	\acmDataLink: Introduced macro . . .	81
\@translatedkeywords: Added macro	48	Now this is the same as the code link	81
\@translatedsubtitle: Added macro	48	\if@ACM@acmcp: Introduced macro . .	42
\@translatedtitle: Added macro . .	48	firstpagestyle: Special treatment for acmcp	109
\ACM@lang@check: Added macro . . .	48	\maketitle: CC license are allowed for non-acm publications	88
\ACM@languages: Introduced macro . .	42	\set@ACM@acmcpbox: Added macro . .	91
\maketitle: I13n	88	standardpagestyle: Special treatment of ACMCP	106
translatedabstract: Added macro . .	49	v1.88	
\translatedkeywords: Added macro . .	48	\@mktitle@i: Moved down title for acmcp	95
\translatedsubtitle: Added macro . .	48	v1.89	
\translatedtitle: Added macro . .	48	General: Added ‘zref-savepos’ package for acmcp	45
v1.85		\@ACM@acmcp@delta: Added dimen . .	45
General: Added CC licenses	82	\@ACM@badge@skip: Redefined macro	77
\@acmengagemetadata: Introduced macro	88	\@ACM@color@frame: Added saving the position of the bottom	81
\@copyrightowner: Added CC licenses	83	\@acmBadgeL: Introduced macro . . .	77
\@copyrightpermission: Added CC licenses	84	\@acmBadgeR: Introduced macro . . .	77
\@parfont: Added: acmengage	112	\@mktitle@i: Moved badges to top header	95
\@typesetengagemetadata: Introduced macro	104	Moved up acmcp title	95
\if@ACM@engage: Introduced macro . .	42	\@mktitle@iii: Moved badges to top header	95
\maketitle: acmengage	88	\acmBadge: Added macro	78
\setcctype: Added macro	83	\acmBadgeL: Redefined macro	78
\setengagemetadata: Introduced macro	88	\acmBadgeR: Redefined macro	78
v1.86		\if@ACM@badge: Deleted macro	77
\country: Error if country is empty .	96	firstpagestyle: Moved badges here	109
standardpagestyle: Special treatment of ACM Engage materials	106	\set@ACM@acmcpbox: Added logo . . .	91
v1.87		Added zref position of the bottom	91
General: Added ‘framed’ package for acmcp	45	v1.91	
No numbering sections for ACMCP	113	\acmContributions: Restored the macro	81
\@ACM@color@frame: Introduced macro	81	standardpagestyle: Redesigned ACMP labels	106
\@ACM@copyright@check@cc: Added macro	84	v1.93	
\@copyrightpermission: CC licenses now are allowed only for non-acm or ACM engage publications	84	General: Added PACMSE journal option	38
\@mkauthors@i: Special treatment for acmcp	99	v2.00	
\@mktitle@i: Special treatment for acmcp	95	General: Deleted acmPrice	82
\acmArticleType: Introduced macro . .	80	\@ACM@copyright@check@cc: Moved to warning	84
\acmCodeDataLink: Added possibility to have several links	81	\@copyrightowner: Changed wording	83
Introduced macro	81	\@folio@ht: Obsoleted macro	109
		\@folio@max: Obsoleted macro	109
		\@folio@voffset: Obsoleted macro . .	109
		\@folio@wd: Obsoleted macro	109
		\@folioblob: Obsoleted macro	109
		\@sect@kernel: Added macro	44
		\@sect@kernel: Added macro	44

\@startsection@kernel: Added macro	44	v2.10	General: Added tocindent for paragraphs and subparagraphs . .	113
\@xsect@kernel: Added macro	44			
\acmPrice: Deleted macro	77	v2.11	\@copyrightpermission: Changed URL for CC licenses	84
firstpagestyle: Deleted blobs	109		\maketitle: Deleted conference date from bibstrip	88
\maketitle: Added “ACM” for papers	88			
Deleted acmPrice	88			
v2.01		v2.12	General: Removed ‘textcase’ package	45
General: Documentation update: POMACS uses acmsmall	38		Use unicode-math and libertine for Unicode engines	56
\acmBooktitle: Moved default conference title to conference setting	72	v2.13	General: Added mathcal in unicode mode	56
\acmPrice: Now the macro produces a warning	77		New warnings about lack of descriptions	57
v2.02		v2.14	\ACM@origparagraph: Added check whether the user changed \paragraph	120
\acmConference: Moved here setting the conference title for bibstrip . .	71		\ACM@origsection: Added check whether the user changed \section	120
v2.03			\ACM@origsubsection: Added check whether the user changed \subsection	120
General: Added documentation about samples	38		\ACM@origsubsubsection: Added check whether the user changed \subsubsection	120
\acmConference: Made setting bibstrip overriding journal	71	v2.15	General: Added symbols in unicode mode	56
\if@ACM@journal@bibstrip@or@tog: Introduced macro	42		\@@authornotemark: Suppressed spurious overfull warning	103
firstpagestyle: Special treatment of tog	109		\@authornotemark: Suppressed spurious overfull warning	102
\maketitle: Special processing for conferences in TOG	88	v2.16	\IfDocumentMetadata: Provided stubs for old kernels	38
standardpagestyle: Special treatment of ACM TOG	106	v2.17	General: Added tagging documentation	38
v2.04			Moved tagged code to the main class	44
General: eISSN is printed, rather than pSSN	71		Rearranged documentation	38
\@copyrightowner: Changed wording	83		\@startsection: Moved tagging code to the main class	49
v2.06		v2.18	General: Added pifont	57
\@mkauthorsaddresses: \streetaddress and \postcode now produce warnings	102		Deleted IMS designation	38
\postcode: We now do not collect postal information anymore	96		Keywords are now deprecated. Deleted them from the samples . .	38
\streetaddress: We now do not collect postal information anymore	96		Merged tagged branch	38
v2.08				
\@parfont: Deleted uppercasing	112			
v2.09				
General: Saved the old definitions (Ulrike’s code	44			
\@ACM@copyright@check@cc: Deleted macro	84			
\@copyrightpermission: Deleted checking whether CC licenses are allowed	84			

Moved note before url (suggested by David F. Kotz)	38		about keywords. Re-added them	38
New code for captions	59	v2.20		
Revised documentation	38		\acmConference: Proceedings headers, footers and folios for acmsmall used for conference proceedings	71
Rewrote caption code for tagging	58			
Switched off warning in tagged mode	57	v2.21		
\@ACM@addtoaddress: Deleted the cleaders code	96		General: Deleted pifont	57
\@correspondingauthormark: Added macro	76		\@correspondingauthormark: Print asterisk	76
\@mkauthors@iii: Email is now clickable	100		\@titlenotes: Added corresponding author note	75
\@mkauthors@iv: Email is now clickable	101		\correspondingauthor: Throw error if more than two corresponding authors	76
\@mkauthorsaddresses: Email is now clickable	102		\if@ACM@corresponding@present: Added macro	76
\correspondingauthor: Added macro	76		\maketitle: Start footnotes from 2	88
v2.19			\showISBNxiii: Added macros	119
General: ACM cleared their position				

Index

Numbers written in *italic* refer to the page where the corresponding entry is described; numbers underlined refer to the code line of the definition; numbers in *roman* refer to the code lines where the entry is used.

Symbols	
\-	540
\@	1975, 1978, 1990, 1995, 2000, 2084, 2121
\@authornotemark	1696, <u>2855</u>
\@ACM@acmcp@delta	<u>288</u> , 2351, 2355
\@ACM@acmcp@box	2338, 2342, 3138, 3223
\@ACM@acmcp@false	174
\@ACM@acmcp@true	232
\@ACM@addtoaddress	<u>2554</u> , 2584, 2585, 2586
\@ACM@author@sperrrow	<u>1801</u> , 2730, 2731, 2783, 2784
\@ACM@badge@skip	<u>1726</u> , 1738, 1751
\@ACM@badge@width	<u>1723</u> , 1735, 1741, 1748, 1754
\@ACM@balance@false	3475, 3499, 3501, 3503, 3515, 3519
\@ACM@check@affil	2309, 2668, 2699, 2704, 2766, 2771, 2811, 2816
\@ACM@city@present@false	2665
\@ACM@city@present@true	2571, 2584, 2660
\@ACM@color@frame	<u>1858</u> , 2335
\@ACM@comm@box	2554
\@ACM@copyright@check@cc	<u>2011</u>
\@ACM@corresponding@present@false	1678
\@ACM@corresponding@present@true	1685
\@ACM@country@present@false	2666
\@ACM@country@present@true	2576, 2586, 2662
\@ACM@engage@false	172
\@ACM@engage@true	229
\@ACM@institution@separator\@DeclareACMFormat	42, 2836, 2837, 2842
\@ACM@inst@present@false	2664
\@ACM@inst@present@true	2569, 2580, 2658
\@ACM@journal@bibstrip@or@tog@false	799, 802
\@ACM@journal@bibstrip@or@tog@true	1528
\@ACM@journal@bibstrip@or@tog@true	215, 237, 1512
\@ACM@journal@bibstrip@false	2275, 2895, 2896, 3024, 3027, 3047, 3057, 3069, 3142, 3165, 3175, 3186, 3227
\@ACM@journal@true	208, 210, 212, 214, 231
\@ACM@label@width	982, 985, 993, 998, 1024
\@ACM@maketitle@typeset@false	1031
\@ACM@maketitle@typeset@true	2184
\@ACM@manuscript@false	203
\@ACM@manuscript@true	201
\@ACM@newfont@false	761, 764, 767
\@ACM@newfont@true	758
\@ACM@printacm@re@false	99, 233
\@ACM@print@folio@true	82
\@ACM@reset@affil	2663, 2698, 2703, 2765, 2770, 2810, 2815
\@ACM@review@true	144
\@ACM@screen@true	1172, 1223,
\@ACM@sigchiam@mode@false	205
\@ACM@sigchiam@mode@true	226
\@ACM@timestamp@true	143
\@ACM@title@width	<u>1725</u> , 2527, 2529, 2538, 2539
\@Description@present@false	823, 826, 932, 2862
\@Description@present@true	2569, 2580, 2658
\@M	472, 3575
\@abstract	1772, 2870, 2877
\@acmArticle	1703, 1706, 2275, 2895, 2896, 3024, 3027, 3047, 3057, 3069, 3142, 3165, 3175, 3186, 3227
\@acmArticlePage	3023, 3041, 3042, 3051, 3054, 3061, 3065
\@acmArticleSeq	1705
\@acmBadgeL	1729, 1744, 1745, 1750, 3156, 3168, 3169, 3180, 3181, 3193, 3196, 3201, 3222
\@acmBadgeR	1728, 1731, 1732, 1737, 3157, 3170, 3171, 3178, 3179, 3194, 3197, 3202, 3223
\@acmBooktitle	1531, 1541, 2242, 2265, 2911
\@acmCodeDataLink	1865, 1866, 1868, 1870, 2358, 2359
\@acmContributions	1871, 2364, 2365
\@acmDOI	1721, 2267, 2270, 2276, 2279, 2919, 3086, 3088, 3110, 3112, 3143, 3228
\@acmEditors	1544, 1545, 1546, 1549, 2912, 2913
\@acmISBN	1719, 2278
\@acmMonth	1709, 1711, 2275
\@acmNumber	1701, 2908, 3047, 3057, 3069, 3141, 3165, 3175, 3186, 3226

\@acmPubDate	\@article@string . . .	1902, 2293, 2295, 2313
<u>1711</u> , 2908, 3047,	. . . 2895, 2908, 2916	\@copyrightowner . . .
3057, 3070, 3143,	\@author	 <u>1972</u> , 2249
3166, 3176, 3187, 3228	1569, 1575, 1582,	\@copyrightpermission
\@acmSubmissionID . .	2684, 2749, 2796, 2843	. . . <u>2011</u> , 2225, 2237
. <u>1570</u> ,	\@authorfont <u>2616</u> , 2621,	\@copyrightyear
1571, 1590, 1591,	2625, 2628, 2631,	2156, 2249, 2251, 2252
1717, 3004, 3005, 3015	2634, 2637, 2640,	\@correspondingauthormark
\@acmVolume	2643, 2646, 2686, 2722	 <u>1686</u> , <u>1689</u>
1699, 2908, 3046,	\@authornotemark . . .	\@currbox . . . 961, 962, 963
3056, 3069, 3141,	. . . <u>1673</u> , <u>1694</u> , <u>2853</u>	\@current@float@struct
3164, 3174, 3186, 3226	\@authornotes	 900, 905
\@acmYear . . 1707, 1714,	. . . <u>1670</u> , <u>1674</u> , 2205	\@currentaffiliation
2157, 2242, 2275, 2899	\@authorsaddresses .	2723, 2726, 2748,
\@acmdefinitionbodyfont	 <u>1641</u> ,	2755, 2756, 2759,
. . . <u>3386</u> , 3398, 3412	2214, 2367, 2848, 2929	2762, 2763, 2768,
\@acmdefinitionheadfont	\@auxout . . 576, 2354, 3343	2795, 2802, 2803,
. . . <u>3388</u> , 3400, 3414	\@begindocumenthook 572	2805, 2807, 2808, 2813
\@acmdefinitionindent	\@beginmaketitlehook	\@currentaffiliations
. . . <u>3387</u> , 3399, 3413	. . . <u>2174</u> , 2177, 2183	 2652,
\@acmdefinitionnotefont	\@botnum 2328	2653, 2654, 2656,
. . . <u>3389</u> , 3401, 3417	\@citestyle 322	2682, 2694, 2695, 2701
\@acmengagemetadata	\@classname 25, 27, 36,	\@currentauthors . . .
. . . <u>2178</u> , 2180, 2932	39, 42, 55, 57, 59,	1689, 2650, 2651,
\@acmownedfalse	60, 63, 65, 67, 68,	2681, 2685, 2686,
. <u>1926</u> , 1929,	71, 73, 75, 76, 79,	2688, 2722, 2725,
1932, 1937, 1940,	81, 84, 85, 88, 90,	2747, 2749, 2750,
1943, 1946, 1949,	92, 93, 96, 98, 104,	2752, 2794, 2796,
1955, 1958, 1961, 1964	105, 108, 109, 112,	2797, 2799, 2853, 2855
\@acmownedtrue	113, 116, 118, 120,	\@currentlabel . 490, 504
. . . <u>1913</u> , 1922, 1952	121, 124, 126, 128,	\@dblfloat <u>944</u>
\@acmplainbodyfont .	129, 132, 134, 136,	\@department . . 1620, 1621
. . . <u>3354</u> , 3366, 3380	137, 140, 142, 146,	\@dotsep <u>3353</u>
\@acmplainheadfont .	147, 165, 176, 179,	\@editorsAbbrev
. . . <u>3356</u> , 3368, 3382	182, 185, 188, 271,	. . . <u>1543</u> , 1548, 2913
\@acmplainindent . . .	433, 457, 759, 762,	\@empty 159, 245,
. . . <u>3355</u> , 3367, 3381	765, 805, 831, 834,	392, 432, 1523,
\@acmplainnotefont .	935, 1173, 1224,	1531, 1545, 1567,
. . . <u>3357</u> , 3369, 3385	1231, 1248, 1255,	1570, 1588, 1590,
\@adddotafter	1262, 1506, 1508,	1595, 1596, 1602,
3242, 3243, 3245,	1560, 1615, 1627,	1731, 1744, 1762,
3257, 3275, 3280, <u>3348</u>	1682, 1715, 1759,	1763, 1819, 1820,
\@additionalaffiliation	1765, 1770, 1777,	1826, 1833, 1865,
. <u>1616</u> , <u>1617</u>	1779, 1780, 1783,	1876, 1902, 2175,
\@addpunct . 2840, 2928,	1785, 1786, 1789,	2209, 2214, 2251,
2929, 3348, 3349, 3469	1797, 1799, 1800,	2267, 2278, 2279,
\@addspaceafter <u>3349</u>	1802, 1803, 1851,	2290, 2293, 2299,
\@affiliationfont . .	1904, 2316, 2561,	2303, 2358, 2361,
. <u>2617</u> , 2622,	2564, 2670, 2673,	2364, 2367, 2403,
2626, 2629, 2632,	2676, 2829, 2832,	2532, 2541, 2550,
2635, 2638, 2641,	2849, 2865, 2892,	2575, 2652, 2685,
2644, 2647, 2699, 2722	3579, 3582, 3589,	2693, 2694, 2749,
\@afterheading 2331	3594, 3599, 3604, 3609	2755, 2762, 2796,
\@afterindentfalse . 2330	\@concepts . <u>1876</u> , 1888,	2802, 2807, 2848,

2858, 2895, 2900, 2912, 2919, 3004, 3015, 3086, 3088, 3110, 3112, 3142, 3143, 3227, 3228, 3357, 3389, 3538, 3555	\@insideauthorgroupttrue 1565	1393, 1398, 1403, 1408, 1413, 1418, 1423, 1428, 1433, 1438, 1443, 1448, 1453, 1458, 1463, 1468, 1473, 1478, 1483, 1488, 1493, 1498, 1503, 1515, 2907, 3046, 3056, 3068, 3164, 3174, 3185
\@endfloatbox 947	\@journalCode .. 1032, 1173, 1224, 1231, 1248, 1255, 1262, 1506	\@keywords 1761, 1762, 1763, 2299, 2301, 2314, 2361, 2362
\@endpefalse 3471	\@journalCode@nr 1032, 1111, 1513	\@latex@error 366
\@endwdefloatbox 947, 955	\@journalName 1113, 1118, 1123, 1128, 1132, 1137, 1142, 1147, 1152, 1157, 1162, 1167, 1175, 1180, 1185, 1190, 1195, 1200, 1204, 1209, 1214, 1219, 1226, 1233, 1238, 1243, 1250, 1257, 1264, 1269, 1274, 1279, 1284, 1289, 1294, 1299, 1304, 1309, 1313, 1318, 1323, 1328, 1333, 1338, 1343, 1347, 1352, 1357, 1362, 1367, 1372, 1377, 1382, 1387, 1392, 1397, 1402, 1407, 1412, 1417, 1422, 1427, 1432, 1437, 1442, 1447, 1452, 1457, 1462, 1467, 1472, 1477, 1482, 1487, 1492, 1497, 1502, 1514, 1515, 2263, 3141, 3226	\@listI 1006, 1010 \@listI 1003, 1010 \@listII 1011 \@listIII 1014 \@listIV 1016 \@listV 1018 \@listVI 1020 \@makefnmark ... 485, 510, 2197, 2201, 2206 \@makefnmtxt 485, 494, 507, 2208 \@marginfigure . 927, 928 \@margintable .. 938, 939 \@minipagefalse . 719, 957 \@minus 3263, 3268, 3273, 3278, 3283, 3288, 3377, 3379, 3409, 3411 \@mkabstract .. 2289, 2869 \@mkauthors 2198, 2200, 2588 \@mkauthors@i 2592, 2594, 2596, 2598, 2612, 2680 \@mkauthors@ii 2715 \@mkauthors@iii 2600, 2602, 2604, 2606, 2610, 2727 \@mkauthors@iv 2608, 2780 \@mkauthorsaddresses 1642, 2824 \@mkbibcitation 2324, 2886 \@mkteasers ... 2198, 2857 \@mkttitle 2198, 2447 \@mkttitle@i 2450, 2452, 2454, 2456, 2470, 2523 \@mkttitle@ii 2536 \@mkttitle@iii 2458, 2460, 2462, 2464, 2468, 2536 \@mkttitle@iv .. 2466, 2546
\@finalstrut ... 495, 508	\@journalNameShort . 1114, 1119, 1124, 1129, 1133, 1138, 1143, 1148, 1153, 1158, 1163, 1169, 1176, 1181, 1186, 1191, 1196, 1201, 1205, 1210, 1215, 1220, 1227, 1234, 1239, 1244, 1251, 1258, 1265, 1270, 1275, 1280, 1285, 1290, 1295, 1300, 1305, 1310, 1314, 1319, 1324, 1329, 1334, 1339, 1344, 1348, 1353, 1358, 1363, 1368, 1373, 1378, 1383, 1388,	\@mktitle@iv .. 2466, 2546
\@firstofone 32		
\@float 949		
\@flushglue 3291		
\@fnsymbol 2194		
\@folio@ht 3148		
\@folio@max 3148		
\@folio@voffset 3148		
\@folio@wd 3148		
\@folioblob 3148		
\@footnotemark . 511, 2192		
\@footnotemark@nolink 511, 2192		
\@footnotetext 486, 512, 2193		
\@footnotetext@nolink 512, 2193		
\@for 298, 3335, 3342		
\@formatdoi 2010, 2270, 2276, 2279, 2919, 3086, 3088, 3110, 3112, 3143, 3228		
\@fps 948		
\@fpsadddefault 948		
\@gobble 31, 320, 364		
\@headfootfont 3017, 3041, 3042, 3043, 3044, 3050, 3053, 3060, 3064, 3078, 3079, 3085, 3087, 3091, 3094, 3102, 3103, 3109, 3111, 3115, 3118		
\@idxitem 3572		
\@ifnextchar 1620		
\@ifnotempty 2162		
\@ifnum 336, 338		
\@ifundefined .. 3336, 3422, 3425, 3428, 3431, 3434, 3438, 3441		
\@iiiparbox 722		
\@indextitlestyle .. 3571		
\@input 474		
\@insideauthorgroupfalse 1558, 1610		

\@mktranslatedabstract	1222, 1229, 1236,	\@setaddresses	2852
..... 455, 2879	1241, 1246, 1253,	\@setauthorsaddresses	
\@mktranslatedkeywords	1260, 1267, 1272,	... 2218, 2367, 2929	
..... 452, 1643	1277, 1282, 1287,	\@setengagemetadata	
\@mpargs	1292, 1297, 1302,	 2181, 2931	
\@mparswitchfalse	1307, 1316, 1321,	\@setthanks	2211, 2928
\@mpfootins	1326, 1331, 1336,	\@shortauthors	
499, 500, 714, 715, 717	1341, 1350, 1355,	3012, 3035, 3043,	
\@mpfootnotetext	1360, 1365, 1370,	3051, 3061, 3079, 3103	
\@ne	1375, 1380, 1385,	\@sidebar	923, 924
324, 328, 332, 334,	1390, 1395, 1400,	\@specialsection	
562, 2946, 2952, 2956	1405, 1410, 1415,	1645, 2294, 2300, 2372	
\@nil	1420, 1425, 1430,	\@essect	244, 466
\@nobreakfalse	1435, 1440, 1445,	\@essect@kernel	244, 466
\@pages@word	1450, 1455, 1460,	\@startPage	1757,
... 2889, 2909, 2916	1465, 1470, 1475,	2403, 2404, 2406, 3009	
\@par	1480, 1485, 1490,	\@startsection	
\@parboxrestore	1495, 1500, 1517, 2275	 241, 461,	
\@parfont	\@plus	472, 3262, 3267,	
3280, 3285, 3290, 3295	 472, 480,	3272, 3277, 3282, 3287	
\@permissionCodeOne	723, 3263, 3268,	\@startsection@kernel	
1115, 1120, 1125,	3273, 3278, 3283,	 241, 461	
1130, 1134, 1139,	3288, 3376, 3378,	\@starttoc	469
1144, 1149, 1154,	3408, 3410, 3466, 3573	\@subparfont	3320
1159, 1164, 1170,	\@printcopyrightfalse	\@subsecfont	3270, 3295
1177, 1182, 1187,	 1925, 1936	\@subsubsecfont	3275, 3295
1192, 1197, 1202,	\@printcopyrighttrue	\@subtitle	1551, 1662,
1206, 1211, 1216,	 1909, 1921	2532, 2533, 2541,	
1221, 1228, 1235,	\@printendtopmatter	2542, 2550, 2551, 2900	
1240, 1245, 1252,	 2329, 2921	\@subtitlefont	
1259, 1266, 1271,	\@printpermissionfalse	 2498, 2533, 2542, 2551	
1276, 1281, 1286,	 1924	\@subtitlenotes	
1291, 1296, 1301,	\@printpermissiontrue	1660, 1664, 1667, 2204	
1306, 1311, 1315,	... 1911, 1920, 1935	\@teaser	2160, 2859
1320, 1325, 1330,	\@printtopmatter	\@teaserfigures	
1335, 1340, 1345,	 2199, 2402	2158, 2160, 2858, 2863	
1349, 1354, 1359,	\@proofindent	\@tempa	298,
1364, 1369, 1374,	... 3448, 3458, 3468	299, 301, 303, 305,	
1379, 1384, 1389,	\@proofnamefont	307, 309, 311, 313,	
1394, 1399, 1404,	... 3447, 3457, 3469	315, 317, 319, 321,	
1409, 1414, 1419,	\@received	323, 325, 327, 329,	
1424, 1429, 1434,	1821, 1823, 1826,	331, 333, 335, 337,	
1439, 1444, 1449,	1827, 1829, 1833, 1834	339, 341, 343, 346,	
1454, 1459, 1464,	\@restonecolfalse	348, 350, 352, 354,	
1469, 1474, 1479,	... 3569	356, 1594, 1595,	
1484, 1489, 1494,	\@restonecoltrue	1818, 1819, 2574,	
1499, 1504, 1516, 1517	... 3569	2575, 2693, 3335,	
\@permissionCodeTwo	\@rightskip	3336, 3337, 3342,	
..... 1116,	... 3291, 3292	3344, 3345, 3538,	
1121, 1126, 1135,	\@saveabstract	3554, 3555, 3557, 3561	
1140, 1145, 1150,	1768, 1769	\@tempb	
1155, 1160, 1165,	\@save teaser	299, 301, 303, 305,	
1171, 1178, 1183,	... 2159, 2160	307, 309, 311, 313,	
1188, 1193, 1198,	\@savetranslatedabstract	315, 317, 319, 321,	
1207, 1212, 1217,	 455, 456		
	\@seccntformat		
	 468		
	\@secfont		
	 3265, 3295		
	\@secondoftwo		
	 30		
	\@sect		
	 243, 462		
	\@sect@kernel		
	... 243, 465		

323, 325, 327, 329, 331, 333, 335, 337, 339, 341, 346, 348, 350, 352, 354, 356, 3556, 3557, 3560, 3561	857, 924, 929, 939, 945, 2190, 2373, 2399, 2887, 2922, 2925, 3577, 3579, 3585	\acksname 401, 412, 420, 428, 3528, 3531, 3532
\@tempboxa . 722, 2412, 2414	\@vspace@acm	\ACM@adddotafter 3243, 3260
\@tempc . 344, 345, 347, 349, 351, 353, 355, 357	. . . 2399, 2925, 3585	\ACM@ArticleType 1836, 3136, 3221
\@tempdima	\@vspace@orig . . 824, 827, 835, 837, 839, 841, 843, 845, 847, 849, 851, 853, 855, 857, 924, 929, 939, 945, 2190, 2373, 2887, 2922, 3577	\ACM@ArticleType@nr 1836, 1838, 3134, 3219
2349, 2352, 2353, 2355, 2408, 2409, 2410, 2416, 2417, 2418	\@vspace@orig . . 824, 827, 835, 837, 839, 841, 843, 845, 847, 849, 851, 853, 855, 857, 924, 929, 939, 945, 2190, 2373, 2887, 2922, 3577	\ACM@cc@type . . . 1970, 2139, 2141, 2142, 2145, 2146, 2147, 2148, 2149, 2150, 2151
\@textbottom 723	\@vspace@orig . . 824, 827, 835, 837, 839, 841, 843, 845, 847, 849, 851, 853, 855, 857, 924, 929, 939, 945, 2190, 2373, 2887, 2922, 3577	\ACM@CC@Url 2140, 2141, 2142, 2143
\@textsuperscript 510, 2197, 2201	\@vspace@orig . . 824, 827, 835, 837, 839, 841, 843, 845, 847, 849, 851, 853, 855, 857, 924, 929, 939, 945, 2190, 2373, 2887, 2922, 3577	\ACM@cc@version 1969, 2141, 2152
\@texttop 724	\@vspace@orig . . 824, 827, 835, 837, 839, 841, 843, 845, 847, 849, 851, 853, 855, 857, 924, 929, 939, 945, 2190, 2373, 2887, 2922, 3577	\acm@copyrightinput 1915
\@thefnmark . 491, 505, 510, 2197, 2201, 2207	\@vspace@orig . . 824, 827, 835, 837, 839, 841, 843, 845, 847, 849, 851, 853, 855, 857, 924, 929, 939, 945, 2190, 2373, 2887, 2922, 3577	\acm@copyrightmode 1915, 1923, 1928, 1931, 1934, 1939, 1942, 1945, 1948, 1951, 1954, 1957, 1960, 1963, 1973, 2012, 2224
\@title . . . 1653, 2312, 2531, 2540, 2549, 2899	\@vspace@orig . . 824, 827, 835, 837, 839, 841, 843, 845, 847, 849, 851, 853, 855, 857, 924, 929, 939, 945, 2190, 2373, 2887, 2922, 3577	\ACM@fontsize 150, 245, 248, 250, 252, 254, 256, 258, 260, 262, 264, 266, 268, 271, 272
\@titlefont 2473, 2528, 2530, 2537, 2539, 2547	\@vspace@orig . . 824, 827, 835, 837, 839, 841, 843, 845, 847, 849, 851, 853, 855, 857, 924, 929, 939, 945, 2190, 2373, 2887, 2922, 3577	\ACM@format 36, 165
\@titlenotes 1647, 1655, 1658, 2203	\@vspace@orig . . 824, 827, 835, 837, 839, 841, 843, 845, 847, 849, 851, 853, 855, 857, 924, 929, 939, 945, 2190, 2373, 2887, 2922, 3577	\ACM@format@nr . . 36, 165, 175, 181, 187, 200, 206, 246, 523, 608, 687, 725, 869, 965, 1527, 2375, 2422, 2448, 2474, 2499, 2590, 2618, 2967, 3031, 3154, 3206, 3300, 3358, 3390, 3449, 3476, 3497
\@tochangmeasure . . . 464	\@vspace@orig . . 824, 827, 835, 837, 839, 841, 843, 845, 847, 849, 851, 853, 855, 857, 924, 929, 939, 945, 2190, 2373, 2887, 2922, 3577	\ACM@lang@check 431, 445, 448, 451, 459
\@toclevel . 3261, 3266, 3271, 3276, 3281, 3286	\@vspace@orig . . 824, 827, 835, 837, 839, 841, 843, 845, 847, 849, 851, 853, 855, 857, 924, 929, 939, 945, 2190, 2373, 2887, 2922, 3577	\ACM@languages 157, 392, 394, 432
\@tocline . . . 482, 483, 484	\@vspace@orig . . 824, 827, 835, 837, 839, 841, 843, 845, 847, 849, 851, 853, 855, 857, 924, 929, 939, 945, 2190, 2373, 2887, 2922, 3577	\ACM@linecount 2945, 2946, 2951, 2952, 2955, 2956
\@topnum 2326	\@vspace@orig . . 824, 827, 835, 837, 839, 841, 843, 845, 847, 849, 851, 853, 855, 857, 924, 929, 939, 945, 2190, 2373, 2887, 2922, 3577	\ACM@linecount@bx 2942, 2962, 2992
\@translatedabstracts . 453, 460, 2290, 2291	\@vspace@orig . . 824, 827, 835, 837, 839, 841, 843, 845, 847, 849, 851, 853, 855, 857, 924, 929, 939, 945, 2190, 2373, 2887, 2922, 3577	\ACM@linecount@bxbht 2944, 2950, 2953, 2954
\@translatedkeywords . 450, 452, 2303, 2304	\@vspace@orig . . 824, 827, 835, 837, 839, 841, 843, 845, 847, 849, 851, 853, 855, 857, 924, 929, 939, 945, 2190, 2373, 2887, 2922, 3577	\ACM@linecountL 2958, 3033, 3036, 3041, 3044, 3050, 3052, 3060,
\@translatedsubtitle 447, 449, 2533, 2542, 2551	\@vspace@orig . . 824, 827, 835, 837, 839, 841, 843, 845, 847, 849, 851, 853, 855, 857, 924, 929, 939, 945, 2190, 2373, 2887, 2922, 3577	
\@translatedtitle 444, 446, 2531, 2540, 2549	\@vspace@orig . . 824, 827, 835, 837, 839, 841, 843, 845, 847, 849, 851, 853, 855, 857, 924, 929, 939, 945, 2190, 2373, 2887, 2922, 3577	
\@typeset@author@bx 2719, 2773, 2779, 2818, 2823	\@vspace@orig . . 824, 827, 835, 837, 839, 841, 843, 845, 847, 849, 851, 853, 855, 857, 924, 929, 939, 945, 2190, 2373, 2887, 2922, 3577	
\@typeset@author@line 2649, 2683, 2707, 2712	\@vspace@orig . . 824, 827, 835, 837, 839, 841, 843, 845, 847, 849, 851, 853, 855, 857, 924, 929, 939, 945, 2190, 2373, 2887, 2922, 3577	
\@typesetengagemetadata 2287, 2930	\@vspace@orig . . 824, 827, 835, 837, 839, 841, 843, 845, 847, 849, 851, 853, 855, 857, 924, 929, 939, 945, 2190, 2373, 2887, 2922, 3577	
\@undefined 572, 792, 2174	\@vspace@orig . . 824, 827, 835, 837, 839, 841, 843, 845, 847, 849, 851, 853, 855, 857, 924, 929, 939, 945, 2190, 2373, 2887, 2922, 3577	
\@undescribed@imagesfalse 801	\@vspace@orig . . 824, 827, 835, 837, 839, 841, 843, 845, 847, 849, 851, 853, 855, 857, 924, 929, 939, 945, 2190, 2373, 2887, 2922, 3577	
\@undescribed@imagetrue . . 830, 833, 934, 2864	\@vspace@orig . . 824, 827, 835, 837, 839, 841, 843, 845, 847, 849, 851, 853, 855, 857, 924, 929, 939, 945, 2190, 2373, 2887, 2922, 3577	
\@vspace 824, 827, 835, 837, 839, 841, 843, 845, 847, 849, 851, 853, 855,	\@vspace@orig . . 824, 827, 835, 837, 839, 841, 843, 845, 847, 849, 851, 853, 855, 857, 924, 929, 939, 945, 2190, 2373, 2887, 2922, 3577	

3062, 3078, 3081, 3085, 3091, 3102, 3105, 3109, 3115, 3137, 3156, 3168, 3169, 3180, 3181, 3193, 3196, 3201, 3222	\ACM@linecountR . 2965, 3063, 3065, 3079, 3082, 3089, 3096, 3103, 3106, 3113, 3120, 3194, 3197, 3202	\ACM@mk@linecount 2947, 2960, 2975, 2977, 2979, 2981, 2983, 2985, 2987	\ACM@NR@adjustedSectionFormat 3240, 3249	\ACM@NR@just 3231, 3265, 3270, 3275, 3280, 3285, 3290	\ACM@orig@baselinestretch 3587	\ACM@orig@paragraph . 3607	\ACM@orig@section . . . 3592	\ACM@orig@subsection 3597	\ACM@orig@subsubsection 3602	\ACM@ps@headings 2936, 2940	\ACM@ps@myheadings 2935, 2939	\ACM@ps@plain . 2934, 2938	\ACM@restore@pagestyle 2934	\ACM@sect@format 3250, 3252	\ACM@sect@format@ 3239, 3252, 3254	\ACM@sect@format@null 3253, 3254	\ACM@timestamp 2995, 3128, 3207, 3212	\acmArticle 14, 1703	\acmArticleSeq . . . 14, 1705	\acmArticleType . . . 19, 1836	\acmBadge 14, 1756	\acmBadgeL 1743	\acmBadgeR 1730, 1756	\acmBooktitle 9, 1532, 1541	\acmCodeDataLink . . . 1864	\acmCodeLink 19, 1864, 1873	\acmConference . . . 9, 1518	\acmConference@date 1521, 3073, 3093, 3096, 3117, 3120, 3190	\acmConference@name ... 1520, 1532, 1533	\acmConference@shortname 1519, 1523, 1524, 1533, 1534, 2244, 3072, 3092, 3095, 3116, 3119, 3189	\acmConference@venue 1522, 2244, 3073, 3093, 3096, 3117, 3120, 3190	\acmContributions . . 1871	\acmDataLink 19, 1873	\acmdefinition 3407	\acmDOI 14, 1721	\acmISBN 14, 1719	\acmJournal 9, 1510	\acmMonth 14, 1709	\acmNumber 14, 1701	\acmplain 3375	\acmPrice 1715	\acmSubmissionID 14, 1717	\acmVolume 14, 1699	\acmYear 14, 1707	\addcontentsline 294, 2871, 3532	\additionalaffiliation 11, 1616	\addresses 1567, 1569, 1574, 1581, 1612, 1630, 1673, 1686, 1694, 1696, 2712, 2779, 2823, 2845	\addto . . . 395, 406, 414, 422	\addtocounter . . . 516, 1896	\addtolength 991, 992, 993, 997, 998, 2953	\addvspace 480	\advance 1007, 1012, 1015, 1017, 1019, 1021, 1562, 1564, 1690, 1861, 2409, 2417, 2525, 2709, 2729, 2746, 2782, 2793, 2854, 2856, 2952, 2956	\affiliation 10, 1609, 2692, 2762, 2807	\and 1549, 1582, 1584, 1599, 1605, 2683, 2688, 2690,	2701, 2707, 2720, 2754, 2773, 2779, 2801, 2818, 2823, 2842	\andify 2319, 2320, 2650, 2653, 2913	\anon 24, 2168	anonsuppress (env.) . . . 24	\apptocmd 3579, 3582	\AssignSocketPlug 914, 915	\AtBeginDocument . . 82, 99, 233, 565, 573, 983, 2941, 3018, 3148, 3255, 3334	\AtBeginEnvironment 823, 826, 835, 837, 839, 841, 843, 845, 847, 849, 851, 853, 855, 857	\AtBeginMaketitle 33, 2177	\AtEndDocument 804, 1763, 1787, 1832, 1902, 2334, 2847, 3523, 3588, 3593, 3598, 3603, 3608	\AtEndEnvironment 829, 832	\AtEndPreamble . . 538, 793, 3419, 3473, 3539	\author 10, 1559, 2186, 2187, 2893	\author@bx 2715, 2721, 2724	\author@bx@sep 2717, 2724, 2729, 2746, 2782, 2793	\author@bx@wd 2716, 2719, 2728, 2729, 2731, 2737, 2739, 2741, 2743, 2746, 2781, 2782, 2784, 2790, 2793	\authornote 13, 1616, 1671, 2894	\authornotemark . . 13, 1691	\authors 1572, 1577, 1584, 2307, 2319, 2321, 2899	\authorsaddresses 13, 1641
---	--	--	---	---	---	----------------------------	------------------------------	---------------------------	---------------------------------------	--	--	----------------------------	--------------------------------------	--	---	---	--	--------------------------------	-------------------------------	--------------------------------	------------------------------	---------------------------	---------------------------------	-----------------------------	-----------------------------	-----------------------------	------------------------------	--	---	--	--	----------------------------	-------------------------------	-----------------------------	----------------------------	-----------------------------	-------------------------------	------------------------------	-------------------------------	--------------------------	--------------------------	---------------------------	-------------------------------	-----------------------------	--	--	---	---------------------------------	-------------------------------	---	--------------------------	--	--	--	--	---	--------------------------	------------------------------	--------------------------------	----------------------------	--	--	----------------------------	--	----------------------------	--	---	-----------------------------	--	--	---	------------------------------	---	----------------------------

B

\balance 3526	\baselineskip . . 2235, 2953, 3263, 3264, 3268, 3269, 3273, 3278, 3283, 3376,
-------------------------	--

3377, 3378, 3379, 3408, 3409, 3410, 3411	\captionsspanish . . . 422	\color 2170, 2233, 2951, 2955, 3135, 3220
\baselinestretch . . .	\ccdesc@parse@finish 1907	\color@begingroup 493, 506
. . . 3587, 3588, 3590	\ccsdesc 15, 1878	\color@endbox 960
\begin 20, 2961, 2991	\ccsdesc@parse 1879, 1880	\color@endgroup
\begingroup	\ccsdesc@parse@end .	 496, 509, 720
. . 469, 561, 2189, 2588, 3232, 3341, 3530	 1879, 1907	\colorbox . 1860, 3135, 3220
\bfseries . . 1022, 2231, 2476, 2478, 2480, 2484, 2486, 2488, 2490, 2492, 2494, 2496, 2637, 2640, 2897, 3295, 3296, 3310, 3311, 3313, 3314, 3316, 3317, 3318, 3319, 3323, 3324, 3327, 3328, 3330, 3331, 3368, 3400	\ccsdesc@parse@finish 1901, 1907	\colorlet 1840, 1842, 1844, 1846, 1848
\bgroup . 460, 924, 928, 939, 945, 1617, 1643, 2293, 2299, 2340, 2719, 2756, 2759, 2820, 2828, 2859, 2869, 2886, 2898	CCSXML (env.) 15	\columnsep 3570
\bibfont 296	\centering . . 473, 506, 592, 601, 932, 941, 2537, 2539, 2721, 2778	\columnseprule 3570
\bibliofont 296	\changes 150	\columnwidth . . . 502, 703, 946, 2781, 2820
\bibliographystyle . 571	\cite . . . 554, 555, 566, 570	\contentsnamefont . . 473
\bibsection 292	\citeA 560	\copyright 2249
\bibstyle 320, 322, 364, 576	\citealt 559	\copyrightpermissionfootnoterule 702
\bibstyle@acmauthoryear 370	\citeANP 556	\copyrightyear . 17, 2156
\bibstyle@acmnumeric 376	\citeauthor 556	\correspondingauthor 13, 1679
\bigskip 1834, 2358, 2359, 2361, 2364, 2367, 2440, 2535, 2545, 2553, 2779, 2823, 2859, 2924	\citeN 554	\country 11, 1624, 2561, 2841
\box 962, 2424, 2426, 2428, 2430, 2432, 2434, 2436, 2438, 2440, 2442, 2444, 2724, 3138, 3223	\citename 567	\csname . 364, 464, 468, 476, 477, 491, 505, 1576, 1583, 1635, 1638, 1883, 1884, 1886, 1887, 1888, 1889, 1890, 1893, 1895, 2316, 2318, 3233, 3256, 3337, 3345
C	\citeNP 559	\CurrentOption
\c@footnote 2194	\citep 555	 152, 153, 154, 155, 156, 162, 163
\c@secnumdepth 463	\citestyle . . . 25, 362, 382	D
\caption@beginex@hook 912	\citeyear 558	\day 3007
\captionssenglish . . . 395	\citeyearNP 558	\DeclareNewFootnote 706, 708
\captionsetup . . 861, 863, 865, 867, 876, 878, 881, 883, 886, 888, 925, 931, 940, 2860	\citeyearpar . . . 557, 570	\DeclareOptionX
\captionsfrench 406	\city . . 11, 1622, 2561, 2838	 42, 152, 153, 154, 155, 156, 158, 162, 163
	\ClassError . . 39, 433, 457, 1508, 1615, 1682, 1770, 1780, 1786, 1800, 1851, 2676, 2892, 3589, 3594, 3599, 3604, 3609	\DeclareRobustCommand 560, 3252
	\ClassInfo	\DeclareTextFontCommand 794
	. . 165, 271, 1506, 1777, 1779, 1783, 1785, 1797, 1799, 1802	\def 25, 42, 151, 157, 248, 250, 252, 254, 256, 258, 260, 262, 264, 266, 268, 299, 301, 303, 305, 307, 309, 311, 313, 315, 317, 319, 321, 323, 324, 325, 326, 327, 328, 329, 330, 331, 332, 333,
	\ClassWarning . . . 27, 176, 182, 188, 759, 762, 765, 831, 834, 935, 1560, 1627, 1715, 1759, 1803, 2561, 2564, 2829, 2832, 2865, 3579, 3582	
	\ClassWarningNoLine . 805, 1765, 1789, 1904, 2670, 2673, 2849	
	\clearpage 2415	
	\CodelineIndex 17	
	\Collect@Body 454, 923, 927, 938, 1768, 2159	

335, 337, 339, 341,	1276, 1277, 1279,	1479, 1480, 1482,
342, 346, 348, 350,	1280, 1281, 1282,	1483, 1484, 1485,
352, 354, 356, 444,	1284, 1285, 1286,	1487, 1488, 1489,
447, 450, 453, 456,	1287, 1289, 1290,	1490, 1492, 1493,
462, 468, 469, 482,	1291, 1292, 1294,	1494, 1495, 1497,
483, 484, 485, 487,	1295, 1296, 1297,	1498, 1499, 1500,
498, 510, 516, 517,	1299, 1300, 1301,	1502, 1503, 1504,
518, 519, 520, 540,	1302, 1304, 1305,	1510, 1513, 1514,
571, 702, 709, 711,	1306, 1307, 1309,	1515, 1516, 1517,
723, 924, 928, 939,	1310, 1311, 1313,	1541, 1543, 1544,
945, 948, 952, 955,	1314, 1315, 1316,	1545, 1551, 1594,
973, 974, 975, 976,	1318, 1319, 1320,	1616, 1617, 1618,
1006, 1011, 1014,	1321, 1323, 1324,	1619, 1620, 1621,
1016, 1018, 1020,	1325, 1326, 1328,	1632, 1633, 1641,
1113, 1114, 1115,	1329, 1330, 1331,	1647, 1652, 1660,
1116, 1118, 1119,	1333, 1334, 1335,	1661, 1670, 1671,
1120, 1121, 1123,	1336, 1338, 1339,	1689, 1699, 1701,
1124, 1125, 1126,	1340, 1341, 1343,	1703, 1705, 1707,
1128, 1129, 1130,	1344, 1345, 1347,	1709, 1711, 1715,
1132, 1133, 1134,	1348, 1349, 1350,	1717, 1719, 1721,
1135, 1137, 1138,	1352, 1353, 1354,	1728, 1729, 1757,
1139, 1140, 1142,	1355, 1357, 1358,	1759, 1761, 1769,
1143, 1144, 1145,	1359, 1360, 1362,	1774, 1805, 1817,
1147, 1148, 1149,	1363, 1364, 1365,	1818, 1853, 1859,
1150, 1152, 1153,	1367, 1368, 1369,	1864, 1870, 1871,
1154, 1155, 1157,	1370, 1372, 1373,	1880, 1881, 1907,
1158, 1159, 1160,	1374, 1375, 1377,	1966, 1969, 1970,
1162, 1163, 1164,	1378, 1379, 1380,	1972, 2010, 2011,
1165, 1167, 1169,	1382, 1383, 1384,	2140, 2156, 2158,
1170, 1171, 1175,	1385, 1387, 1388,	2160, 2177, 2178,
1176, 1177, 1178,	1389, 1390, 1392,	2179, 2183, 2197,
1180, 1181, 1182,	1393, 1394, 1395,	2201, 2211, 2218,
1183, 1185, 1186,	1397, 1398, 1399,	2309, 2339, 2372,
1187, 1188, 1190,	1400, 1402, 1403,	2402, 2447, 2473,
1191, 1192, 1193,	1404, 1405, 1407,	2498, 2523, 2536,
1195, 1196, 1197,	1408, 1409, 1410,	2546, 2555, 2561,
1198, 1200, 1201,	1412, 1413, 1414,	2564, 2568, 2569,
1202, 1204, 1205,	1415, 1417, 1418,	2571, 2572, 2574,
1206, 1207, 1209,	1419, 1420, 1422,	2579, 2580, 2584,
1210, 1211, 1212,	1423, 1424, 1425,	2586, 2588, 2616,
1214, 1215, 1216,	1427, 1428, 1429,	2617, 2621, 2622,
1217, 1219, 1220,	1430, 1432, 1433,	2625, 2626, 2628,
1221, 1222, 1226,	1434, 1435, 1437,	2629, 2631, 2632,
1227, 1228, 1229,	1438, 1439, 1440,	2634, 2635, 2637,
1233, 1234, 1235,	1442, 1443, 1444,	2638, 2640, 2641,
1236, 1238, 1239,	1445, 1447, 1448,	2643, 2644, 2646,
1240, 1241, 1243,	1449, 1450, 1452,	2647, 2649, 2651,
1244, 1245, 1246,	1453, 1454, 1455,	2656, 2663, 2668,
1250, 1251, 1252,	1457, 1458, 1459,	2680, 2681, 2682,
1253, 1257, 1258,	1460, 1462, 1463,	2684, 2691, 2692,
1259, 1260, 1264,	1464, 1465, 1467,	2693, 2719, 2720,
1265, 1266, 1267,	1468, 1469, 1470,	2727, 2749, 2755,
1269, 1270, 1271,	1472, 1473, 1474,	2762, 2780, 2796,
1272, 1274, 1275,	1475, 1477, 1478,	2802, 2807, 2824,

environments:	3194, 3196, 3197,	2221, 2222, 2226,
abstract	17 3201, 3202, 3218, 3223	2234, 2237, 2238,
acks	24 \fancyheadoffset . . .	2245, 2246, 2247,
anonsuppress	24 . . . 3125, 3132, 3217	2253, 2254, 2266,
CCSXML	15 \fancyhf 3020, 3150	2271, 2278, 2279,
marginfigure	36 \fancypagestyle 3019, 3149	2280, 2281, 2282,
marginfigure	36 \fi 60, 68, 76, 85, 93, 105,	2283, 2284, 2285,
printonly	24 121, 129, 137, 147,	2287, 2291, 2296,
screenonly	24 160, 180, 186, 199,	2297, 2302, 2305,
sidebar	36 204, 234, 240, 269,	2306, 2325, 2336,
teaserfigure	17 270, 282, 287, 300,	2356, 2357, 2360,
translatedabstract		2363, 2366, 2367,
.	20 310, 312, 314, 316,	2398, 2407, 2419,
\everypar	2873, 2881 318, 320, 322, 324,	2445, 2471, 2497,
\excludecomment .	1875, 326, 328, 330, 332,	2522, 2526, 2534,
3542, 3544, 3548, 3549	334, 336, 338, 340,	2543, 2552, 2559,
\ExecuteOptionsX . . .	342, 347, 349, 351,	2576, 2577, 2579,
. . . 54, 62, 70, 78,	353, 355, 357, 358,	2581, 2583, 2587,
87, 95, 107, 111,	367, 369, 383, 388,	2613, 2648, 2655,
115, 123, 131, 139, 149	391, 400, 404, 411,	2671, 2674, 2678,
\expandafter 343, 345,	419, 427, 430, 443,	2689, 2705, 2706,
573, 722, 1574,	458, 464, 478, 497,	2710, 2721, 2744,
1575, 1581, 1582,	537, 541, 550, 568,	2745, 2753, 2761,
1635, 1638, 1883,	570, 574, 577, 604,	2772, 2791, 2792,
1884, 1886, 1887,	605, 606, 684, 700,	2800, 2806, 2817,
1893, 1895, 3233, 3256	718, 744, 756, 789,	2826, 2851, 2866,
\ExplSyntaxOff	917 790, 791, 797, 821,	2868, 2876, 2878,
\ExplSyntaxOn	894 831, 834, 892, 893,	2884, 2889, 2896,
	936, 950, 954, 981,	2900, 2914, 2917,
	1505, 1525, 1530,	2918, 2919, 2957,
F		2964, 2990, 2994,
\familydefault	796 1534, 1535, 1540,	3006, 3011, 3015,
\fancyfoot	1550, 1566, 1571,	3016, 3025, 3027,
3038, 3046, 3056,	1578, 1585, 1586,	3028, 3033, 3034,
3068, 3072, 3077,	1591, 1592, 1600,	3039, 3048, 3058,
3101, 3128, 3139,	1606, 1607, 1608,	3074, 3075, 3077,
3140, 3158, 3160,	1613, 1631, 1650,	3086, 3088, 3097,
3164, 3174, 3185,	1659, 1669, 1676,	3098, 3099, 3101,
3189, 3198, 3203,	1684, 1688, 1697,	3110, 3112, 3121,
3207, 3212, 3224, 3225	1698, 1714, 1742,	3122, 3123, 3126,
\fancyhead . 3033, 3034,	1755, 1767, 1771,	3129, 3142, 3143,
3035, 3036, 3041,	1780, 1786, 1794,	3144, 3158, 3161,
3042, 3043, 3044,	1800, 1814, 1815,	3167, 3177, 3191,
3050, 3052, 3053,	1824, 1830, 1831,	3192, 3198, 3199,
3060, 3062, 3063,	1835, 1849, 1856,	3203, 3204, 3210,
3064, 3078, 3079,	1869, 1891, 1899,	3213, 3214, 3227,
3081, 3082, 3085,	1900, 1906, 1927,	3228, 3229, 3247,
3087, 3091, 3094,	1930, 1933, 1938,	3258, 3260, 3333,
3102, 3103, 3105,	1941, 1944, 1947,	3352, 3374, 3406,
3106, 3109, 3111,	1950, 1953, 1956,	3444, 3463, 3494,
3115, 3118, 3133,	1959, 1962, 1965,	3495, 3520, 3521,
3138, 3156, 3157,	2009, 2155, 2167,	3527, 3538, 3546,
3168, 3169, 3170,	2173, 2176, 2187,	3552, 3555, 3565,
3171, 3178, 3179,	2188, 2198, 2200,	3566, 3569, 3591,
3180, 3181, 3193,	2212, 2219, 2220,	

3596, 3601, 3606, 3611	\gdef .. 160, 1519, 1520,	\hsize . 502, 1861, 2196,
\firstpagestyle 3148	1521, 1522, 1524,	2341, 2344, 2523,
\floatname 922	1541, 1546, 1548,	2525, 2527, 2536,
\flushbottom 730, 732,	1569, 1572, 1574,	2538, 2546, 2589,
734, 736, 738, 741, 743	1577, 1589, 1597,	2709, 2719, 2775, 2820
\footnote 2892	1603, 1635, 1638,	\hskip 3468
\footnotemark 520, 1653,	1732, 1745, 1772,	\hspace 1738,
1662, 2854, 2856, 2890	1804, 1821, 1827,	1751, 1859, 2022,
\footnoterule 709	1866, 1871, 1884,	2033, 2042, 2049,
\footnotesep ... 495, 508	1886, 1887, 1893,	2056, 2070, 2082,
\footnotesize .. 501,	2404, 2686, 2690,	2090, 2104, 2112,
3003, 3017, 3038,	2695, 2725, 2726,	2119, 2126, 2724, 2778
3046, 3056, 3068,	2747, 2748, 2750,	\ht 2408, 2416
3072, 3077, 3085,	2754, 2756, 2763,	\Huge . 2482, 2484, 2486,
3091, 3101, 3140,	2794, 2795, 2797,	2488, 2490, 2492, 2494
3160, 3164, 3174,	2801, 2803, 2808,	\hypersetup
3185, 3189, 3198,	2836, 2837, 2842, 3337	.. 543, 549, 551, 2310
3203, 3209, 3225, 3576	\geometry ... 610, 613,	\hyphenpenalty 3575
\footnotetext	620, 627, 634, 641,	
.... 1649, 1656,	648, 655, 662, 671, 678	
1658, 1665, 1668, 1675	\getrefnumber .. 1764,	I
\footnotetextauthorsaddresses	1788, 1903, 2346, 2889	\if 344
..... 2210, 2217	\global . 479, 499, 802,	\if@ACM@acmcp 173, 284,
\footnotetextcopyrightpermission	830, 833, 934, 961,	1854, 2213, 2228,
..... 2225, 2229	1511, 1512, 1526,	2298, 2332, 2524,
\footrulewidth	1528, 1562, 1564,	2708, 3130, 3215, 3350
3022, 3131, 3152, 3216	1565, 1610, 2321,	\if@ACM@acmthm . 71, 3420
\foreignlanguage 446, 449	2322, 2326, 2328,	\if@ACM@affiliation@obeypunctuation
\format 36	2342, 2355, 2569,	 1614, 2557,
\FrameCommand 1859	2571, 2576, 2580,	2577, 2579, 2581, 2582
\FrameRestore 1862	2584, 2586, 2664,	\if@ACM@anonymous 124,
\FrameSep 2350	2665, 2666, 2683,	1568, 1580, 1587,
\frenchspacing 3612	2707, 2711, 2721,	1611, 1629, 1654,
\fulltextwidth	2773, 2776, 2818,	1663, 1672, 1680,
..... 942, 946, 962	2861, 2864, 2952,	1692, 2163, 2169,
	2956, 3475, 3499,	2185, 2215, 3013, 3547
	3501, 3503, 3515, 3519	\if@ACM@authordraft
G	\grantnum 25, 3537	 140, 588, 2230
\g@addto@macro . 161,	\grantsponsor .. 25, 3536	\if@ACM@authorversion
446, 449, 452,		 88, 2236, 2258
460, 1549, 1581,	H	\if@ACM@badge 1723
1584, 1599, 1605,	\hbox 510, 2197, 2201	\if@ACM@balance
1612, 1630, 1653,	\headrulewidth 3021, 3151	 108, 3496, 3524
1655, 1658, 1662,	\height	\if@ACM@citypresent
1664, 1667, 1673,	1734, 1740, 1747, 1753	 2659, 2672
1674, 1686, 1689,	\hfill 2548, 2955	\if@ACM@corresponding@present
1694, 1696, 1737,	\hfuzz ... 1690, 2854, 2856	... 1648, 1677, 1681
1750, 1823, 1829,	\href 1637, 1640,	\if@ACM@countrypresent
1868, 1888, 1895,	1735, 1741, 1748,	 2661, 2675
2160, 2164, 2166,	1754, 2142, 2143,	\if@ACM@engage
2177, 2180, 2688,	2757, 2760, 2803,	... 171, 389, 402,
2701, 2752, 2759,	2805, 2844, 3558, 3562	2241, 2287, 3084, 3108
2768, 2799, 2805,	\hrule 703, 710, 2548	\if@ACM@instpresent
2813, 2853, 2855		 2657, 2669

\if@ACM@journal	\if@printcopyright .	341, 346, 348, 350,
. . . <u>167</u> , 235, 384,	 <u>1908</u> , 2248	352, 354, 356, 392,
396, 407, 415, 423,	\if@printpermission	432, 572, 1523,
862, 1537, 1810,	 <u>1910</u> , 2237	1531, 1533, 1545,
2567, 2847, 2872, 2880	\if@twocolumn . 3525, 3569	1567, 1570, 1588,
\if@ACM@journal@bibstrip	\if@undescrbed@images	1590, 1595, 1596,
. <u>168</u> , 2240, 2262,	 <u>800</u> , 804	1602, 1693, 1731,
2274, 2906, 3067, 3184	\IfBeginWith 1634	1744, 1763, 1819,
\if@ACM@journal@bibstrip@origcase	<u>169</u> , 2216, 3030, 3153	1820, 1826, 1833,
<u>169</u> , 2216, 3030, 3153	523, 608, 687, 725,	1865, 1883, 1892,
\if@ACM@maketitle@typeset	869, 965, 1111,	1902, 2174, 2209,
. . . . <u>456</u> , <u>1030</u> , 1769	1711, 1838, 1973,	2214, 2251, 2267,
\if@ACM@manuscript .	2012, 2375, 2422,	2278, 2279, 2290,
. <u>166</u> ,	2448, 2474, 2499,	2293, 2299, 2303,
279, 1807, 2239, 2255	2590, 2618, 2733,	2358, 2361, 2364,
\if@ACM@natbib	2786, 2967, 3031,	2367, 2403, 2532,
. . . <u>116</u> , 290, 381, 553	3154, 3300, 3358,	2541, 2550, 2575,
\if@ACM@natbib@override	3390, 3449, 3476, 3497	2652, 2685, 2693,
. <u>116</u>	\ifcsname 363	2694, 2749, 2755,
\if@ACM@newfonts <u>757</u> , <u>768</u>	\ifdim 2352, 2410, 2418, 2954	2762, 2796, 2802,
\if@ACM@nonacm . . <u>96</u> ,	\IfDocumentMetadata . <u>30</u>	2807, 2848, 2858,
597, 1763, 1787,	\IfDocumentMetadataF	2870, 2895, 2900,
1902, 2223, 2273,	 32, 803	2912, 2919, 3004,
2847, 2901, 3037,	\IfDocumentMetadataT	3015, 3024, 3086,
3045, 3055, 3066,	 31, 895, 918	3088, 3110, 3112,
3080, 3104, 3159,	\IfDocumentMetadataTF 30	3142, 3143, 3227,
3163, 3173, 3183, 3208	\IfEq 2139, 2145,	3228, 3233, 3256,
\if@ACM@pbalance <u>112</u> , 3474	2146, 2147, 2148,	3260, 3538, 3555,
\if@ACM@printacmref	2149, 2150, 2151, 2152	3557, 3561, 3588,
. <u>1781</u> , 2323	\IfFileExists 759, 762, 765	3593, 3598, 3603, 3608
\if@ACM@printccs . .	\IfInteger 1802	\ifxetex 769
. <u>1775</u> , 2292	\ifluatex 777	\ignorespaces . . 495,
\if@ACM@printfolios	\ifNAT@numbers 570	508, 802, 1618,
<u>1795</u> , 3025, 3027,	\ifnum 175,	1619, 1621, 1632,
3033, 3034, 3077,	181, 187, 200, 463,	2563, 2566, 2568,
3101, 3158, 3198, 3203	1527, 1764, 1788,	2570, 2571, 2572,
\if@ACM@review	1897, 1898, 1899,	2573, 2577, 2831,
. <u>79</u> , 2942, 2959, 2966	1903, 1923, 1928,	2834, 2835, 2877,
\if@ACM@screen <u>55</u> , <u>542</u> , 3540	1931, 1934, 1939,	2885, 2891, 2899, 3469
\if@ACM@sighiamode	1942, 1945, 1948,	\immediate . 477, 576, 3343
. <u>170</u> , 486,	1951, 1954, 1957,	\includecomment
596, 795, 944, 951,	1960, 1963, 2187,	. . . 3541, 3545, 3551
2198, 2200, 2721, 3124	2224, 2346, 2730,	\includegraphics . .
\if@ACM@timestamp . .	2783, 2825, 2889, 3206	 1735, 1741,
<u>132</u> , 2995, 3127, 3205	\ifPDFTeX 745	1748, 1754, 2142, 2344
\if@ACM@urlbreakonhyphens	\IfSubStr 1560, 1627	\indent 3448
. <u>63</u> , 539	\ifvmode 2556	\input 746
\if@acmowned <u>1912</u>	\ifvoid 714	\InputIfFileExists . . 26
\if@Description@present	\ifx 159, 245, 299,	\institution
. <u>798</u> ,	301, 303, 305, 307,	. <u>11</u> , 1619, <u>2561</u> , 2837
829, 832, 933, 2863	309, 311, 313, 315,	\interfootnotelinepenalty
\if@filesw 475, 575	317, 319, 321, 323,	 489
\if@insideauthorgroup	325, 327, 329, 331,	\interlinepenalty . . 489
. <u>1557</u> , 1563	333, 335, 337, 339,	\item 3468, 3572

\itshape	\leftmarginv	\LTX@adddotafter . . .
2242, 2244, 3297,	1001, 1002, 1018, 1019	 3257, 3260
3298, 3319, 3320,	\leftmarginvi	
3354, 3366, 3388, 3457	. . . 1002, 1020, 1021	
	\leftskip . 2547, 2821, 3293	
J	\let 241, 242,	M
\jobname 474, 477	243, 244, 320, 322,	\makeatletter 469
	334, 336, 338, 364,	\makebox 2231,
K	461, 466, 467, 511,	3133, 3138, 3218, 3223
\kern 702, 703, 709, 710, 2548	512, 554, 555, 556,	\MakeFramed 1861
\keywords 15, 1761	557, 558, 559, 562,	\makelabel 1025
\keywordsname	566, 724, 792, 824,	\maketitle 18, 2183
385, 387, 397, 399,	825, 827, 828, 835,	\MakeUppercase
408, 410, 416, 418,	836, 837, 838, 839,	. . . 2686, 2688, 3244
424, 426, 1645, 2300	840, 841, 842, 843,	\marginfigure 927
	844, 845, 846, 847,	marginfigure(env.) . . . 36
	848, 849, 850, 851,	\marginpar 487, 926, 932, 941
	852, 853, 854, 855,	\marginparsep . . 943, 3125
	856, 857, 858, 924,	\marginparwidth . 943, 3125
	925, 929, 930, 939,	\margintable 938
	940, 945, 946, 947,	margintable(env.) . . . 36
	1010, 1025, 1029,	\mathchardef . . 2757, 2760
	1622, 1623, 1624,	\mbox 342
	1756, 1762, 1873,	\mdseries . . 2501, 2503,
	1876, 2175, 2190,	2505, 2509, 2511,
	2191, 2192, 2193,	2513, 2515, 2517,
	2206, 2207, 2208,	2519, 2521, 2638, 2641
	2211, 2218, 2307,	\medskip . . 2377, 2379,
	2321, 2322, 2373,	2381, 2383, 2711,
	2374, 2399, 2400,	2713, 2777, 2867, 2897
	2585, 2683, 2707,	\MessageBreak
	2773, 2779, 2818,	. . . 177, 183, 189,
	2823, 2887, 2888,	190, 191, 192, 193,
	2922, 2923, 2925,	194, 195, 196, 197,
	2926, 2934, 2935,	434, 435, 436, 437,
	2936, 2938, 2939,	439, 441, 806, 815,
	2940, 3238, 3240,	1790, 1791, 1792, 1851
	3242, 3243, 3245,	\metadata@authors . .
	3251, 3254, 3257,	. . . 2307, 2308, 2311
	3572, 3577, 3578,	\mktitle@bx 2182, 2408,
	3585, 3586, 3587,	2409, 2412, 2416,
	3592, 3597, 3602, 3607	2417, 2424, 2426,
\lastbox 2873, 2881	\libertineLF 794	2428, 2430, 2432,
\leaders 2548	\lineskip 2778, 2822	2434, 2436, 2438,
\leavevmode . . . 2548, 2777	\linespacing . . . 472, 473	2440, 2442, 2444,
\leftmargin 1006, 1011,	\liningnums 792	2528, 2537, 2547,
1014, 1016, 1018, 1020	\list 1023	2711, 2776, 2777, 2861
\leftmargini 990, 991,	\listisep . 1004, 1005, 1009	\month 1710, 3007
992, 993, 1006, 1007	\listparindent 1008	\moveleft 961
\leftmarginii 996, 997,	\LoadClass 272	
998, 999, 1011, 1012	\long 456, 487, 498, 924,	N
\leftmarginiii	928, 939, 1769,	\NAT@setcites 360
999, 1000, 1014, 1015	1772, 1774, 2160, 2928	\NAT@aysep 351
\leftmarginiv	\loop 2411, 2951	\NAT@citetp 562
1000, 1001, 1016, 1017		\NAT@close
		300, 302, 304, 306, 349

<code>\NAT@cmprs</code> ..	328, 330, 332	<code>\newtheoremstyle</code> ...	727, 728, 729, 731,
<code>\NAT@cmt</code>	355		3375, 3407
<code>\NAT@ctype</code>	562	<code>\newwrite</code>	476
<code>\NAT@find@eq</code>	343	<code>\next</code>	3235, 3237, 3247
<code>\NAT@fullfalse</code>	562	<code>\nobreak</code>	2897
<code>\NAT@longnametrue</code> .	340	<code>\noindent</code> ..	485, 1646,
<code>\NAT@merge</code> ..	334, 336, 338	1834, 2208, 2295,	971, 972, 977, 978,
<code>\NAT@nmfmt</code>	342	2301, 2377, 2379,	979, 980, 1117,
<code>\NAT@numbersfalse</code> ..	314	2381, 2383, 2414,	1122, 1127, 1131,
<code>\NAT@numberstrue</code> 316, 318		2421, 2528, 2530,	1136, 1141, 1146,
<code>\NAT@open</code>	300,	2533, 2542, 2548,	1151, 1156, 1161,
302, 304, 306, 347, 562		2549, 2551, 2650,	1166, 1174, 1179,
<code>\NAT@partrue</code>	562	2711, 2712, 2776,	1184, 1189, 1194,
<code>\NAT@rem@eq</code>	345	2822, 2861, 2862,	1199, 1203, 1208,
<code>\NAT@sep</code> 308, 310, 312, 357		2897, 2898, 2931, 3458	1213, 1218, 1225,
<code>\NAT@sort</code> ...	324, 326, 332	<code>\noindentparagraph</code> .	1232, 1237, 1242,
<code>\NAT@superfalse</code>	316		2389, 3281
<code>\NAT@supertrue</code>	318	<code>\normalbaselines</code> ...	2720
<code>\NAT@swafalse</code>	561	<code>\normalcolor</code>	716
<code>\NAT@up</code>	342	<code>\normalfont</code>	510,
<code>\NAT@yrsep</code>	353	1834, 2201, 2617,	1298, 1303, 1308,
<code>\NeedsTeXFormat</code>	1	2622, 2631, 2632,	1312, 1317, 1322,
<code>\newblock</code>	563	2634, 2635, 2647,	1327, 1332, 1337,
<code>\newbox</code>		2948, 3369, 3386,	1342, 1346, 1351,
2182, 2338, 2554, 2715		3398, 3401, 3466, 3612	1356, 1361, 1366,
<code>\newcommand</code> .	370, 376,	<code>\normalparindent</code> ...	701
431, 445, 448, 451,		<code>\normalsize</code>	
569, 802, 1518,		2498, 2617, 2631,	1371, 1376, 1381,
1609, 1643, 1679,		2632, 2635, 2949, 3612	1386, 1391, 1396,
1691, 1730, 1743,		<code>\num@authorgroups</code> ..	1401, 1406, 1411,
1818, 1878, 1968,		1553, 1564, 2187,	1416, 1421, 1426,
2168, 2573, 2582,		2733, 2737, 2739, 2786	1431, 1436, 1441,
3002, 3281, 3528,		<code>\num@authors</code>	1446, 1451, 1456,
3537, 3553, 3567, 3568		1555, 1562, 1576,	1461, 1466, 1471,
<code>\newcount</code> .	1553, 1555, 2945	1583, 1636, 1639, 2825	1476, 1481, 1486,
<code>\newcounter</code> 1877, 2998, 3000		<code>\numexpr</code> .	1491, 1496, 1711,
<code>\newdimen</code> 288, 942, 982, 2716		2997, 2999, 3001	1712, 1713, 1841,
<code>\newenvironment</code> .	454,	<code>\nxandlist</code>	1843, 1845, 1847,
927, 938, 1858, 2159		2308	1974, 1976, 1979,
<code>\newfloat</code>	921		1981, 1982, 1984,
<code>\newif</code>	166,		1986, 1988, 1991,
167, 168, 169, 170,			1993, 1996, 1998,
171, 173, 757, 798,			2001, 2004, 2007,
800, 1030, 1557,			2013, 2024, 2035,
1677, 1908, 1910,			2043, 2050, 2057,
1912, 2657, 2659, 2661			2071, 2083, 2091,
<code>\newlabel</code>	3344		2105, 2113, 2120,
<code>\newlength</code>			2127, 2132, 2138,
1723, 1725, 1726, 2944			2378, 2380, 2382,
<code>\newsavebox</code>	2943		2384, 2386, 2388,
<code>\newskip</code>	1004, 2717		2390, 2392, 2394,
<code>\newtheorem</code>			2396, 2425, 2427,
3423, 3426, 3429,			2429, 2431, 2433,
3432, 3435, 3439, 3442			2435, 2437, 2439,
			2441, 2443, 2451,
			2453, 2455, 2457,
			2459, 2461, 2463,

2465, 2467, 2469, 2477, 2479, 2481, 2483, 2485, 2487, 2489, 2491, 2493, 2495, 2502, 2504, 2506, 2508, 2510, 2512, 2514, 2516, 2518, 2520, 2593, 2595, 2597, 2599, 2601, 2603, 2605, 2607, 2609, 2611, 2620, 2623, 2624, 2627, 2630, 2633, 2636, 2639, 2642, 2645, 2735, 2736, 2738, 2740, 2788, 2970, 2972, 2974, 2976, 2978, 2980, 2982, 2984, 2986, 2988, 3040, 3049, 3059, 3162, 3172, 3182, 3302, 3303, 3306, 3309, 3312, 3315, 3321, 3325, 3329, 3332, 3360, 3361, 3362, 3363, 3364, 3365, 3370, 3371, 3372, 3373, 3392, 3393, 3394, 3395, 3396, 3397, 3402, 3403, 3404, 3405, 3451, 3452, 3453, 3454, 3455, 3456, 3459, 3460, 3461, 3462, 3478, 3479, 3480, 3482, 3484, 3486, 3488, 3490, 3491, 3493, 3500, 3502, 3504, 3506, 3508, 3510, 3512, 3514, 3516, 3518	85, 93, 105, 109, 113, 121, 129, 137, 147 \PackageInfo 57, 59, 65, 67, 73, 75, 81, 84, 90, 92, 98, 104, 118, 120, 126, 128, 134, 136, 142, 146, 1173, 1224, 1231, 1248, 1255, 1262 \PageIndex 16 \pageref 3009 \pagestyle 3146 \par . 446, 449, 471, 712, 956, 1646, 1834, 2211, 2218, 2237, 2242, 2244, 2295, 2301, 2344, 2359, 2362, 2365, 2369, 2377, 2379, 2381, 2383, 2424, 2426, 2428, 2440, 2444, 2533, 2535, 2542, 2545, 2548, 2551, 2553, 2579, 2581, 2583, 2650, 2654, 2711, 2713, 2720, 2722, 2723, 2750, 2752, 2759, 2768, 2777, 2779, 2797, 2799, 2805, 2813, 2821, 2823, 2859, 2860, 2861, 2863, 2877, 2897, 2920, 2924, 2928, 2931, 3464 \paragraph 3276, 3607, 3608, 3610 \parbox 592, 601, 2529, 2539, 2948 \parindent . . 685, 701, 991, 2211, 2218, 2235, 2343, 3277, 3294, 3355, 3387, 3573 \parskip . . 685, 2235, 3573 \part 3286 \PassOptionsToClass 162, 163 \pdfgentounicode . . 754 \pdfglyphtounicode 747, 748, 749, 750, 751, 752, 753 \pdfstringdefDisableCommands 515 \phantomsection 294, 2871, 3532	 1623, 1624, 2561, 2835 \postcode 2561, 2832 printonly (env.) 24 \ProcessOptionsX . . . 164 \proof 3464 \proofname 3464 \protected 1633 \protected@edef . 490, 504 \protected@write . . 2354 \providecommand . . 30, 31, 32, 563, 567, 3353 \ProvidesClass 5, 6 \ProvidesFile 3 \ps@headings . . 2936, 2940 \ps@myheadings 2935, 2939 \ps@plain 2934, 2938 \pushQED 3465 \put 2962, 2992 Q \qed 3465 \quad 468, 3051, 3054, 3061, 3065, 3207 R \raggedright 2529, 2547, 2821, 3574 \raisebox 1734, 1740, 1747, 1753, 2231, 3134, 3219 \real@adddotafter 3242, 3245 \realSectionformat 3238, 3250, 3251 \received 17, 1818 \RecordChanges 18 \ref 2909, 2916 \refname 293, 294 \relax 175, 181, 187, 200, 207, 247, 289, 343, 344, 432, 477, 524, 609, 688, 724, 726, 870, 966, 1112, 1527, 1531, 1554, 1556, 1562, 1564, 1690, 1691, 1693, 1764, 1788, 1839, 1862, 1883, 1892, 1902, 1903, 1923, 1928, 1931, 1934, 1939, 1942, 1945, 1948, 1951,
P \p@ 480, 685, 692, 702, 703, 709, 710, 3131, 3216, 3263, 3268, 3273, 3274, 3278, 3279, 3283, 3284, 3288, 3289, 3466, 3570, 3573 \PackageError 60, 68, 76,		

1954, 1957, 1960, 1963, 1973, 2012, 2206, 2207, 2224, 2351, 2352, 2355, 2376, 2423, 2449, 2475, 2500, 2591, 2619, 2718, 2728, 2729, 2730, 2731, 2734, 2737, 2739, 2741, 2743, 2746, 2778, 2781, 2782, 2783, 2784, 2787, 2790, 2793, 2825, 2854, 2856, 2889, 2946, 2968, 2969, 2971, 2973, 2989, 2999, 3001, 3004, 3032, 3155, 3206, 3233, 3256, 3260, 3301, 3359, 3391, 3450, 3466, 3477, 3498, 3557, 3561, 3573	<code>\reset@font</code> 488, 501 <code>\reversemarginpar</code> . . 669 <code>\rightarrow</code> 1880 <code>\rightskip</code> 3292 <code>\rotatebox</code> 3135, 3220 <code>\rule</code> 495, 508 S <code>\savebox</code> 2948 <code>screenonly (env.)</code> 24 <code>\scriptsize</code> 2345, 2951, 2955 <code>\scshape</code> 3356, 3447 <code>\section</code> 293, 2385, 2387, 2391, 2393, 2395, 2397, 2875, 2883, 3261, 3531, 3592, 3593, 3595 <code>\section@raggedright</code> 3291, 3295, 3296, 3304, 3305, 3307, 3308, 3310, 3311, 3313, 3314, 3316, 3317, 3318, 3323, 3324, 3327, 3328, 3330, 3331 <code>\Sectionformat</code> 3234, 3238, 3240, 3251 <code>\SelectFootnoteRule</code> 705, 707 <code>\selectlanguage</code> 1644, 2879 <code>\set@ACM@acmcpbobox</code> 2333, 2338 <code>\setbox</code> 499, 961, 2342, 2412, 2528, 2537, 2547, 2711, 2721, 2776, 2861, 2873, 2881 <code>\setccotype</code> 16, 1968 <code>\setcitestyle</code> 26, 297, 371, 377 <code>\setcopyright</code> 15, 1966 <code>\setcounter</code> 2195, 2202, 2288, 2406, 2999, 3001, 3299, 3322, 3326, 3351 <code>\setengagemetadata</code> 18, 2179 <code>\setkeys</code> . 42, 179, 185, 1510, 1805, 1853, 1966, 2696, 2697, 2702, 2703, 2764, 2765, 2769, 2770, 2809, 2810, 2814, 2815 <code>\setlength</code> 685, 686, 692, 701, 984,	985, 990, 996, 999, 1000, 1001, 1002, 1724, 1727, 2343, 2950 <code>\setmathfont</code> 772, 773, 780, 781 <code>\setmonofont</code> . . . 775, 783 <code>\settopmatter</code> . . . 17, 102, 1529, 1793, 1805, 1806, 1808, 1811, 1813, 1816, 2905 <code>\setTrue</code> 470 <code>\SetWatermarkColor</code> 591, 600 <code>\SetWatermarkFontSize</code> 590, 599 <code>\SetWatermarkText</code> 592, 601 <code>\sfdefault</code> 796 <code>\sffamily</code> . . 2476, 2478, 2480, 2482, 2484, 2486, 2490, 2494, 2496, 2616, 2621, 2625, 2646, 3017, 3295, 3296, 3297, 3304, 3305, 3307, 3308, 3313, 3314, 3323, 3324, 3327, 3328 <code>\shortauthors</code> 18, 1588, 1589, 1596, 1597, 1599, 1602, 1603, 1605, 2320, 2322, 3016 <code>\shortcite</code> 566, 569 <code>\shorttitle</code> 3036, 3044, 3054, 3065, 3078, 3102 <code>\showeprint</code> 3553 <code>\showISBNx</code> 3567 <code>\showISBNxiii</code> 3567 <code>\sidebar</code> 921 <code>sidebar (env.)</code> 36 <code>\skip</code> 715 <code>\small</code> 488, 926, 933, 941, 1834, 2377, 2379, 2381, 2383, 2622, 2647, 2873, 2881, 2897, 3158 <code>\smallskipamount</code> . . 1005 <code>\smash</code> 1733, 1739, 1746, 1752 <code>\socket</code> 897, 908 <code>\space</code> 1561, 1628, 1793, 2316, 3142, 3227, 3538, 3579, 3582, 3590, 3595, 3600, 3605, 3610 <code>\specialcomment</code> . . . 3529 <code>\standardpagestyle</code> . 3018
---	---	--

<code>\startPage</code>	15, 1757	<code>\theACM@time@hours</code> .		<code>\unvbox</code> .	500, 717, 722,
<code>\state</code> .	11, 1623, 2561, 2839		3001, 3008		2414, 2711, 2777, 2861
<code>\stepcounter</code> 1656, 1658,		<code>\theACM@time@minutes</code> 3008		<code>\upshape</code>	1022
1665, 1668, 1675, 1882		<code>\theenumi</code>	973	<code>\url</code> .	1866, 1868, 2010, 3538
<code>\StrDel</code>	2574	<code>\theenumii</code>	974	<code>\UrlBreakPenalty</code> . .	
<code>\streetaddress</code> 2561, 2829		<code>\theenumiii</code>	975		2757, 2760
<code>\string</code>	434,	<code>\theenumiv</code>	976	<code>\urlstyle</code> . . .	522, 531, 534
435, 438, 440, 576,		<code>\thefootnote</code>	2194	<code>\usebox</code>	2962, 2992
1561, 1628, 1715,		<code>\theindex</code>	3569	<code>\usepackage</code>	
1759, 1793, 2355,		<code>\theoremstyle</code>	3418, 3421, 3437, 3445		10, 11, 12, 13, 14, 15
2893, 2894, 3344,		<code>\thepage</code>		V	
3579, 3582, 3590,			3009, 3025, 3027,	<code>\value</code>	1899
3595, 3600, 3605, 3610			3033, 3034, 3077,	<code>\vbox</code>	499, 961,
<code>\strut</code>	3136, 3221		3101, 3158, 3198, 3203		2342, 2528, 2537,
<code>\strutbox</code>	495, 508	<code>\thispagestyle</code> 2413, 2420			2547, 2711, 2776, 2861
<code>\subsection</code>		<code>\thmname</code>	3385, 3417	<code>\vskip</code> . .	471, 715, 723, 956
	3266, 3597, 3598, 3600	<code>\thmnote</code>	3385, 3417	<code>\vspace</code>	2353, 3577
<code>\subsection</code>		<code>\thmnumber</code>	3385, 3417	<code>\vsplit</code>	2412
	3271, 3602, 3603, 3605	<code>\thr@@</code>	338	<code>\vtop</code>	2721
<code>\subtitle</code>	10, 1551	<code>\time</code>	2996, 2999, 3001	W	
<code>\subtitlenote</code> . .	13, 1661	<code>\title</code>	10, 2893	<code>\wd</code>	963
T		<code>\titlenote</code> . .	13, 1652, 2894	<code>\write</code>	576, 3343
<code>\tag</code>	902, 905, 910	<code>\tl</code>	900	X	
<code>\tagpdfparaOff</code>	913	<code>\topsep</code> . .	1009, 1013, 3466	<code>\xdef</code>	347,
<code>\tagpdfparaOn</code>	899	<code>\translatedabstract</code> 454			349, 351, 353, 355, 357
<code>\tagpdfsetup</code>	919	translatedabstract		Y	
<code>\teaserfigure</code>	2159	(env.)	20	<code>\year</code>	1708, 3007
<code>\teaserfigure (env.)</code> . . .	17	<code>\translatedkeywords</code>		Z	
<code>\terms</code>	1759		20, 451	<code>\ze</code>	326, 330,
<code>\textbf</code> . .	1885, 1897, 2931	<code>\translatedsubtitle</code>			472, 495, 508, 686,
<code>\textbullet</code> 518, 1884,			20, 448		723, 864, 868, 879,
3051, 3054, 3061, 3065		<code>\translatedtitle</code> .	20, 445		990, 996, 1008,
<code>\textheight</code> 2410, 2412,		<code>\trivlist</code>	3467		2211, 2218, 2231,
2418, 2954, 3134, 3219		<code>\tw@</code>	336		2235, 2326, 2328,
<code>\textit</code>	1898,	<code>\two@digits</code> . . .	3007, 3008		2343, 2821, 2873,
2907, 2911, 2912, 2913		<code>\twocolumn</code> .	2430, 2432,		2881, 3021, 3022,
<code>\textrightarrow</code>			2434, 2436, 2438, 2442		3133, 3138, 3151,
519, 1880, 1894		U			3152, 3218, 3223,
<code>\textsuperscript</code> . . .	1690	<code>\unskip</code>	517, 713,		3262, 3267, 3272,
<code>\textwidth</code> 943, 962, 963,			1625, 1632, 2558,		3282, 3287, 3294,
2196, 2523, 2536,			2563, 2566, 2568,		3367, 3399, 3570, 3573
2546, 2589, 2728, 2775			2570, 2571, 2572,	<code>\z@skip</code> 471, 956, 1013, 3293	
<code>\thanks</code>	13, 2161, 2928		2573, 2654, 2831,	<code>\zposy</code>	2349, 2350
<code>\thankses</code>			2834, 2835, 2837,	<code>\zrefused</code>	2347, 2348
	2164, 2166, 2209, 2928		2838, 2839, 2840,	<code>\zsaveposy</code>	1862, 2368
<code>\the</code> . .	1576, 1583, 1636,		2841, 2842, 2844,		
1639, 1708, 1710,			2891, 2899, 2916, 2929		
2355, 2951, 2955, 3007					